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Travel Patterns and Characteristics of the Population from Zero Vehicle Households in New York State



Latif Patwary
Melrose Pan
Majbah Uddin

May 2025

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ZERO VEHICLE HOUSEHOLDS IN NEW YORK STATE**

Latif Patwary
Melrose Pan
Majbah Uddin

May 2025

Prepared for
New York State Department of Transportation

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OAK RIDGE NATIONAL LABORATORY
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ABBREVIATIONS AND ACRONYMS

<i>drvrcnt</i>	driver count
HH	household
<i>hhvehcnt</i>	household vehicle count
ICT	information and communication technology
MOE	margin of error
Nass-Suff	Nassau and Suffolk counties, New York
Putn-Rock-West	Putnam, Rockland, and Westchester counties, New York
NHTS	National Household Travel Survey
NYC	New York City
NYMTC	New York Metropolitan Transportation Council
NYS	New York State
PMT	person miles of travel
POV	privately owned vehicle
VMT	vehicle miles of travel

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EXECUTIVE SUMMARY

This study explores how zero vehicle and vehicle deficit (i.e., fewer cars than licensed drivers) households manage their daily travel needs and overall mobility decisions. Using the 2009, 2017, and 2022 National Household Travel Survey, the research team investigates key patterns such as trip rates, trip distances, travel modes, and trip purposes, as well as evaluates the impact of information and communication technologies like online shopping and telework. Additionally, the team evaluates how the COVID-19 pandemic has influenced travel behavior, with a focus on New York State, where a significant number of zero vehicle and vehicle deficit households offer unique perspectives on the challenges and opportunities associated with limited vehicle access.

Sociodemographic Characteristics

- In both 2009 and 2017, about two-thirds of Manhattan’s population ages five and older belonged to zero vehicle households—by far the largest share of any region. The rest of New York City (NYC) saw a decline in zero vehicle and vehicle deficit households over that period. For the entire state, the zero vehicle household population share was 23%, whereas the share of vehicle deficit households showed a slight decline (from 22% in 2009 to 18% in 2017).
- Manhattan’s vehicle deficit households typically shared just one vehicle among multiple drivers (100% in 2009 and 96% in 2017), whereas suburban areas (Nassau, Suffolk, Putnam, Rockland, and Westchester) more often had multiple vehicles but still not enough for every driver.
- Zero vehicle and vehicle deficit households in Manhattan reported higher employment rates (63% and 72%, respectively) than those with enough vehicles for every driver, likely owing to dense public transit networks and the concentration of jobs. Elsewhere, particularly in the suburbs and rural regions, car access appeared more critical for securing employment.
- In 2017, zero vehicle households in dense urban areas had a wide range of annual incomes (e.g., one-fifth earning more than \$100,000), whereas those in rural and smaller urban areas skewed heavily to lower income brackets (more than 70%–80% below \$25,000). These zero vehicle households were predominantly renters, whereas vehicle deficit and not-deficit households showed substantially higher homeownership—especially outside Manhattan.
- About half of households that own more than one vehicle yet remain in deficit were multiadult families with children. In contrast, one-vehicle-multiple-driver households were more evenly divided, with 35% composed of adults without children and another 35% composed of multiadult families with children.

Trip Patterns

- Over time, zero vehicle households showed relatively stable average daily person-trip rates (3.1 trips) in both NYS and the rest of the US. In contrast, a significant decline has been observed among vehicle deficit and not-deficit households since 2009, yet they had significantly higher trip rates than zero vehicle households.
- In 2017, zero vehicle households in Manhattan made significantly more trips per day (around 3.7 trips) compared with those in other NYS regions. They even made significantly more trips than nonurban households having enough vehicles, indicating that car-free living does not always translate to fewer trips—especially where transit and walk options are available.

- From 2009 to 2022, daily person trips declined in the Middle Atlantic census division for all household types. However, zero vehicle households in the region still averaged slightly more daily trips than their counterparts in the rest of the US (though this difference was not always statistically significant).
- Households without vehicles consistently registered shorter trip distances than those of car-owning households, although the difference in significance varied by region. In suburban areas, not-deficit drivers generally traveled the longest distances.
- Zero vehicle households relied heavily on walking and public transit, especially in NYC (e.g., 59% walked in Manhattan and 41% used public transit in the rest of NYC in 2017), but vehicle deficit and not-deficit groups in suburban and rural regions leaned more toward private vehicle trips.
- Zero vehicle households, particularly in dense urban areas, devoted a higher share of trips to earning a living and to social and recreational activities. They also tended to cluster more trips in late morning and early afternoon, whereas vehicle-owning households distributed travel more evenly, peaking during commuter rush hours.
- In NYS, as expected, vehicle deficit households had significantly fewer vehicle miles of travel than those of not-deficit vehicle households (17 and 26 miles, respectively) in 2017. However, no significant difference was observed between these households in Manhattan.

Use of Transportation-Related Technologies

- In 2017, zero vehicle households in NYC relied on e-commerce (38% shopped online 1–5 times and 12% shopped online 6–10 times) more than their counterparts in other urban or nonurban regions. For example, 23% shopped online 1–5 times in other urban regions, and 75% still do not shop online in nonurban regions. Meanwhile, vehicle-owning households, especially in suburban or rural settings, reported fewer “no purchase” shares.
- Although 21% of zero vehicle households in NYC used rideshare services 1–5 times per month in 2017, nearly 69% never used them. Outside NYC, nearly all zero vehicle households refrained from ride-sharing. Even among vehicle-owning households, rideshare adoption was rare in suburban and nonurban areas.
- Car-sharing use was minimal across all regions, though zero vehicle households in NYC reported slightly higher adoption (7% used it at least once per month). Bikeshare was more widely used in NYC, with nearly 29% of zero vehicle households using it at least twice per month. However, in suburban and rural regions, bikeshare adoption was negligible across all vehicle ownership categories.

Views of Transportation Services

- Nearly 18% of zero vehicle households in NYC and the New York Metropolitan Transportation Council (NYMTC) region stated that fuel costs affected their travel. In contrast, 69% of nonurban vehicle deficit households and 59% of not-deficit households reported that gas prices significantly influenced their travel decisions.
- Walking to save money was more common among zero vehicle households in urban areas, with 65% in NYC and 71% in other urban areas choosing to walk.

- Public transit was a primary cost-saving measure for zero vehicle households, with 75% in NYC and 74% in NYMTC using it to reduce expenses. In contrast, public transit was far less relevant as a cost-saving option in nonurban areas, where only 4% of vehicle deficit and 6% of not-deficit households relied on it because of limited transit availability.

Travel Behavior During COVID-19

- During the COVID-19 pandemic, zero vehicle households in the Middle Atlantic census division saw a notable increase in online shopping, with 34% reporting that they ordered online more frequently than before. However, this was lower than vehicle deficit households (47%) and not-deficit households (39%).
- Public transit use declined substantially among zero vehicle households, with 50% in the Middle Atlantic and 43% in the rest of the US reporting reduced transit usage compared with that of prepandemic levels. Despite this decline, zero vehicle households remained more dependent on public transit than car-owning groups. Notably, 12% of zero vehicle households in the Middle Atlantic and 5% in the rest of the US reported using transit more frequently during COVID-19.
- Work commute patterns among zero vehicle households in the Middle Atlantic showed substantial shifts, with 13% commuting more often than before the pandemic and 35% commuting less frequently. This level of volatility was higher than that of vehicle-owning households, the majority of which followed prepandemic work travel patterns. In the rest of the US, 7% of zero vehicle households commuted more frequently, and 27% reduced their commute trips.

1. INTRODUCTION

1.1 BACKGROUND

Zero vehicle households and those with fewer vehicles than drivers (“vehicle deficit” households) represent a growing share in many urban and suburban areas (Li, 2025). According to Blumenberg et al. (2020), approximately 7% of US households have no personal vehicles, and another 15% have fewer vehicles than licensed drivers in the households. The travel choices of these households have important implications for transportation planning because these groups often rely on alternative modes, such as public transit, active transportation, or shared mobility, to meet their travel needs. Understanding how and why households manage without enough cars is essential for designing resilient infrastructure, particularly given that vehicle deficit households can be found across a wide range of incomes and residential contexts (Blumenberg & Smart, 2014). Their experience highlights gaps in current transportation systems and opportunities to improve mobility through targeted investments.

At the same time, the growing prevalence of information and communication technologies (ICTs), including online shopping, teleworking (i.e., work from home), and on-demand services, has added a new dimension to household travel behavior. Researchers have noted that e-commerce can substitute some physical trips (Mokhtarian, 2002; Patwary & Khattak, 2022), and telecommuting can reduce or shift travel demand to nonpeak hours (Faber et al., 2024). These changes may be especially meaningful for zero vehicle and vehicle deficit households, which must balance limited car access with potentially burdensome travel requirements for errands, work, and family obligations. If households can rely on home delivery, remote work, or other ICT-facilitated options, they may be better able to compensate for their limited vehicle availability. Alternatively, lack of reliable broadband or insufficient familiarity with new technologies may hinder the adoption of these potentially beneficial practices (Etminani-Ghasrodashti & Hamidi, 2020).

Beyond ICT use, zero vehicle and vehicle deficit households frequently turn to shared mobility solutions such as bike-sharing, car-sharing, and ride-sharing (Blumenberg et al., 2020). In dense urban centers, these services can fill mobility gaps by offering access without car ownership, and in less dense areas, they may provide critical last-mile connections (Shaheen & Chan, 2016). Analyzing how different household types use such options sheds light on the evolving role of shared modes in the broader transportation ecosystem. Overall, focusing on zero vehicle and vehicle deficit households helps clarify how innovations in ICT and shared mobility can transform travel behavior, reduce car dependence, and improve accessibility for people who, by choice or necessity, live with limited vehicle access.

New York State (NYS) offers a particularly compelling context in which to explore zero vehicle and vehicle deficit households. Its spectrum of environments—from the dense core of New York City (NYC), where transit usage is among the highest in the nation, to sprawling suburban and rural communities—illustrates the varied realities faced by households lacking sufficient cars. NYS also ranks among the states with the highest proportion of zero vehicle households (TitleMax, 2019; Tomer, 2011), underscoring the relevance of exploring how these groups adapt their travel choices in areas with markedly different levels of transit service, population density, and economic opportunity. By investigating the constraints and adaptations of these households across the state, policymakers and planners can better understand how to bridge mobility gaps, support better mobility access, and develop infrastructure that accommodates a wide range of travel behaviors and needs.

This study addresses the pressing need to anticipate and understand the evolving travel preferences of zero vehicle and vehicle deficit households in NYS. By comparing these groups’ travel behaviors with those of households that have enough vehicles, the study attempts to provide valuable insights into where and how to target investments, refine policy initiatives, and design services to better serve diverse

populations. Moreover, this focus complements NYS’s broader transportation goals—reducing carbon emissions, promoting accessibility and affordability, and enhancing overall quality of life—by investigating the mobility patterns of households that are inherently less car-dependent. This report continues a series of special population analyses that use the 2017 National Household Travel Survey (NHTS) data to investigate distinct demographic and mobility segments. Previous installments examined rural residents, older adults, low-income communities, and millennial populations (Liu et al., 2022; Pan et al., 2024; Patwary et al., 2025; Uddin et al., 2024). Building on that foundation, the current study’s emphasis on zero vehicle and vehicle deficit households contributes to a more complete understanding of how contemporary lifestyle changes, alternative transportation modes, and technology innovations can shape the future of travel in NYS.

1.2 OBJECTIVES

This study examines the travel behaviors, demographics, and mobility choices of zero vehicle and vehicle deficit households, with the goal of understanding how their limited car access influences overall transportation patterns. In particular, it seeks to evaluate the roles of sociodemographic factors, emerging technologies, and the COVID-19 pandemic in shaping these households’ daily travel. Drawing on NHTS data across multiple years (2009, 2017, and 2022) and regions (New York City, New York State, Middle Atlantic census division, and the rest of the US), this analysis offers a comprehensive view of how constraints on vehicle ownership affect mobility decisions. The specific objectives are as follows:

- Examine sociodemographic characteristics (e.g., gender, household status, and geographic location) of zero vehicle and vehicle deficit households.
- Assess travel behaviors of these households in terms of daily trip rates, trip lengths, travel purposes, and mode choices.
- Compare these households’ travel patterns to those of households with enough vehicles to identify key differences and potential mobility gaps.
- Investigate the role of various influences—such as gas prices, travel cost, and walking access—in shaping travel decisions for households with limited vehicle access.
- Evaluate the effects of new transportation-related technologies, including ride-sharing, car-sharing, bike-sharing, online shopping, and telework, on reducing these households’ reliance on privately owned vehicles (POVs).
- Explore how the COVID-19 pandemic has changed travel behaviors, with particular attention to shifts in commuting patterns, online purchases, and the adoption of flexible work arrangements.

1.3 NATIONAL HOUSEHOLD TRAVEL SURVEY (NHTS)

The NHTS is the principal dataset for examining everyday travel across the US. It compiles in-depth information on how individuals make daily trips—including trip frequency, distance, duration, mode, and purpose—as well as personal, household, and vehicle characteristics (NHTS, 2025).

For this study, researchers draw on three series of NHTS: 2009, 2017, and 2022. The 2009 NHTS included approximately 150,000 households nationwide, offering an extensive baseline of US travel behavior at that time. In 2017, the survey included more than 129,000 households—26,000 from the national sample plus additional respondents from 13 state or metropolitan planning organization add-ons. Among these were more than 17,000 households from NYS, which enables a more granular look at zero

vehicle and vehicle deficit households in the region. The 2022 NHTS, part of the Next Generation initiative, captured responses from more than 25,000 households across the country, providing insights into travel patterns during the COVID-19 pandemic. Although NYS did not participate as an add-on in the 2022 survey, the data still reflect broader pandemic-era shifts—such as increased telework and online shopping—and the ways households with limited vehicle access navigated these changes. By comparing findings from all three survey years, this study sheds light on how zero vehicle and vehicle deficit households adapt to evolving transportation trends and how these adaptations vary between NYS and the rest of the US.

1.4 HOUSEHOLD BREAKDOWN BY VEHICLE OWNERSHIP

This study distinguishes household vehicle ownership by comparing the total number of vehicles in the household (i.e., household vehicle count, or *hhvehcnt*) with the total number of licensed drivers (i.e., driver count, or *drvrcnt*). Specifically, vehicle ownership is categorized as follows:

- **Zero vehicle household:** households that own no vehicles at all (*hhvehcnt* = 0)
- **Vehicle deficit household:** households that own at least one vehicle but have fewer vehicles than licensed drivers (*hhvehcnt* > 0 and *hhvehcnt* < *drvrcnt*)
- **Not vehicle deficit household:** households that own enough vehicles for every licensed driver or that have more vehicles than drivers (*hhvehcnt* > 0 and *hhvehcnt* ≥ *drvrcnt*)

Because not all households that fall into the “vehicle deficit” category have the same conditions—some may own only a single vehicle, whereas others possess multiple vehicles yet still not enough for each driver—this study further separates vehicle deficit households into the following:

- **One vehicle multiple drivers in the household:** Only one vehicle is available to multiple drivers in the household.
- **Vehicle deficit with more than one vehicle in the household:** These households own multiple vehicles but not enough for each licensed driver.

1.5 GEOGRAPHICAL REGION BREAKDOWN BY NHTS YEAR

To capture the travel patterns—ranging from urban cores to less dense communities—this study employs a geographical categorization reflecting both regional and national perspectives. Within NYS, multiple subareas are distinguished based on population density and administrative boundaries, and a wider regional context is incorporated through the Middle Atlantic census division.

For the 2009 and 2017 survey periods, this analysis focuses on NYS and several subregions that provide a nuanced view of local travel:

- **Manhattan:** the densely populated core of NYC, characterized by extensive public transit and high levels of walkability
- **Rest of NYC:** the other four boroughs (Bronx, Brooklyn, Queens, and Staten Island), which also feature robust transit networks but vary in density and land use
- **Nass-Suff:** Nassau and Suffolk counties on Long Island, generally suburban in character, with moderate-to-high car dependence

- **Putn-Rock-West:** Putnam, Rockland, and Westchester counties, suburban enclaves north of NYC, where commuting patterns often mix rail, bus, and private vehicles
- **Other Urban:** mid-sized cities and surrounding areas within NYS that exhibit an urban form distinct from that of the NYC region
- **Nonurban:** rural communities with comparatively limited public transportation options and more dispersed land uses
- **NYS:** an aggregate that encompasses all areas within the state, offering a statewide overview
- **Rest of US:** all survey respondents outside NYS, providing a nationwide perspective

Because the 2022 NHTS does not provide state-level information, direct comparisons at the state or subregional level are not feasible. Instead, this study adopts a more aggregated approach by distinguishing the following:

- **Middle Atlantic (Mid-Atlantic) Census Division:** Composed of New York, New Jersey, and Pennsylvania, this division reflects broader regional trends pertinent to NYS.
- **Rest of US:** Comprising all other states outside the Mid-Atlantic, this division serves as a complementary national reference group.

1.6 ORGANIZATION OF REPORT

This report is structured to provide a comprehensive examination of zero vehicle and vehicle deficit households, particularly within NYS. Section 2 describes their demographic characteristics—such as household composition, income, and vehicle-related attributes—while also drawing comparisons across different parts of NYS and, where applicable, the rest of the US. Section 3 focuses on key travel metrics, including person trips, miles traveled, and trip purposes, illustrating how limited vehicle access influences daily mobility. Section 4 investigates these households’ adoption of various transportation-related technologies, from online shopping and telecommuting to ride-, car-, and bike-sharing, and assesses how these innovations may reduce reliance on private car travel. Section 5 explores the views of transportation services by zero vehicle and vehicle deficit households. Section 6 concentrates on changes brought about by COVID-19, specifically shifts in public transit use, commuting to physical workplaces, and online purchasing patterns. Although the 2022 NHTS data do not permit analyses for NYS, this section still offers valuable insights into how the pandemic reshaped travel in the broader Mid-Atlantic census division and the rest of the country. Finally, the report concludes with a discussion of key findings and directions for future research.

2. DEMOGRAPHIC PROFILE OF ZERO VEHICLE AND VEHICLE DEFICIT HOUSEHOLDS

2.1 POPULATION BY HOUSEHOLD VEHICLE OWNERSHIP TYPE

2.1.1 Share of Population

The distribution of population (individuals ages 5 years and older) by vehicle ownership type across NYS regions and the rest of the US is illustrated in Figure 2-1 based on 2009 and 2017 NHTS data. Some interesting trends and findings emerge from the data. Manhattan stands out with by far the largest share of zero vehicle households in both 2009 and 2017, accounting for about two-thirds of the population ages 5 years and older. Meanwhile, the rest of NYC showed a notable shift away from zero vehicle or vehicle deficit households (which together declined from 70% to 60%) and toward “not vehicle deficit” households over that same period. In the suburban counties around NYC (Nassau, Suffolk, Putnam, Rockland, and Westchester), approximately three-quarters or more of households each year were not vehicle deficit. However, the proportion of vehicle deficit households dipped slightly between 2009 and 2017.

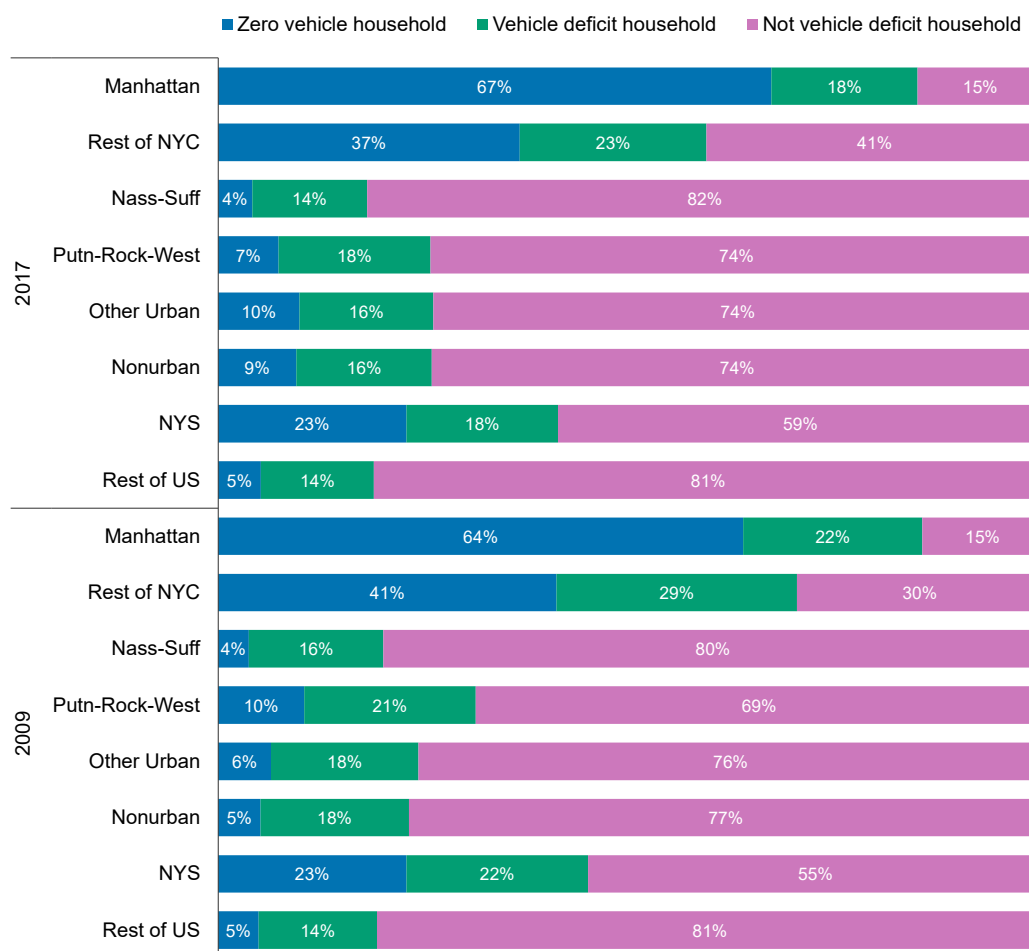


Figure 2-1. Distribution of population age 5 years and older by vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Other urban areas and nonurban parts of the state displayed slight increases in zero vehicle households, though still at much lower levels than those in Manhattan. Statewide, zero vehicle households remained roughly steady at just under a quarter of the population (23%), whereas the share of vehicle deficit households showed a slight decline. Notably, the rest of the US exhibited very little change over time, with around 5% zero vehicle households and over 80% not vehicle deficit in both years. The rest of the US had a lower percentage in the vehicle deficit category compared with that of NYS in 2017.

Population distribution is further explored based on vehicle deficit household type in Table 2-1. In both 2009 and 2017, Manhattan’s vehicle deficit households were almost exclusively those with a single vehicle shared by multiple drivers (100% in 2009, dipping slightly to 96% in 2017). This stands in contrast to suburban counties (Nassau, Suffolk, Putnam, Rockland, and Westchester) where a larger share of vehicle deficit households had more than one vehicle but still not enough to match the number of drivers. For example, in Nassau–Suffolk, the proportion of “more than one vehicle” deficit households rose from about half (51%) to nearly two-thirds (65%) between 2009 and 2017. In the same period, Putnam–Rockland–Westchester also saw a shift from 55% to 59% in that multivehicle deficit category. Other urban and nonurban areas showed relative stability in terms of vehicle deficit categories, with about 50% of vehicle deficit households in these areas consisting of one vehicle for multiple drivers. Similarly, NYS statewide remained stable across the period, with about two-thirds of vehicle deficit households being classified under the “one-vehicle-multiple-drivers” category.

Table 2-1. Distribution of population ages 5 years and older by vehicle deficit household type.

Year	Region	One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household
2009	Manhattan	100%	0%
	Rest of NYC	78%	22%
	Nass-Suff	49%	51%
	Putn-Rock-West	45%	55%
	Other Urban	51%	49%
	Nonurban	50%	50%
	NYS	67%	33%
	Rest of US	52%	48%
2017	Manhattan	96%	4%
	Rest of NYC	78%	22%
	Nass-Suff	35%	65%
	Putn-Rock-West	41%	59%
	Other Urban	61%	39%
	Nonurban	61%	39%
	NYS	67%	33%
	Rest of US	60%	40%

2.1.2 Population Race

The distribution of population by household race and household vehicle ownership type presents several noteworthy observations (Table 2-2). In Manhattan, for instance, zero vehicle households skewed mostly toward Whites (65%), but they were notably more racially diverse than the “vehicle deficit” category, which was overwhelmingly White at 89%. By contrast, in Putnam–Rockland–Westchester, zero vehicle households included a significant share of African Americans (44%), which was higher than the share of Whites (38%). Outside the city, nonurban areas tended to be predominantly White across all household

types, exceeding 90% in some categories. Meanwhile, other urban centers (beyond the NYC metro area) showed fairly balanced mixes of White and African American zero vehicle households. At the state and national levels, zero vehicle households were somewhat more racially distributed than the fully “vehicle-secure” (i.e., not deficit) households, but the gap was not as pronounced as it was in Manhattan or the surrounding suburbs.

Table 2-2. Population distribution by household race and vehicle ownership type (2017 NHTS data).

Region	Vehicle ownership type	White	African American	Hispanic	Other
Manhattan	Zero vehicle household	65%	9%	15%	11%
	Vehicle deficit household	89%	3%	0%	8%
	Not vehicle deficit household	77%	11%	6%	6%
Rest of NYC	Zero vehicle household	39%	29%	17%	15%
	Vehicle deficit household	51%	19%	6%	23%
	Not vehicle deficit household	47%	23%	9%	22%
Nass-Suff	Zero vehicle household	47%	35%	0%	18%
	Vehicle deficit household	72%	10%	1%	17%
	Not vehicle deficit household	82%	6%	4%	8%
Putn-Rock-West	Zero vehicle household	38%	44%	11%	7%
	Vehicle deficit household	62%	11%	10%	17%
	Not vehicle deficit household	70%	14%	6%	9%
Other Urban	Zero vehicle household	45%	40%	8%	7%
	Vehicle deficit household	69%	18%	4%	9%
	Not vehicle deficit household	81%	10%	3%	6%
Nonurban	Zero vehicle household	69%	19%	6%	6%
	Vehicle deficit household	91%	3%	0%	6%
	Not vehicle deficit household	93%	2%	1%	5%
NYS	Zero vehicle household	47%	25%	14%	13%
	Vehicle deficit household	65%	15%	4%	16%
	Not vehicle deficit household	73%	12%	5%	10%
Rest of US	Zero vehicle household	47%	35%	7%	11%
	Vehicle deficit household	67%	14%	6%	13%
	Not vehicle deficit household	77%	10%	5%	8%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

2.1.3 Population Employment

A clear geographic divide emerges in how employment status aligns with vehicle ownership (Figure 2-2). In Manhattan, for example, zero vehicle and vehicle deficit households showed a surprisingly high share of employed individuals (63% and 72%, respectively), even higher than the households with no vehicle deficit (56%). This likely reflects Manhattan’s dense urban environment, high reliance on public transit, and younger workforce more apt to live without owning any vehicles.

Elsewhere, however, having fewer cars than drivers tends to coincide with lower employment rates, especially in suburban or rural areas. In Nassau–Suffolk, for example, only 14% of those in zero vehicle households were employed, in contrast to about half or more of those in households with enough vehicles.

Similarly, in nonurban areas, zero vehicle households had a very low employment share (26%). This pattern suggests that outside Manhattan (and certain parts of NYC), reliable vehicle access may be more critical for holding a job, owing to fewer alternatives like public transit.

Overall, vehicle deficit households and those without deficits typically post higher employment rates than zero vehicle households in most regions, except for Manhattan. This highlights the importance of local transportation infrastructure and urban density in shaping the relationship between car ownership and employment.

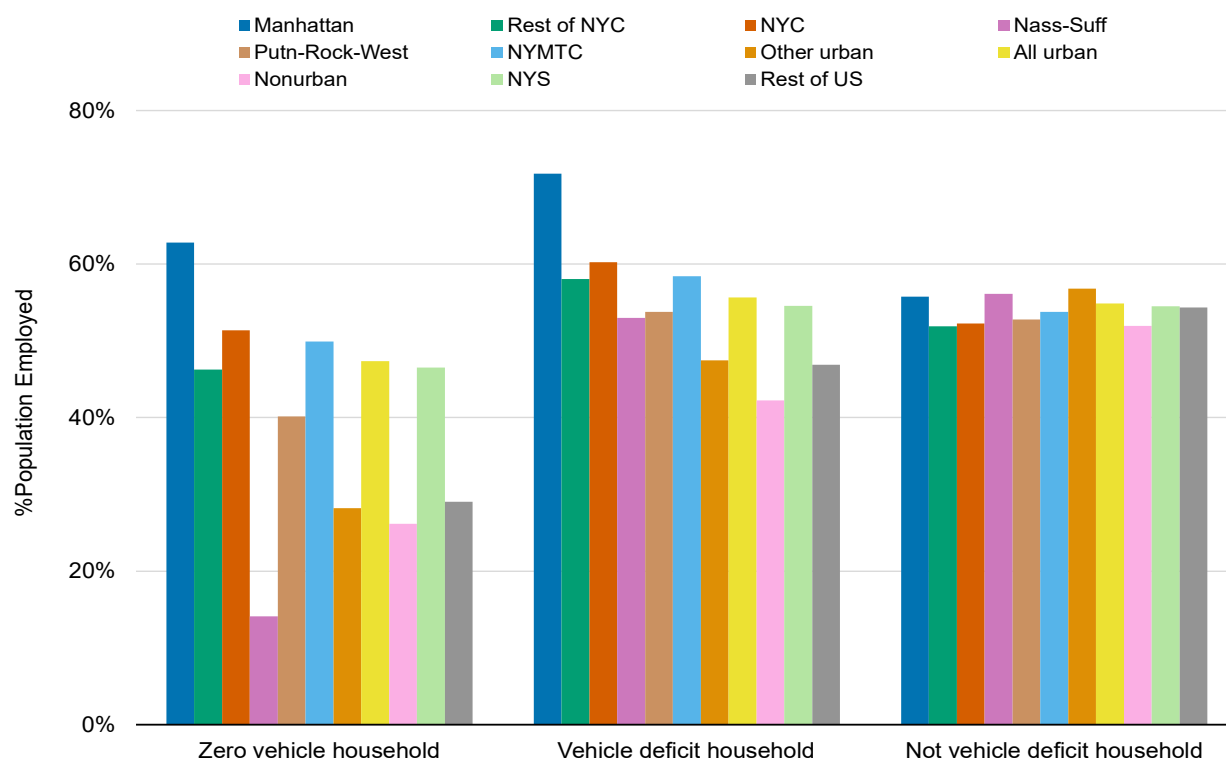


Figure 2-2. Percent of population employed by household vehicle ownership type (2017 NHTS data).

Looking at the Mid-Atlantic census division over time (2009, 2017, and 2022), some noteworthy shifts in employment rates for each vehicle ownership category can be observed, especially compared with the rest of the US (Table 2-3). Zero vehicle households in the Mid-Atlantic saw their employment share climb steadily (from 39% in 2009 to 48% in 2022). This highlights how increased transit accessibility or changing workforce demographics might be boosting workforce participation among those without cars. In contrast, vehicle deficit households in the Mid-Atlantic dipped from 57% employed in 2009 to 49% in 2022. Meanwhile, zero vehicle households' employment rates also rose (from 26% to 38%) in the rest of the country, though they remained lower than rates in the Mid-Atlantic. Vehicle deficit households and those with no deficit outside the Mid-Atlantic saw smaller gains in employment share.

Table 2-3. Percent population employed by household vehicle ownership type.

Year	Region	Zero vehicle household	Vehicle deficit household	Not vehicle deficit household
2009	Mid-Atlantic division	39%	57%	57%
	Rest of US	26%	47%	56%
2017	Mid-Atlantic division	42%	53%	55%
	Rest of US	29%	47%	54%
2022	Mid-Atlantic division	48%	49%	56%
	Rest of US	38%	45%	54%

2.2 HOUSEHOLD CHARACTERISTICS BY VEHICLE OWNERSHIP TYPE

2.2.1 Share of Households

Figure 2-3 illustrates the share of households by vehicle ownership type across regions based on 2017 NHTS data. Similar to the population distribution in Section 2.1.1, Manhattan stands out with an exceptionally high share of zero vehicle households—over 70%—with only 16% not vehicle deficit. By contrast, suburban areas (Nassau, Suffolk, Putnam, Rockland, and Westchester) had very few zero vehicle households, often well under 15%, and the vast majority (around three-quarters or more) had enough vehicles to meet drivers’ needs. In the rest of NYC, almost half of all households were zero vehicle, indicating that many city residents outside Manhattan also rely on transit or other forms of travel.

At the statewide level, nearly one-third of all households had no vehicles, which was heavily influenced by the high zero vehicle share in NYC. Meanwhile, outside NYS, the country overall was dominated by car ownership, with less than 10% of households lacking vehicles. These patterns reflect the huge urban–suburban divide in travel behavior: high reliance on transit in Manhattan and other parts of NYC versus greater auto dependency in surrounding suburbs and much of the rest of the country.

Furthermore, in the Mid-Atlantic census division, which includes NYS, about one-fifth of households consistently reported owning no vehicles over the years (Figure 2-4). This is substantially higher than the 7% zero vehicle rate in the rest of the US. Meanwhile, vehicle deficit households in the Mid-Atlantic region declined slightly over time (from 13% in 2009 and 2017 to 10% by 2022), with a corresponding growth in the share of households that had enough vehicles (from 67% in 2009 to 71% in 2022). By contrast, in the rest of the country, the proportion of households meeting all driver needs stayed high (over 80%) but dipped slightly in 2022. Overall, the Mid-Atlantic had more zero vehicle households than the rest of the nation, while also seeing a gradual reduction in households lacking enough vehicles.

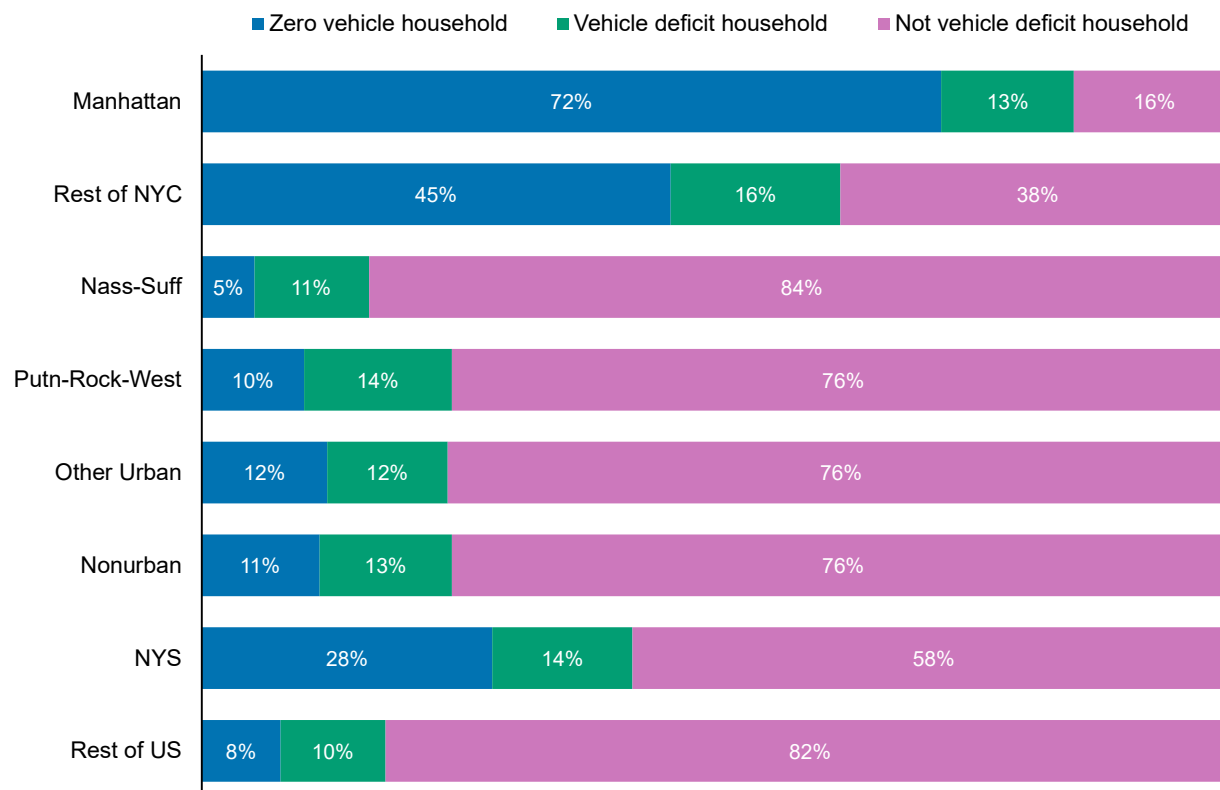


Figure 2-3. Share of households by vehicle ownership type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

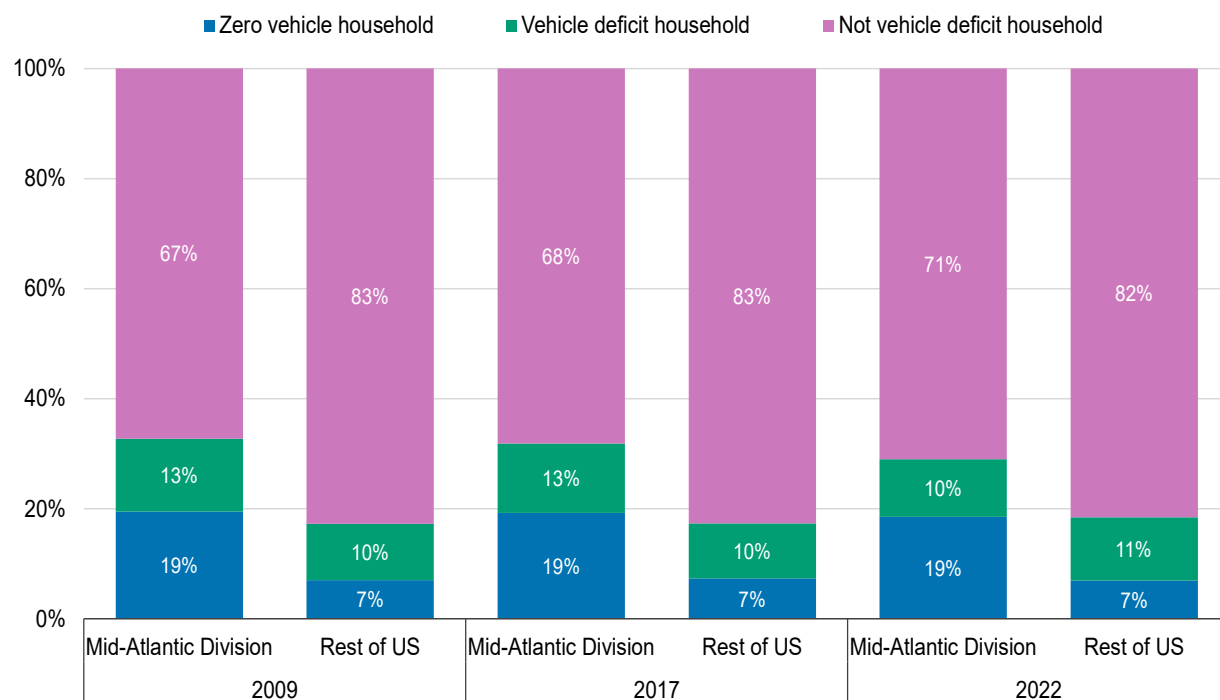


Figure 2-4. Share of households by vehicle ownership type in the Mid-Atlantic. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

2.2.2 Household Size

Figure 2-5 shows interesting patterns in household size distribution by vehicle ownership type across the regions in 2017. A clear pattern emerges in which zero vehicle households in urban areas like Manhattan and the rest of NYC were predominantly one-person units, whereas suburban regions like Nassau–Suffolk or Putnam–Rockland–Westchester reported a surprisingly substantial share of larger families (four or more persons) among those with no cars at all. This suggests that in high-density areas with ample transit, smaller households are more likely to opt out of car ownership; in less transit-friendly suburbs, bigger families may be zero vehicle households by necessity or preference.

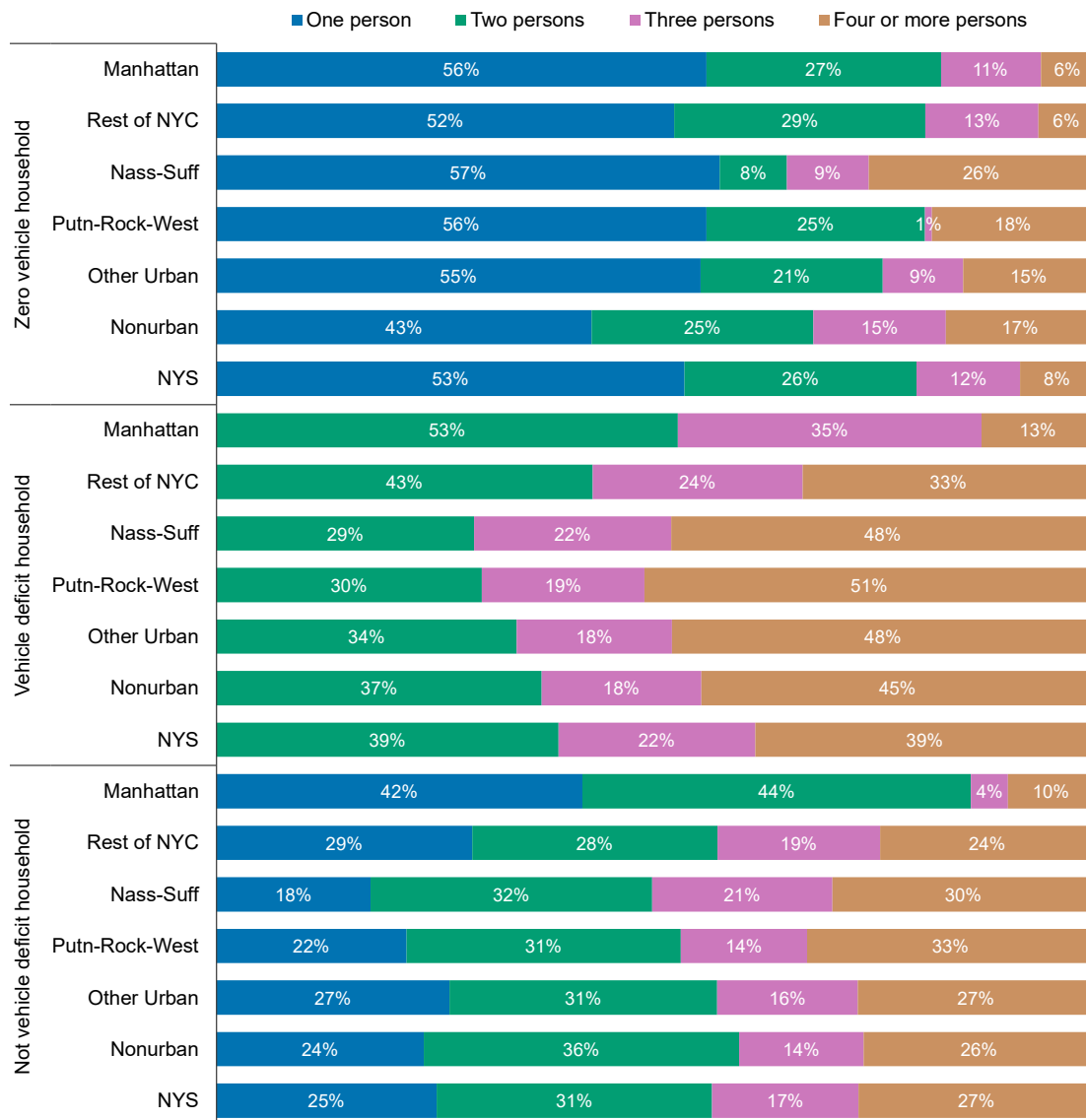


Figure 2-5. Household size distribution by vehicle ownership type and region (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Because vehicle deficit households are multiperson households, the largest suburban and rural families (four or more persons) disproportionately fall into this category. By contrast, households with enough vehicles typically feature more balanced household size distributions in urban areas.

2.2.3 Household Income

Household income distribution by vehicle ownership type across regions in 2017 is illustrated in Figure 2-6. A notable split emerges when looking at zero vehicle households in major urban areas versus outside areas. In NYC and the broader New York Metropolitan Transportation Council (NYMTC) region, zero car households spanned a wide range of incomes, with about one-fifth earning more than \$100,000. However, nearly all zero car households in other urban and especially nonurban areas fell into the lowest income bracket (more than 70% and 80% below \$25,000, respectively). This points to a substantial difference between those who choose to relinquish car ownership in transit-rich places, including some higher earners in NYC, versus those in more remote or smaller cities who may lack both transportation and sufficient financial resources.

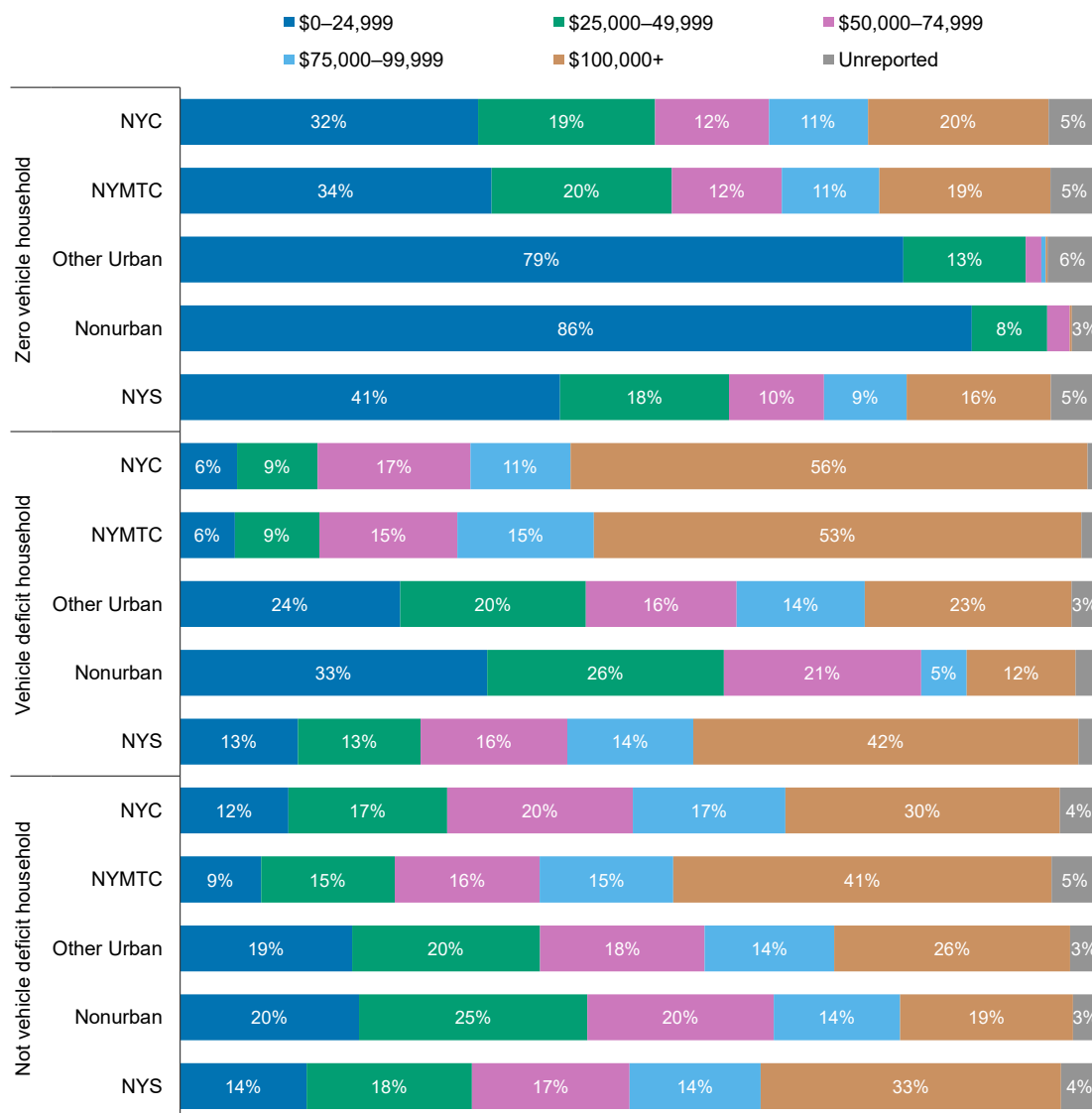


Figure 2-6. Household income distribution by vehicle ownership type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

By contrast, vehicle deficit households in NYC and NYMTC skewed to higher income (over half above \$100,000). These are multidriver households for which one or two cars suffice in an environment with robust transit or walk/bike options. Outside the immediate metro area, however, most vehicle deficit households occupied lower- to middle-income brackets, suggesting that fewer cars than drivers can be an economic constraint in areas with more limited public transportation. Finally, households with enough vehicles also differed regionally. Many households in NYC and NYMTC earned \$100,000 or more; other urban and nonurban regions showed a more balanced distribution of incomes, though still skewing to higher incomes than those of zero vehicle households in those areas.

When household income distribution is compared with vehicle deficit household type, an interesting contrast appears between metropolitan and more rural areas (Table 2-4). For example, in NYC and NYMTC regions, both types of deficit households—whether a single vehicle shared by multiple drivers or multiple vehicles still insufficient for all drivers—tended to have higher income. In these areas, half or more of deficit households earned \$100,000 or above, suggesting that at least some households choose to limit car ownership because of dense urban environments, robust transit, or lifestyle preferences, even though they have the means to purchase additional vehicles.

By comparison, nonurban and many other urban households with only one car serving multiple drivers were clustered heavily in the lower-income brackets (e.g., 30%–40% below \$25,000 in nonurban settings). This pattern likely reflects economic constraints rather than a discretionary choice to have fewer vehicles. Meanwhile, among those who have “vehicle deficit with more than one vehicle,” suburban and rural households showed more variation in incomes but still a noticeable share in the lower- to middle-income brackets.

Table 2-4. Household income distribution by vehicle deficit household type (2017 NHTS data).

Vehicle deficit household type	Region	\$0–24,999	\$25,000–49,999	\$50,000–74,999	\$75,000–99,999	\$100,000+	Unreported
One vehicle with multiple drivers in household	NYC	7%	9%	18%	10%	56%	0%
	NYMTC	8%	10%	17%	12%	52%	1%
	Other Urban	28%	24%	19%	13%	13%	3%
	Nonurban	37%	30%	21%	5%	4%	3%
	NYS	15%	15%	18%	12%	39%	1%
Vehicle deficit with more than one vehicle in household	NYC	0%	8%	11%	14%	60%	7%
	NYMTC	0%	6%	8%	22%	58%	6%
	Other Urban	15%	12%	10%	16%	44%	3%
	Nonurban	23%	16%	22%	6%	33%	0%
	NYS	7%	9%	10%	19%	51%	4%

2.2.4 Home Ownership

Home ownership distribution by vehicle ownership type is presented in Table 2-5. Zero vehicle households were dominated by renters in practically every region, from about 77% in Manhattan and 82% in the rest of NYC to over 90% in other urban areas. Even in suburban or rural contexts, at least three-quarters of zero vehicle households rented their homes. This reinforces the notion that going car-free (by choice or necessity) often aligns with more transient or lower-commitment housing arrangements, particularly in areas with strong rental markets and better transit options. Conversely, Manhattan, vehicle deficit households and especially households with enough vehicles showed much higher homeownership

rates. In Nassau–Suffolk and Putnam–Rockland–Westchester, the majority of vehicle deficit households (up to 87% in Putnam–Rockland–Westchester) and the vast majority of nondeficit households (86% in Nassau–Suffolk and 72%–77% in other suburbs/rural) were homeowners. This is likely tied to suburban homeownership norms and the necessity of owning multiple vehicles for families spread across larger geographies with fewer transit options. Manhattan is an outlier because even the majority of households with a vehicle surplus still rented (around 61%), reflecting Manhattan’s overall high-rental housing market and high cost of home ownership.

Table 2-5. Home ownership distribution by vehicle ownership type (2017 NHTS data).

Household vehicle ownership type	Region	Own	Rent	Some other arrangement	Unreported
Zero vehicle household	Manhattan	22.1%	77.1%	0.8%	0.0%
	Rest of NYC	17.2%	82.1%	0.4%	0.3%
	Nass-Suff	40.3%	59.7%	0.0%	0.0%
	Putn-Rock-West	14.8%	85.2%	0.0%	0.0%
	Other Urban	8.7%	90.7%	0.6%	0.0%
	Nonurban	15.0%	84.8%	0.2%	0.0%
	NYS	17.9%	81.5%	0.5%	0.1%
	Rest of US	15.3%	82.9%	1.8%	0.0%
Vehicle deficit household	Manhattan	36.5%	63.5%	0.0%	0.0%
	Rest of NYC	70.2%	29.8%	0.0%	0.0%
	Nass-Suff	75.8%	24.2%	0.0%	0.0%
	Putn-Rock-West	87.2%	12.3%	0.5%	0.0%
	Other Urban	55.1%	43.8%	1.1%	0.0%
	Nonurban	64.6%	31.4%	4.1%	0.0%
	NYS	64.5%	34.8%	0.7%	0.0%
	Rest of US	54.5%	43.9%	1.7%	0.0%
Not vehicle deficit household	Manhattan	35.8%	60.7%	3.5%	0.0%
	Rest of NYC	55.1%	42.8%	2.0%	0.0%
	Nass-Suff	86.0%	12.9%	1.0%	0.0%
	Putn-Rock-West	72.7%	25.6%	1.7%	0.0%
	Other Urban	68.1%	31.1%	0.8%	0.0%
	Nonurban	77.2%	21.8%	1.0%	0.0%
	NYS	69.3%	29.5%	1.3%	0.0%
	Rest of US	69.3%	29.8%	0.9%	0.0%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

In vehicle deficit households, a clear divide emerges between households sharing one vehicle among multiple drivers and those with more than one vehicle but still a deficit (Figure 2-7). In nearly every region, the multiple-vehicle deficit group exhibited higher homeownership rates, often over 80%, even in NYC and the NYMTC region. This suggests that these households tend to be larger families or higher-income suburban vehicle owners who have not quite matched their vehicle count to the number of drivers. By contrast, one-vehicle-multiple-driver households were more likely to rent, especially in other urban areas and the rest of the US, where over 50% were renters.

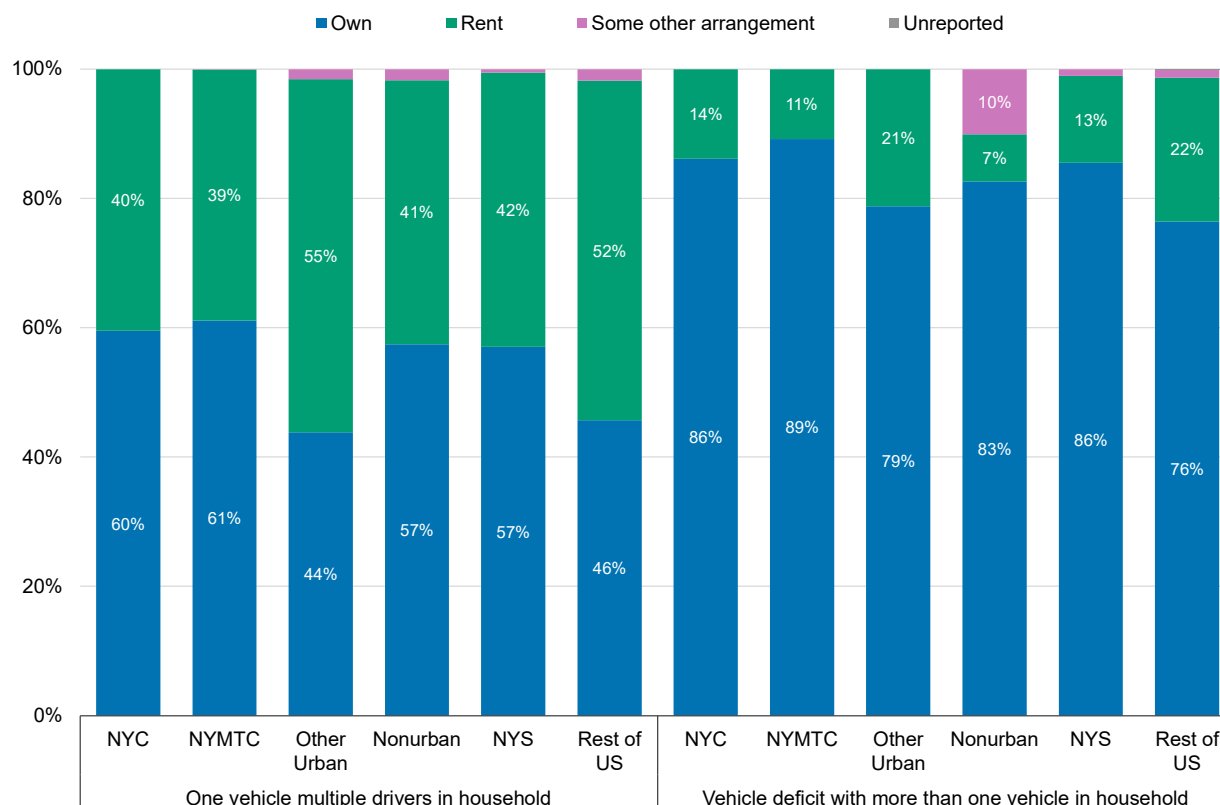


Figure 2-7. Home ownership distribution by vehicle deficit household type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

2.2.5 Vehicle Age

The distribution of vehicle age by vehicle deficit household across regions in 2017 is shown in Figure 2-8. Looking at vehicle age sheds light on how households manage their fleets, particularly when they have fewer cars than drivers. In Manhattan, for instance, the share of vehicles 11 years old or older was relatively high among both vehicle deficit (36%) and nondeficit (36%) households. By contrast, vehicle deficit households in the rest of NYC had somewhat fewer older vehicles (28%) than their nondeficit counterparts (36%). Outside the city, patterns were somewhat mixed. Nonurban areas showed a substantial proportion of older vehicles, with 35% in vehicle deficit households and 39% in those without a deficit. This underscores that rural or small-town residents—who often rely more heavily on personal cars—may keep older vehicles in service longer, regardless of how many cars they own relative to drivers.

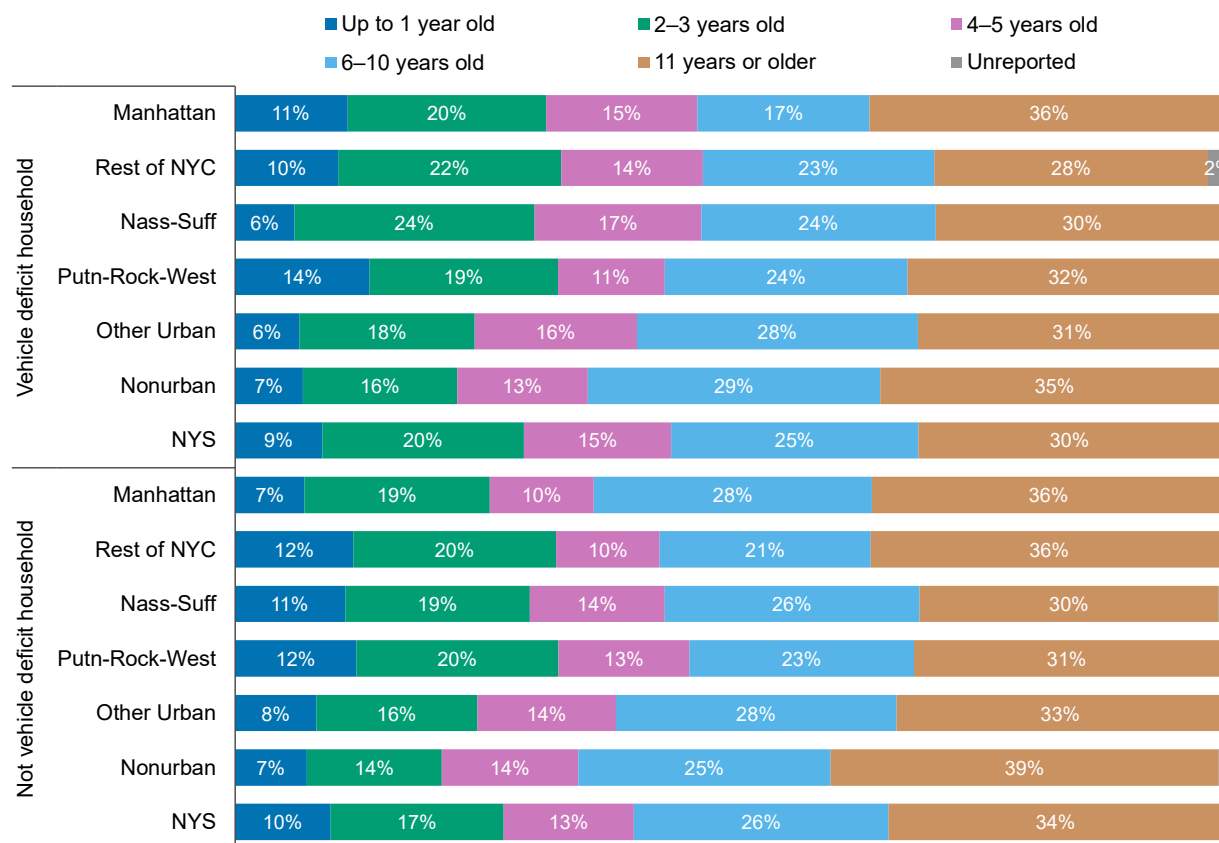


Figure 2-8. Distribution of vehicle age by vehicle deficit household type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 2%.

2.2.6 Household Life Cycle

Figure 2-9 shows the relationships between household life cycle category and vehicle ownership patterns. Households without cars are largely composed of adults without children. In 2017, about 60% were nonretired adults with no children, and another 25% were retirees without children. In contrast, vehicle deficit households were predominantly multiadult families with children, making up nearly 40% of that group. Last, the not vehicle deficit households had a more balanced distribution across life stages but still skewed toward households without children (40%) and retired adults without children (28%).

A clear distinction emerges in the distribution of NYS households by life cycle and vehicle deficit household type in Figure 2-10. An almost even split occurred between adults without children (35%) and multiadult families with children (35%) in one-vehicle-multiple-driver households, with retired adults without children making up nearly 30% of this group. By contrast, nearly half (48%) of households that owned more than one vehicle yet remained in deficit were multiadult families with children. This could suggest that even two or more cars can be insufficient when multiple adults (and children) need regular access to a vehicle.

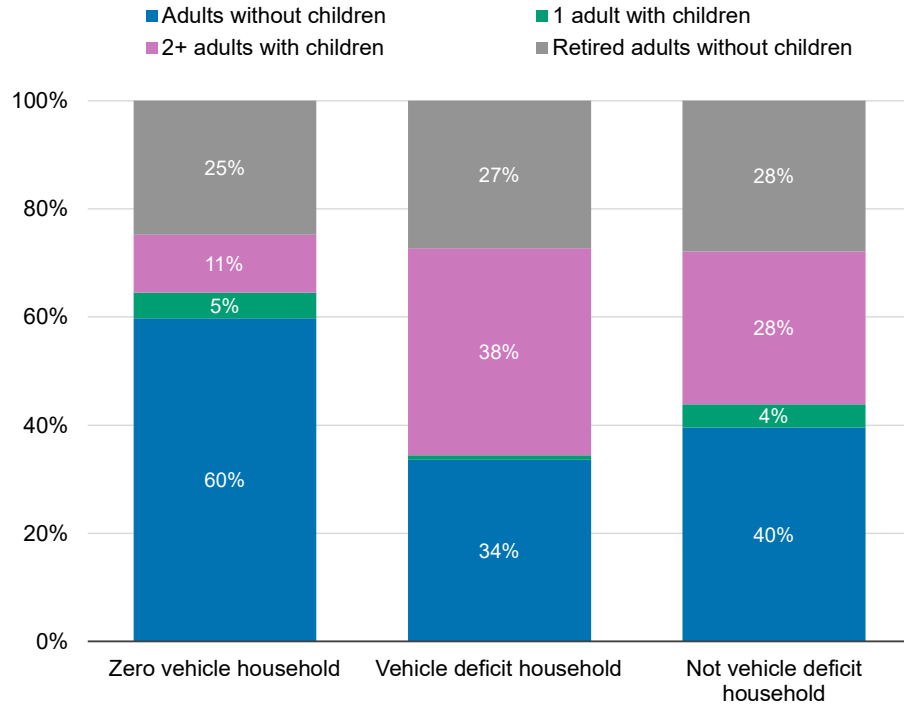


Figure 2-9. Distribution of NYS households by life cycle and vehicle ownership type (2017 NHTS data).
 Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.
 Unlabeled segments represent values lower than 2%.

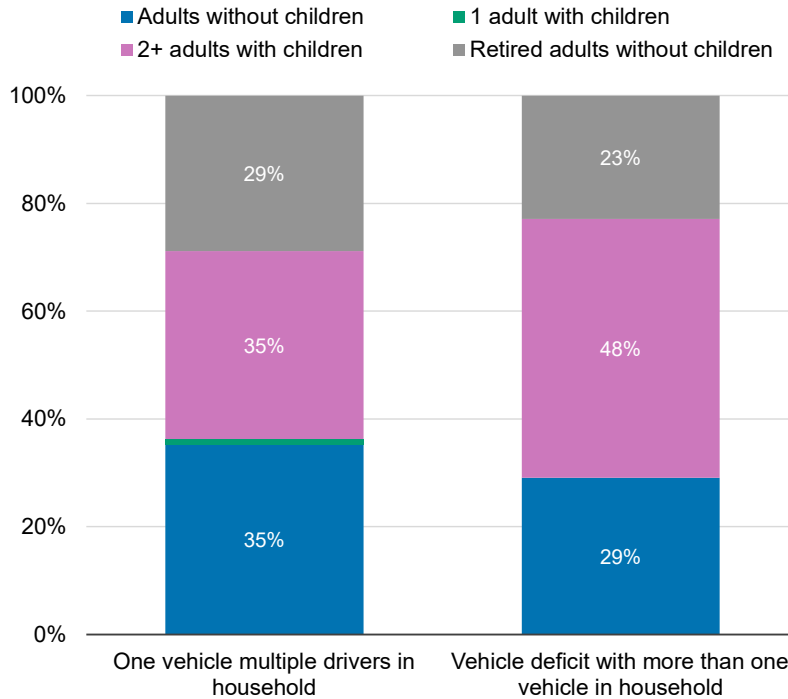


Figure 2-10. Distribution of NYS households by life cycle and vehicle deficit household type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 2%.

3. TRIP PATTERNS BY VEHICLE OWNERSHIP TYPE

3.1 TRAVEL STATISTICS BY PERSON TRIPS

3.1.1 Average Daily Person Trip Rate

Having established the demographic profiles of zero vehicle and vehicle deficit households, the focus now turns to examining their trip patterns. This section begins with a look at average daily person trips per person (e.g., how many trips each person makes on a typical day) across different vehicle ownership categories.

Figure 3-1 illustrates average daily person trips per person by vehicle ownership type in NYS and the rest of the US, comparing NHTS data from 2009 with that of 2017. The figure also factors in the margin of error (MOE), which is critical when assessing whether the observed differences among vehicle ownership categories, regions, and years are statistically significant. The MOE provides a range of potential variation in each estimate and helps determine the reliability of differences between categories. For example, NYS vehicle deficit households had a notable decline in average daily person trips per person from 3.8 trips in 2009 to 3.2 in 2017. After accounting for their MOEs, the difference (i.e., decline) in person trip rates of NYS vehicle deficit households between 2009 and 2017 is statistically significant. Similarly, the dip from 3.8 to 3.4 trips in the households with enough vehicles is statistically significant. In contrast, NYS zero vehicle households maintained roughly the same average daily trip rate (3.1 trips), although the difference is not statistically significant.

A similar pattern appears in the rest of the US, where zero vehicle households remained stable (around 2.6 trips in both years), and vehicle-owning households—whether deficit or not—showed significant decreases over time. Comparing NYS with the rest of the US, zero vehicle households in NYS made significantly more trips (around 3.1) than their counterparts in the rest of the US made (2.6 trips) in both 2009 and 2017. However, the differences between vehicle-owning households across NYS and the rest of the US are not statistically significant. Overall, these patterns suggest that zero vehicle households maintained consistent trip patterns over the study period, whereas those with access to vehicles appear to have reduced their daily trips in 2017 compared with those in 2009.

Examining the data within NYS, as shown in Figure 3-2, zero vehicle households in various regions showed significantly lower average daily person trip rates than those of vehicle-owning groups in 2009, except for in Manhattan. For instance, in the rest of NYC, zero vehicle households averaged 2.8 ± 0.25 trips, and not deficit households were at 3.5 ± 0.20 ; the difference is statistically significant.

By 2017, these gaps narrowed in most parts of the state so that zero vehicle households in the rest of NYC (3.0 ± 0.28 trips), Nassau–Suffolk (2.1 ± 1.12 trips), and other urban areas (2.8 ± 0.29 trips) no longer lay so far below their vehicle-owning counterparts. These ranges indicate no statistically significant difference between zero vehicle and other households. However, exploring the differences among NYS regions further shows that Manhattan’s zero vehicle households remain distinctive. At 3.7 ± 0.30 trips, their average daily person trip rate surpassed zero vehicle households in other regions, such as the rest of NYC or nonurban areas. In other words, Manhattan residents without cars still made more daily trips on average than their zero vehicle counterparts elsewhere, likely reflecting the dense mix of employment, errands, and amenities accessible by foot or transit. Meanwhile, vehicle deficit households in Manhattan (4.0 ± 0.27) also showed relatively high trip rates, underscoring that even a shortfall of cars does not necessarily reduce mobility in a transit-rich setting.

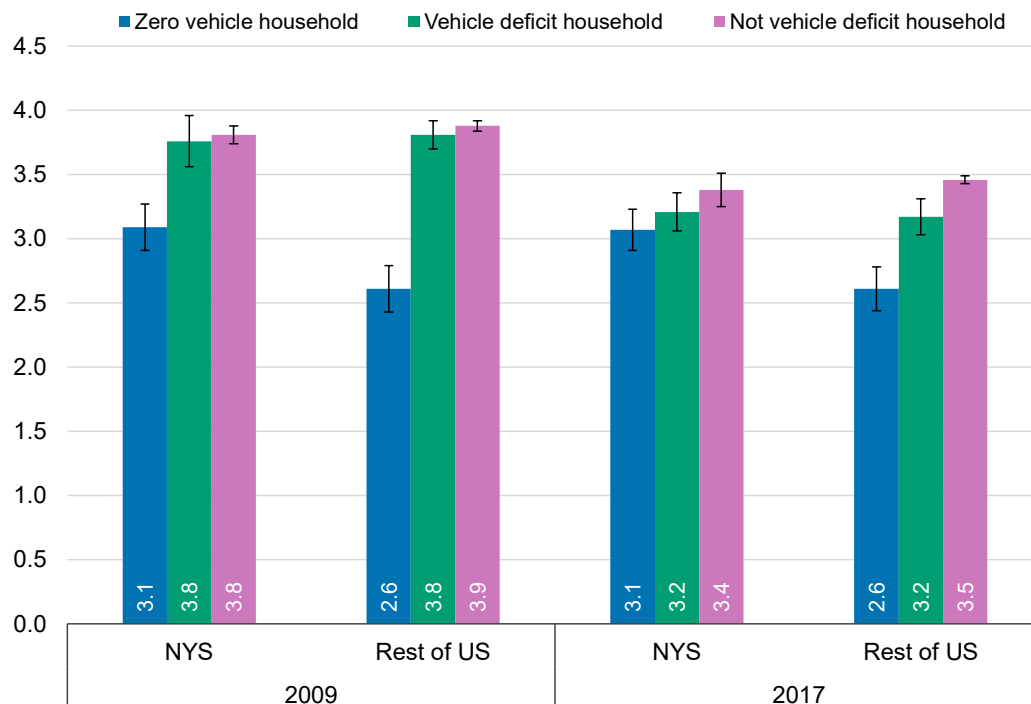


Figure 3-1. Average daily person trips per person by vehicle ownership type.

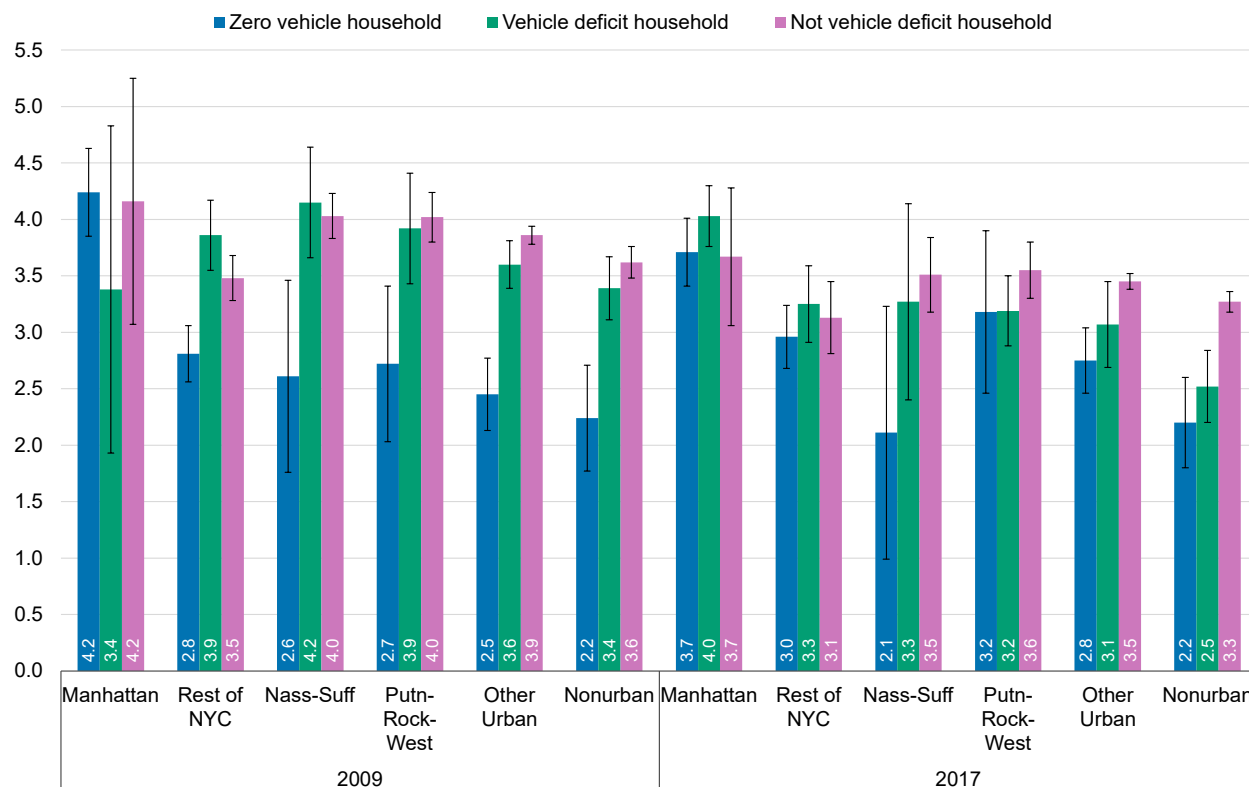


Figure 3-2. Average daily person trips per person by vehicle ownership type and NYS regions.

Figure 3-3 further shows average daily person trips per person by vehicle ownership type in the Mid-Atlantic region and the rest of the US over the years. On the whole, all three vehicle ownership categories in the Mid-Atlantic and the rest of the country showed a downward trend in average daily person trips from 2009 to 2022. The 2009 estimates for both regions tended to hover around four trips per person among vehicle-owning households (whether deficit or not) and around three trips or fewer for zero vehicle households. By 2017, these rates had dropped somewhat, particularly among vehicle deficit households. All categories in both regions saw more pronounced declines by 2022; for instance, the Mid-Atlantic's vehicle deficit group dropped to 1.8 ± 0.28 trips.

Despite the overall decline in average daily person trips per person, zero vehicle households in the Mid-Atlantic remained slightly higher than their counterparts in the rest of the US over the years. However, these differences are not statistically significant. Vehicle deficit and not vehicle deficit households started off at similar or higher levels in 2009 (3.9–3.8 trips) across both regions, then steadily fell to 1.8–2.5 trips by 2022 in the Mid-Atlantic (2.2–2.3 for the rest of the US). These considerable drops in person trips could reflect broader shifts in travel demand, such as telework expansion or changing lifestyle patterns, particularly in 2022.

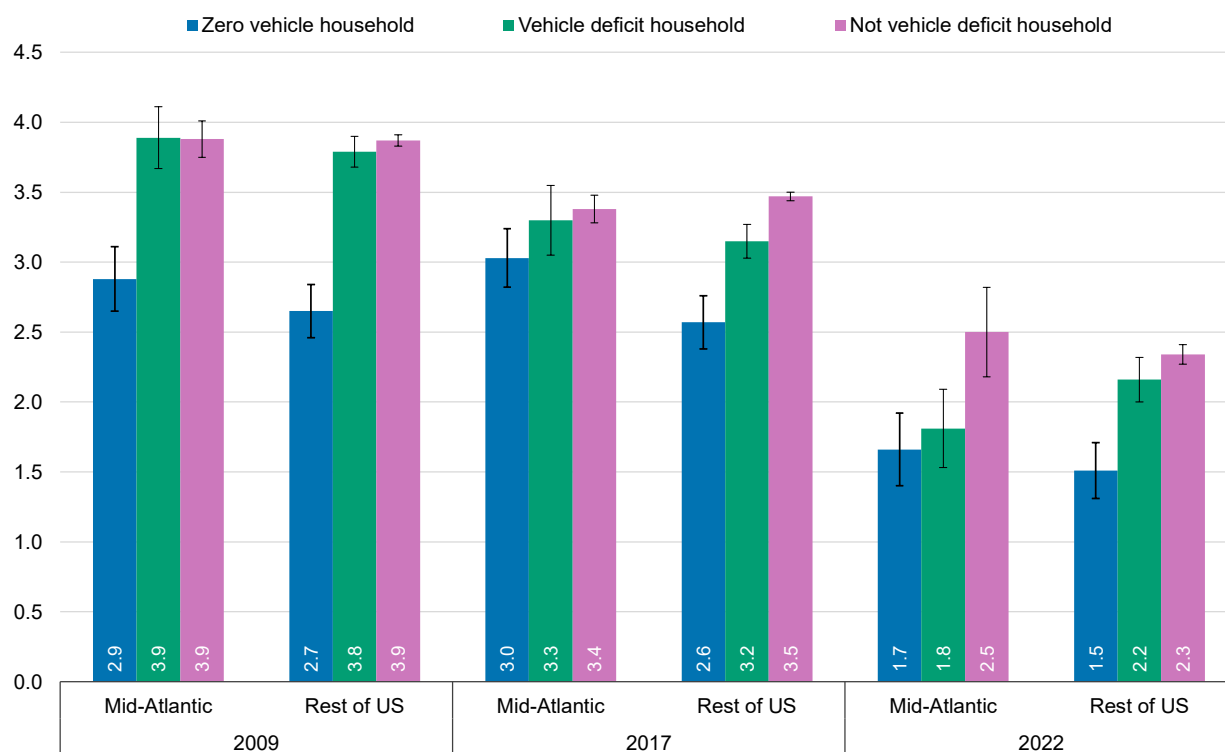


Figure 3-3. Average daily person trips per person by vehicle ownership type in the Mid-Atlantic and the rest of the US.

Figure 3-4 presents the average daily person trips per person by vehicle deficit household type using 2017 NHTS data. In Manhattan, having just one vehicle for multiple drivers yielded an especially high trip rate (4.1 ± 0.31 trips), noticeably greater than that for vehicle deficit households with more than one vehicle (2.1 ± 1.52 trips). In fact, the upper bound for the multivehicle deficit category (3.62 trips) remained below the lower bound of the one-vehicle deficit group (3.79 trips), indicating a statistically significant difference. Elsewhere in NYS, the differences between these two vehicle deficit household types are not significant.

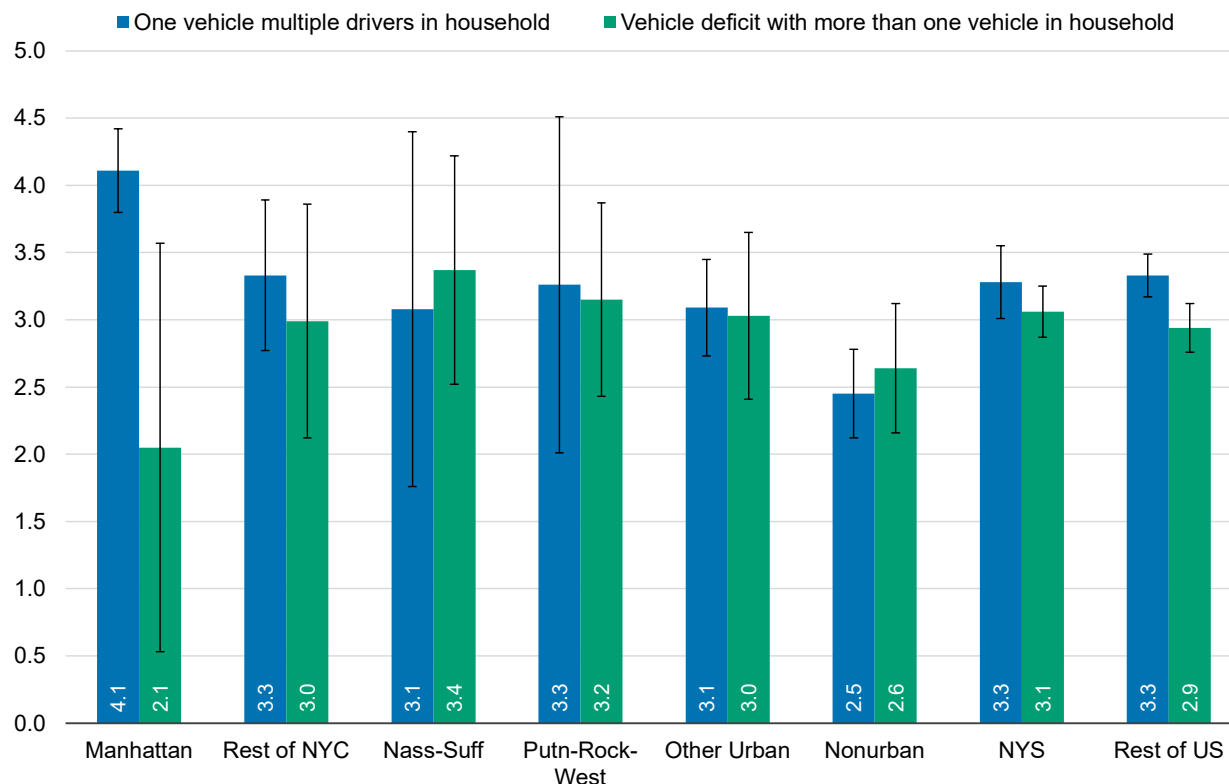


Figure 3-4. Average daily person trips per person by vehicle deficit household type (2017 NHTS data).

3.1.2 Average Daily Person-Trip Length

Figure 3-5 illustrates the average person-trip length by vehicle ownership type over time in NYS and the rest of the US. Note that air trips are removed from the trip length estimations. Between 2009 and 2017, average person-trip lengths remained more or less similar among vehicle ownership categories in NYS. However, zero vehicle households traveled significantly fewer miles than other household types in both regions over the years. In NYS, not-deficit vehicle households traveled significantly more than others. This is statistically significant for 2009 but not for 2017. In both years, trip lengths of all vehicle ownership types were lower in NYS than in the rest of the US. The trip length difference in not-vehicle deficit households is significant for 2017, and it is significant for other households for 2009.

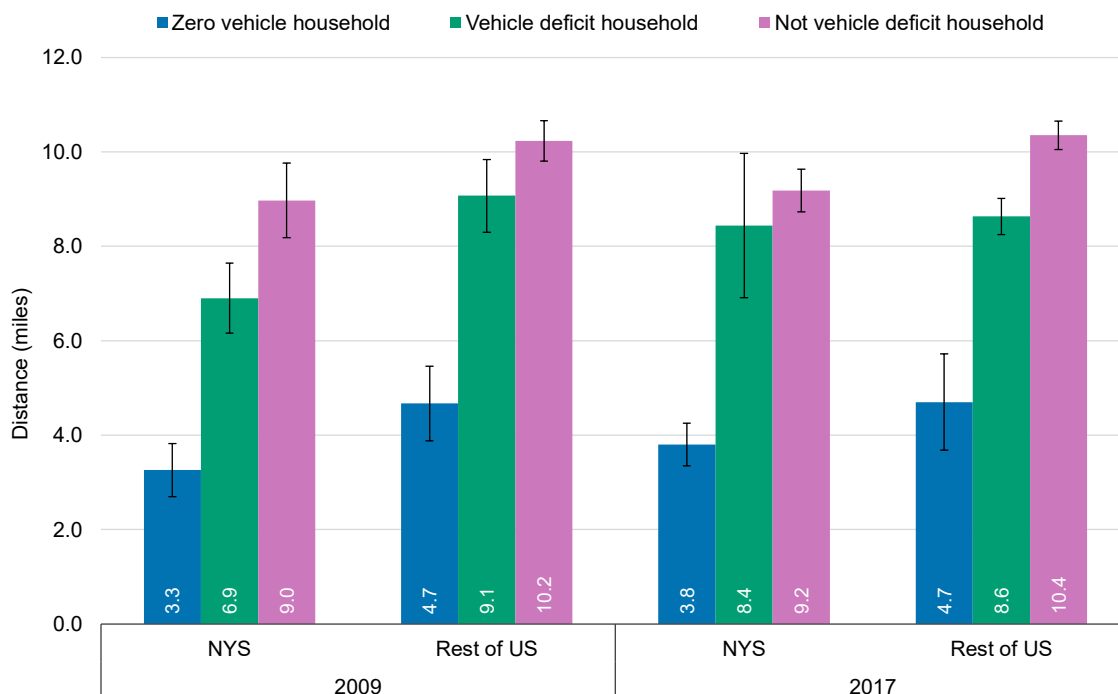


Figure 3-5. Average person-trip length by vehicle ownership type.

Considering the distribution across NYS regions, zero vehicle households traveled less than not vehicle deficit households in 2009 (Table 3-1). Vehicle deficit households generally were somewhere in between. For instance, in Manhattan, zero vehicle trips averaged about 3.0 miles (± 1.1) compared with 7.95 miles (± 4.8) for vehicle deficit and 9.2 miles (± 5.1) for not-deficit households. Similar patterns appeared in the suburbs (Nassau–Suffolk and Putnam–Rockland–Westchester) and in nonurban areas, where not-deficit households consistently recorded trip lengths around 8–11 miles, with zero vehicle excursions closer to 2–5 miles.

By 2017, several regions showed increases in trip lengths for certain groups, most notably Manhattan vehicle deficit households, which jumped from an average of 8.0 to 9.7 miles. Other areas displayed more moderate gains or remained stable within overlapping confidence intervals. For instance, zero vehicle households' trip lengths in Putnam–Rockland–Westchester rose from 1.8 (± 0.5) to 4.8 (± 2.5) miles; this is not statistically significant, however. Likewise, trip lengths for nonurban travelers from zero vehicle households declined from 5.0 (± 2.2) to 4.0 (± 2.2) miles.

The Mid-Atlantic region showed similar findings over the years (Figure 3-6). In 2009, zero vehicle households traveled the fewest miles (2.9 miles), whereas vehicle deficit and not-deficit households were identical (3.9 miles). By 2017, all three vehicle ownership categories in the Mid-Atlantic had higher averages, suggesting these households were taking longer trips than they had in 2009. The rest of the US followed a similar pattern, albeit with zero vehicle households often traveling even farther (4.7 miles in 2017) than their Mid-Atlantic counterparts (4.0 miles). However, this difference in trip length is not statistically significant. In 2022, the Mid-Atlantic vehicle deficit group showed a long average person trip (16.3 miles)—but also a very wide MOE—that was significantly higher than that of zero vehicle households. Averaging around 11.3 miles per person trip, the not-deficit households fell in between the other two groups.

Table 3-1. Average person-trip length (miles) by vehicle ownership type.

Year	Region	Estimate			Margin of error		
		Zero vehicle household	Vehicle deficit household	Not vehicle deficit household	Zero vehicle household	Vehicle deficit household	Not vehicle deficit household
2009	Manhattan	3.0	8.0	9.2	1.1	4.8	5.1
	Rest of NYC	3.3	5.3	7.2	0.8	1.2	2.8
	Nass-Suff	3.0	7.5	8.2	1.8	2.7	1.8
	Putn-Rock-West	1.8	8.1	8.7	0.5	2.5	1.2
	Other Urban	4.2	8.4	9.5	1.1	1.0	0.9
	Nonurban	5.0	8.8	11.2	2.2	1.3	1.1
2017	Manhattan	3.3	9.7	5.4	0.8	5.0	1.6
	Rest of NYC	4.3	6.0	6.7	0.7	1.6	1.4
	Nass-Suff	3.4	11.9	9.4	3.4	3.3	1.5
	Putn-Rock-West	4.8	9.0	9.5	2.5	3.2	2.0
	Other Urban	2.8	10.5	10.0	0.9	6.0	1.0
	Nonurban	4.0	8.9	11.9	2.2	1.8	1.2

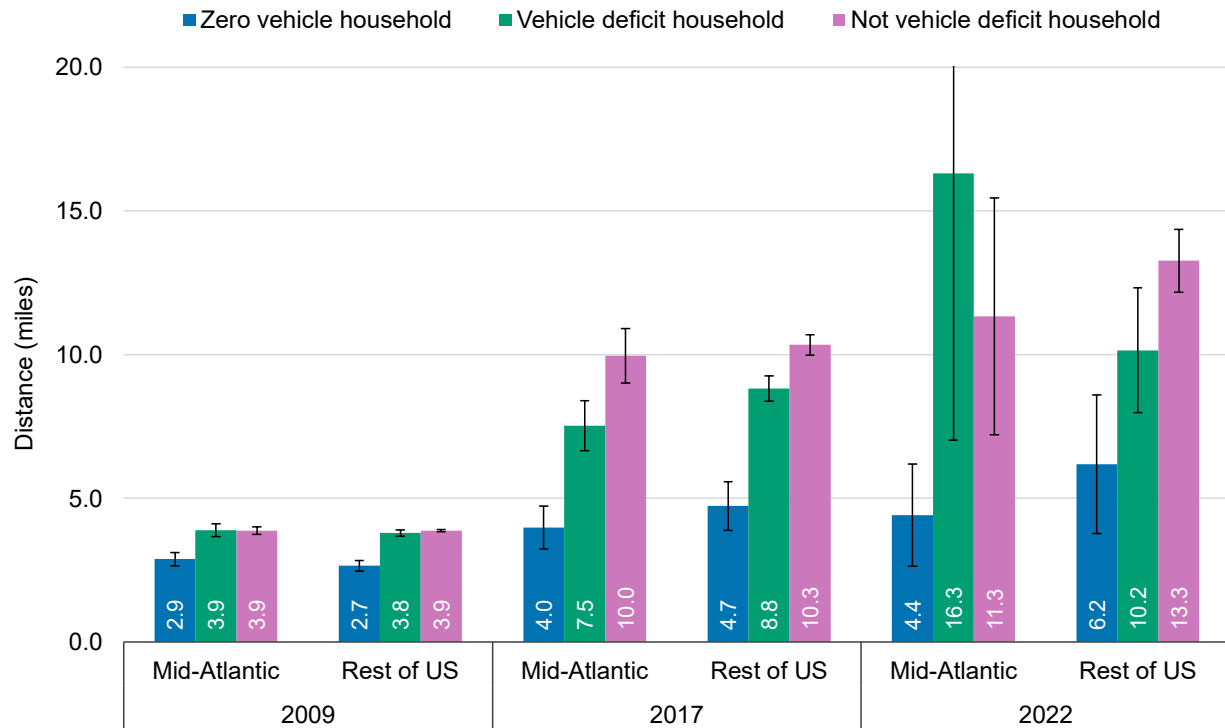


Figure 3-6. Average person-trip length (miles) by vehicle ownership type in the Mid-Atlantic and the rest of the US.

3.1.3 Mode Share

When exploring the mode share distribution, a few interesting contrasts emerge, as shown in Figure 3-7. For zero vehicle households in Manhattan and the rest of NYC, walking and public transit dominate. Over half of Manhattan's zero-car travelers choose to walk, and more than a quarter use transit. In the rest of

NYC, transit (41%) and walking (49%) are similarly high. By contrast, residents of zero vehicle households in nonurban parts of NYS rely more heavily on being a passenger in someone else's vehicle (24%) and show relatively modest transit shares (10%), suggesting more limited transit service outside major metros. Interestingly, in Nassau–Suffolk, 43% of zero vehicle households still walk as their primary mode. But 20% ride as passengers, reflecting a suburban environment that remains somewhat car-oriented yet walkable in certain areas. Note that the existence of privately owned vehicle driver (POV-driver) mode for zero vehicle households could indicate their use of rental cars or driving someone else's car.

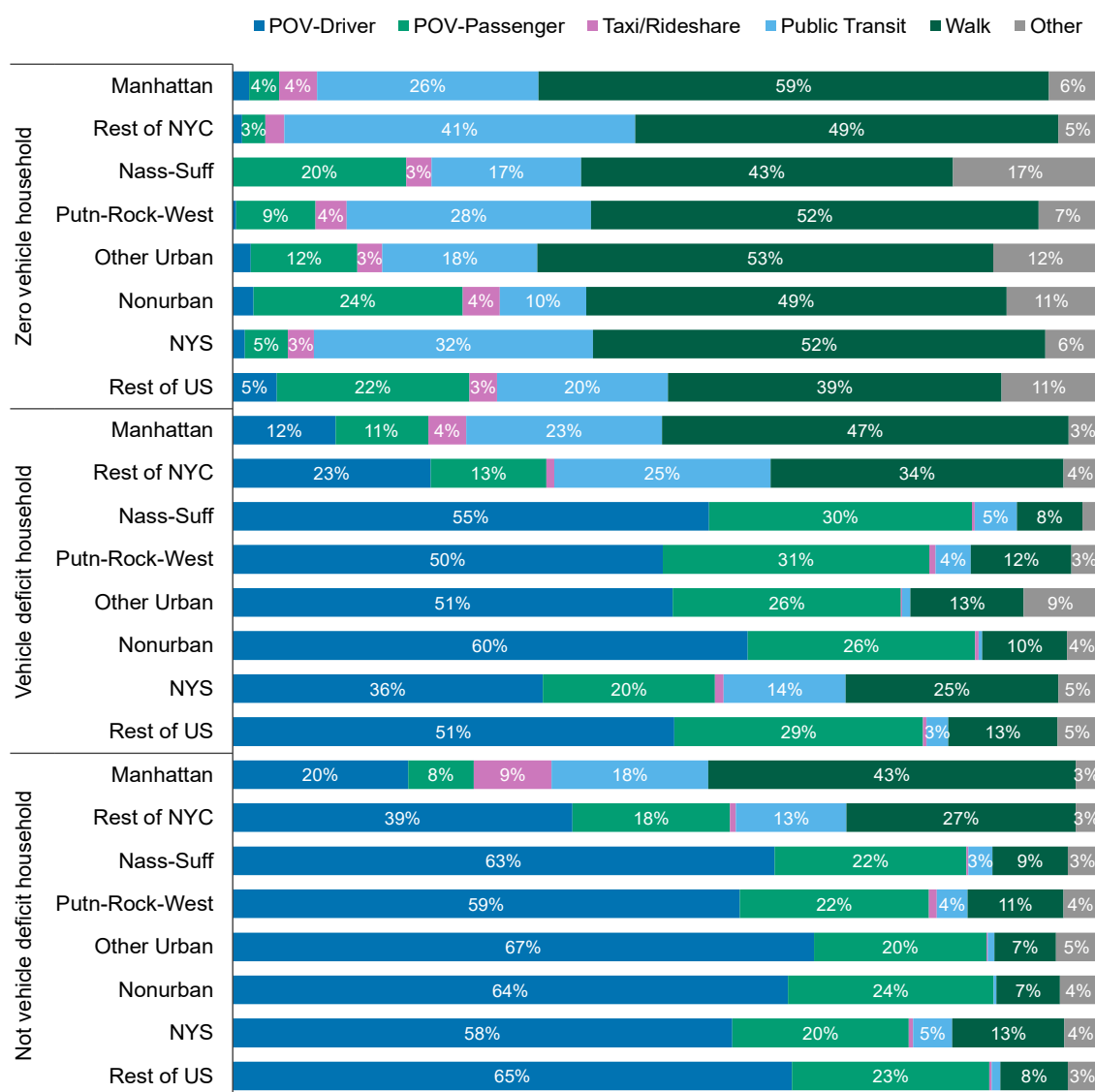


Figure 3-7. Mode share distribution by vehicle ownership type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%. POV-Driver = privately owned vehicle driver; POV-Passenger = privately owned vehicle passenger.

For vehicle deficit households, Manhattan and the rest of NYC again show heavy use of public transit and walking. However, driving now makes up a notable share (12%–23%), implying these households do have at least a few cars, just not enough for every driver. In suburban and nonurban regions, vehicle

deficit households are strongly POV-driver oriented (50%–60% or more). Meanwhile, households with enough vehicles are overwhelmingly driver oriented outside major cities. In Nassau–Suffolk, for example, 63% of not-deficit households drive, whereas Manhattan’s fully motorized households still choose to walk nearly half the time.

Comparing vehicle deficit households in NYS with those in the rest of the US reveals some clear differences in how people travel, especially when exploring the two vehicle deficit categories (Figure 3-8). For one-vehicle-multiple-driver households in NYS, driving alone accounted for only about one-third of trips in both 2009 and 2017, with walking and transit making up a sizable share. By 2017, walking increased to 30% and transit to 17%. In the rest of the country, such households relied far more heavily on driving (nearly half of trips), and public transit remained negligible at just 3%. Over time, NYS showed a modest shift away from private-auto modes toward more active or public transportation, whereas the rest of the US showed little change and remained largely automobile oriented.

Driving is the dominant mode in vehicle deficit households with more than one vehicle, and members of those households drive more than those from one-vehicle-multiple-driver households. NYS travelers still walked or took transit more often than their counterparts in the rest of the US. For instance, in 2017, multivehicle deficit households in NYS made 8% of their trips by public transit compared with only 1% in the rest of the US. Notably, taxi and rideshare usage remained small across the regions, hovering around 1% or less. Overall, these differences highlight how NYS’s more robust transit networks and walkable environments foster alternatives to driving, even among households that own at least one car but remain short of vehicles for every driver.

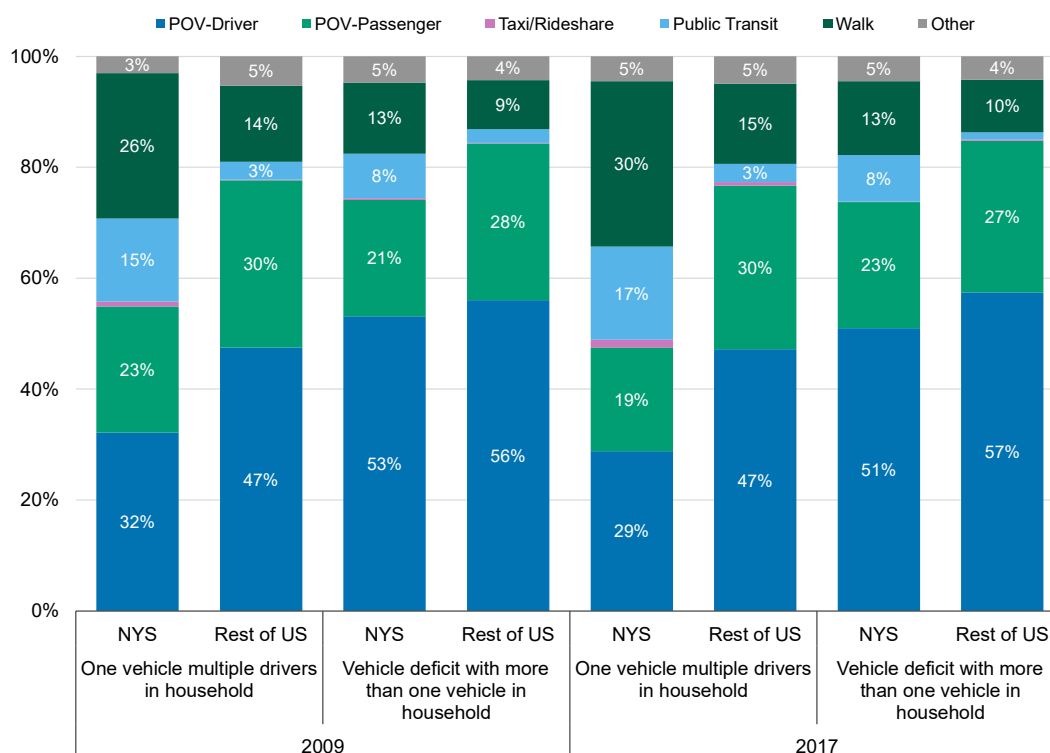


Figure 3-8. Mode share distribution by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values that are 3% or lower.

3.1.4 Trip Purpose

Figure 3-9 shows the distribution of average daily person trips by trip purpose and vehicle ownership type. Several regional and ownership-based differences arise on why people travel. For zero vehicle households in dense urban areas like Manhattan, earning a living and social and recreational trips each make up roughly one-fifth to one-third of daily travel. This may reflect a balance between commuting and leisure activities in a location with ample transit and walk options. In suburbs such as Nassau–Suffolk, family and personal business trips (e.g., shopping and errands) dominate zero vehicle travel at over half of all trips, suggesting that even without a car, everyday errands remain a key driver of travel demand in suburban environments.

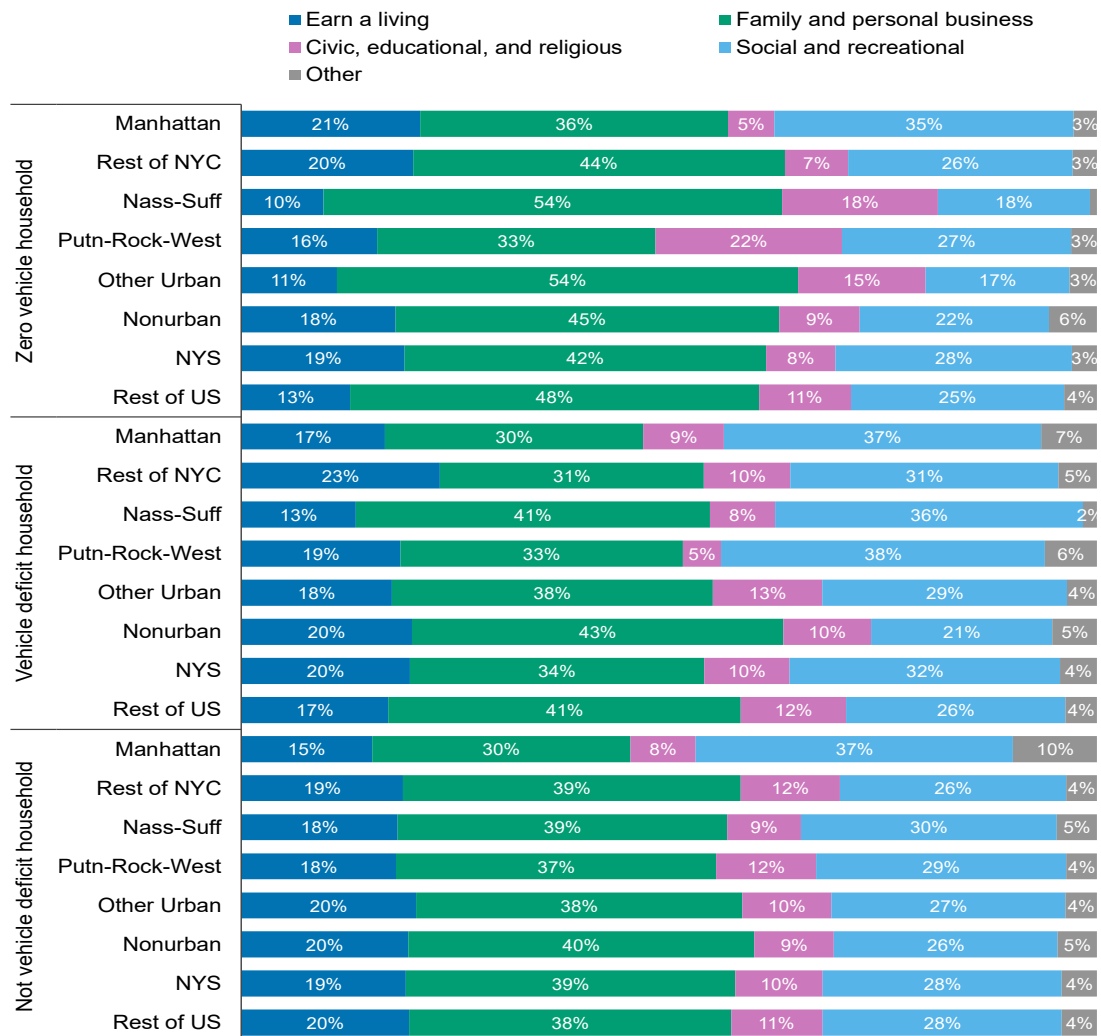


Figure 3-9. Share of person trips by trip purpose and vehicle ownership type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 2%.

Among vehicle deficit households, Manhattan again shows a high share of social and recreational trips (37%), whereas suburban and nonurban areas lean toward family and personal business (over 40%). This could indicate that when households are short on vehicles, city dwellers can still readily access leisure activities, whereas suburban and rural residents devote a large portion of their limited trips to essential

errands. Finally, households with enough vehicles exhibit somewhat more uniform trip purposes across the board: about 30%–40% for family and personal business, around 20% for earning a living, and a quarter to a third for social or recreational.

In vehicle deficit households, Manhattan stands out for dedicating nearly 40% of trips to social and recreational purposes by one-vehicle-multiple-driver households, which is higher compared with the rest of NYC or statewide (both 31%) (Table 3-2). In contrast, family and personal business (e.g., errands and shopping) dominates one-vehicle travel in more suburban areas like Nassau–Suffolk (51%) and nonurban regions (49%). This suggests that, with only one car to go around, Manhattan households still manage more leisure outings, which could be attributed to dense and transit-friendly infrastructure, whereas suburban or rural households prioritize essential errands when mobility is limited.

By comparison, multivehicle deficit households show different patterns. In Manhattan, they overwhelmingly focus on family and personal business (85% of trips), which is an outlier given the lower sample size (N = 10). The shares are more balanced among earning a living, family and personal business, and social and recreational trips elsewhere. For instance, in the rest of NYC, work-related trips (28%) and social and recreational outings (34%) both feature prominent shares, with family and personal business following closely with about 25%.

Table 3-2. Share of person trips by trip purpose and vehicle deficit household type (2017 NHTS data).

Vehicle deficit household type	Region	Earn a living	Family and personal business	Civic, educational, and religious	Social and recreational	Other
One vehicle multiple drivers in household	Manhattan	17%	29%	10%	38%	7%
	Rest of NYC	22%	32%	10%	31%	5%
	Nass-Suff	15%	51%	1%	33%	0%
	Putn-Rock-West	15%	37%	5%	39%	5%
	Other Urban	15%	39%	14%	28%	4%
	Nonurban	18%	49%	7%	22%	4%
	NYS	19%	35%	10%	31%	5%
	Rest of US	15%	45%	11%	25%	4%
Vehicle deficit with more than one vehicle in household	Rest of NYC	28%	25%	9%	34%	4%
	Nass-Suff	13%	37%	11%	37%	2%
	Putn-Rock-West	21%	30%	4%	37%	7%
	Other Urban	22%	35%	10%	29%	3%
	Nonurban	24%	35%	15%	20%	7%
	NYS	21%	32%	10%	33%	4%
	Rest of US	20%	34%	15%	27%	4%

Notes: Manhattan is excluded in the “vehicle deficit with more than one vehicle in household” category because of low sample size. Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Looking at the one-vehicle-multiple-driver deficit households exclusively, several notable shifts have been observed in the broader geography level (i.e., Mid-Atlantic division), as shown in Figure 3-10. In 2009, nearly half of all trips were devoted to family and personal business (48%), and only about a quarter were for social and recreational purposes. By 2022, those proportions essentially reversed: social and recreational outings rose to 37%, and family and personal errands fell to 24%. At the same time, trips to earn a living increased from 17% to 25%. Comparing the Mid-Atlantic with the rest of the US

highlights a different dynamic. Outside the region, family and personal business remained the dominant activity in all years studied. Meanwhile, social and recreational travel edged upward from 25% to 30% in 2022.

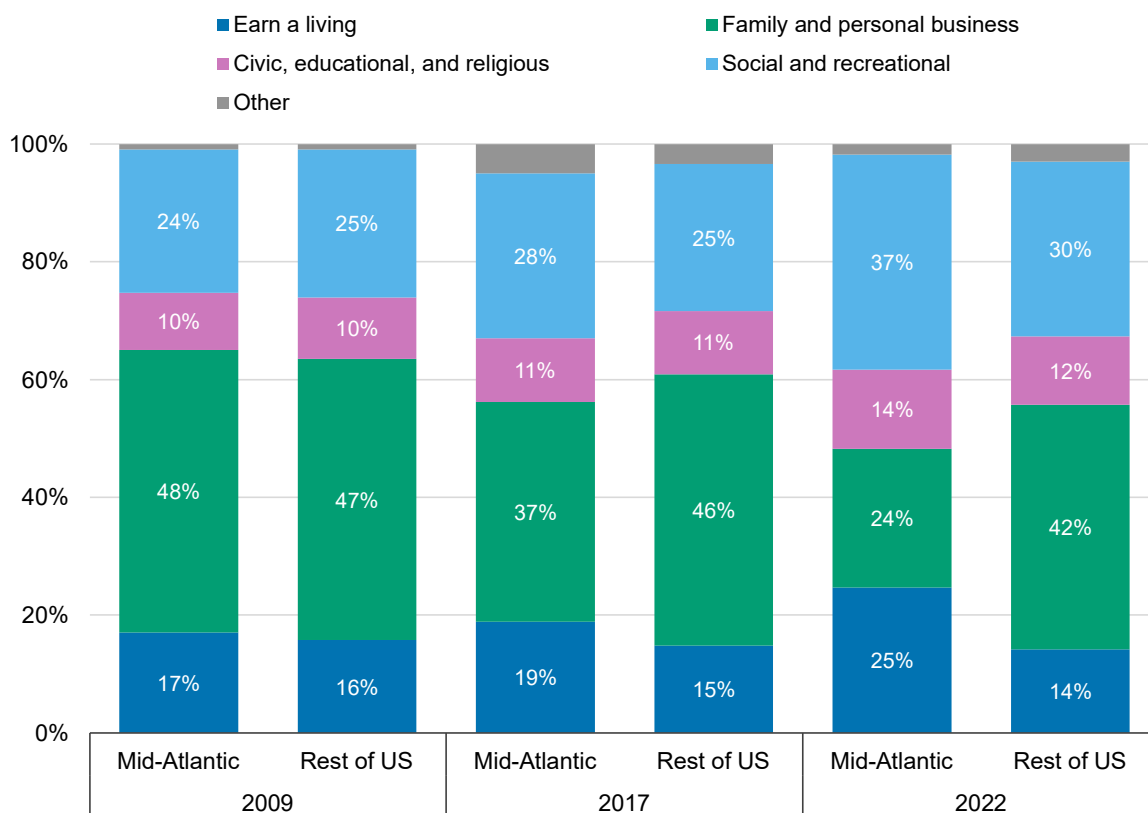


Figure 3-10. Share of person trips by trip purpose in one-vehicle-multiple-driver households. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 6%.

3.1.5 Time of Travel

The temporal distribution of person trips by time of travel, as illustrated in Figure 3-11, shows several interesting trends. Across all regions, zero vehicle households tended to cluster more of their trips during the late morning to early afternoon. For instance, in Nassau–Suffolk, 24% of zero vehicle travel occurred between 9 a.m. and noon, and 33% took place between noon and 3 p.m. Putnam–Rockland–Westchester zero vehicle travelers made more than a quarter (29%) of their trips between 3 and 6 p.m. This midday concentration may reflect greater reliance on transit schedules or the ability to schedule trips around peak commuting times, especially where transit service frequency can be lower.

In contrast, vehicle deficit and not vehicle deficit households—particularly in more suburban or rural areas—show a more even spread of travel throughout the day, with a higher percentage of trips during the morning rush (6–9 a.m.) and late afternoon rush (3–6 p.m.). In Manhattan, even deficit households made a noticeable share of trips in the evening (6–9 p.m. and 9 p.m.–midnight). The presence of nightlife, cultural events, or flexible working hours may explain why a sizable share of trips occurred outside the typical “9 to 5” time window.

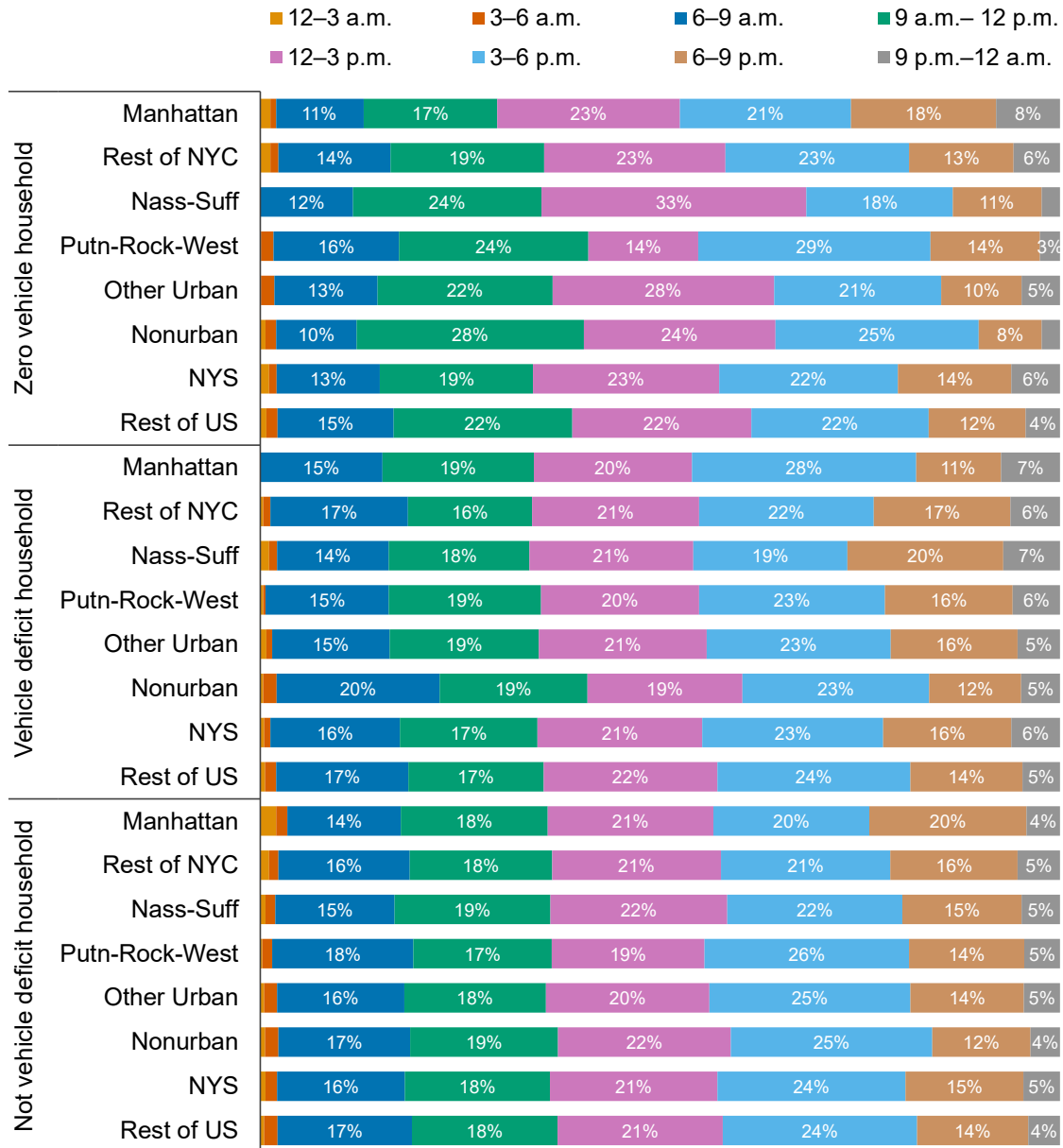


Figure 3-11. Temporal distribution of person trips by time of travel and vehicle ownership type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

In Table 3-3, a noteworthy contrast emerges between one-vehicle-multiple-driver households and those with more than one vehicle but still a deficit. In Manhattan, for example, one-vehicle-multiple-driver households tended to distribute their trips more evenly throughout the workday, with a sizable peak in the late afternoon (3–6 p.m. at 28%) and a fair share in the evening (6–9 p.m. at 11%). By contrast, vehicle deficit with more than one vehicle households in Manhattan showed a higher concentration of trips in the late morning (9 a.m.–noon) and early afternoon (noon–3 p.m.). Elsewhere, the two deficit types look more similar.

Table 3-3. Temporal distribution of person trips by time of travel and vehicle deficit household type (2017 NHTS data).

Vehicle deficit household type	Region	12–3 a.m.	3–6 a.m.	6–9 a.m.	9 a.m.–12 p.m.	12–3 p.m.	3–6 p.m.	6–9 p.m.	9 p.m.–12 a.m.
One vehicle multiple drivers in household	Manhattan	0%	0%	16%	18%	19%	28%	11%	8%
	Rest of NYC	0%	1%	17%	15%	23%	22%	18%	5%
	Nass-Suff	0%	0%	16%	23%	25%	14%	20%	3%
	Putn-Rock-West	0%	0%	12%	21%	19%	22%	22%	4%
	Other Urban	1%	1%	13%	21%	21%	23%	15%	6%
	Nonurban	0%	1%	22%	19%	19%	24%	10%	5%
	NYS	0%	1%	16%	17%	22%	23%	16%	5%
	Rest of US	1%	1%	16%	18%	23%	24%	13%	4%
Vehicle deficit with more than one vehicle in household	Manhattan	0%	0%	2%	43%	43%	10%	2%	0%
	Rest of NYC	1%	1%	18%	19%	14%	22%	15%	11%
	Nass-Suff	2%	2%	13%	15%	18%	22%	19%	9%
	Putn-Rock-West	1%	1%	18%	18%	21%	24%	12%	7%
	Other Urban	1%	1%	18%	15%	21%	23%	17%	5%
	Nonurban	1%	3%	18%	17%	19%	23%	14%	5%
	NYS	1%	1%	17%	17%	19%	22%	16%	8%
	Rest of US	1%	2%	17%	16%	20%	24%	15%	6%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

3.1.6 Person Miles of Travel

As expected, zero vehicle households had lower average person miles of travel (PMT) than those of their car-owning counterparts across most regions, although the statistical significance of these differences varies after accounting for MOEs (Figure 3-12). For instance, in Manhattan, zero vehicle households traveled about 12 miles (± 2.69), whereas vehicle deficit households averaged 39 miles (± 21.18); the difference is statistically significant. Meanwhile, zero vehicle households versus not-deficit households showed a small interval overlap, so a significant gap between those two groups cannot be confirmed.

In Nassau–Suffolk, the difference is more substantial: zero vehicle households traveled on average 7 miles (± 2.68), which is significantly lower than that of vehicle deficit (39 ± 14.05) and not-deficit (33 ± 6.12) households. By contrast, in some regions the picture is less clear. In the rest of NYC, for example, zero vehicle PMT is 13 (± 2.09), and not-deficit is 21 (± 4.19), a significant difference. However, zero vehicle and vehicle deficit (20 ± 5.31) are borderline. Finally, at the state level, NYS overall shows a zero vehicle average PMT of 12 (± 1.28) versus 31 (± 1.49) for not-deficit households—again, a clear gap with no overlapping intervals. These results underscore that having no car typically corresponds to noticeably lower daily mileage, though the size of the gap could depend on the specific region’s density, transit availability, and travel options.

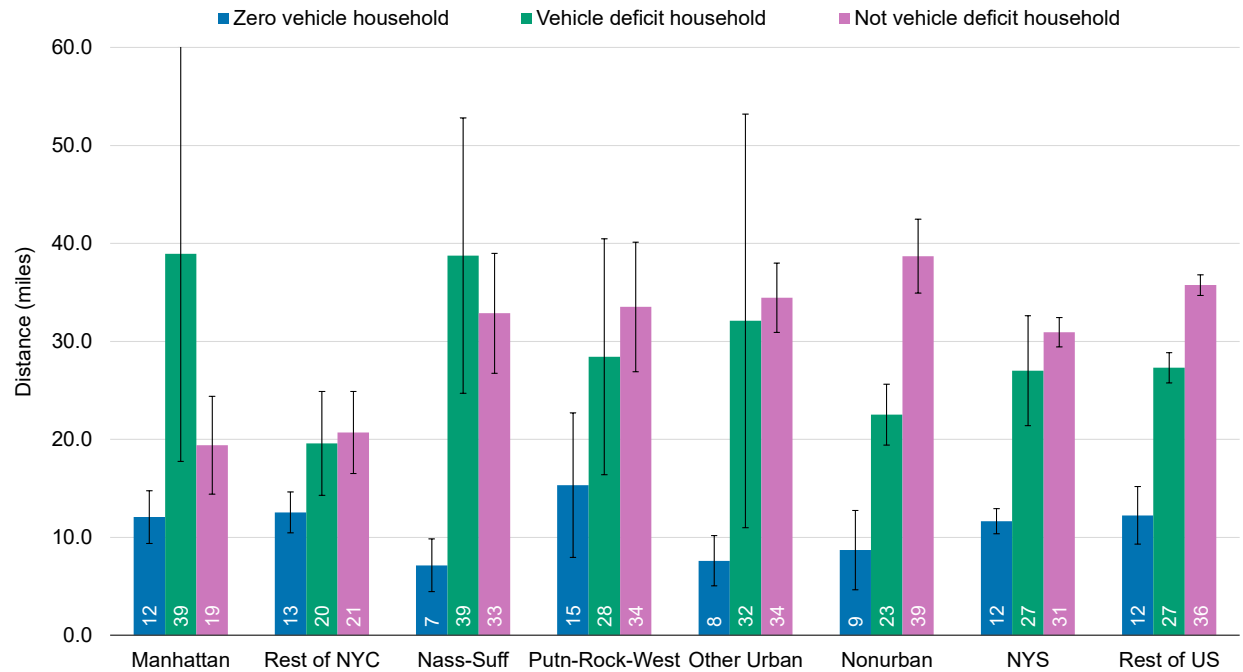


Figure 3-12. Average Daily person miles of travel by vehicle ownership type (2017 NHTS data).

Among vehicle deficit households, having just one car or having multiple cars yet still fewer vehicles than drivers does not necessarily yield dramatic differences in daily PMT across most regions (Table 3-4). In the rest of NYC, for example, the values for one-vehicle (19) and multivehicle deficits (20) are nearly identical, and given the relatively wide MOEs, no statistically significant gap is present. Similarly, in Putnam–Rockland–Westchester, both deficit types had 28 PMT, also within overlapping intervals. In short, although these two deficit groups differ in how many cars they own, their overall person distance traveled does not consistently diverge in a statistically meaningful way.

Table 3-4. Average daily PMT by vehicle deficit household type (2017 NHTS data).

Region	Estimate		Margin of error	
	One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household	One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household
Rest of NYC	19	20	7	7
Nass-Suff	40	38	29	17
Putn-Rock-West	28	28	18	14
Other Urban	34	30	28	12
Nonurban	21	24	4	6
NYS	26	29	7	5
Rest of US	26	30	2	3

Note: Manhattan is excluded because of low sample size.

3.2 VEHICLE TRAVEL

3.2.1 Average Daily Vehicle Trip Rate

Across both 2009 and 2017, zero vehicle households registered a small (but nonzero) number of daily vehicle trips per driver, which was around 0.1 in NYS and 0.5 in the rest of the US (Figure 3-13). This could stem from situations in which people without a personal vehicle borrowed a friend's car, rented a vehicle, or otherwise gained access to one for occasional trips on travel days. Even so, these rates are significantly lower than those for vehicle deficit or not-deficit households, highlighting that those lacking a car at home rarely drove. Meanwhile, vehicle deficit households in NYS dropped from 1.8 daily car trips per driver in 2009 to 1.5 in 2017, which appears statistically significant. Deficit households in the rest of the US dropped from 2.4 to 2.0, again a statistically significant decline. Households with enough vehicles (not-deficit) also drove less in 2017 (2.7 in NYS and 2.9 in the rest of the US) than in 2009 (3.1 and 3.3, respectively). All told, although zero vehicle households remained distinct in their near-minimal driving, all groups showed a downward trend in daily car trips over time. This trend could suggest shifts in travel behavior, possibly due to changes in telework, ride-hailing, or lifestyle preferences.

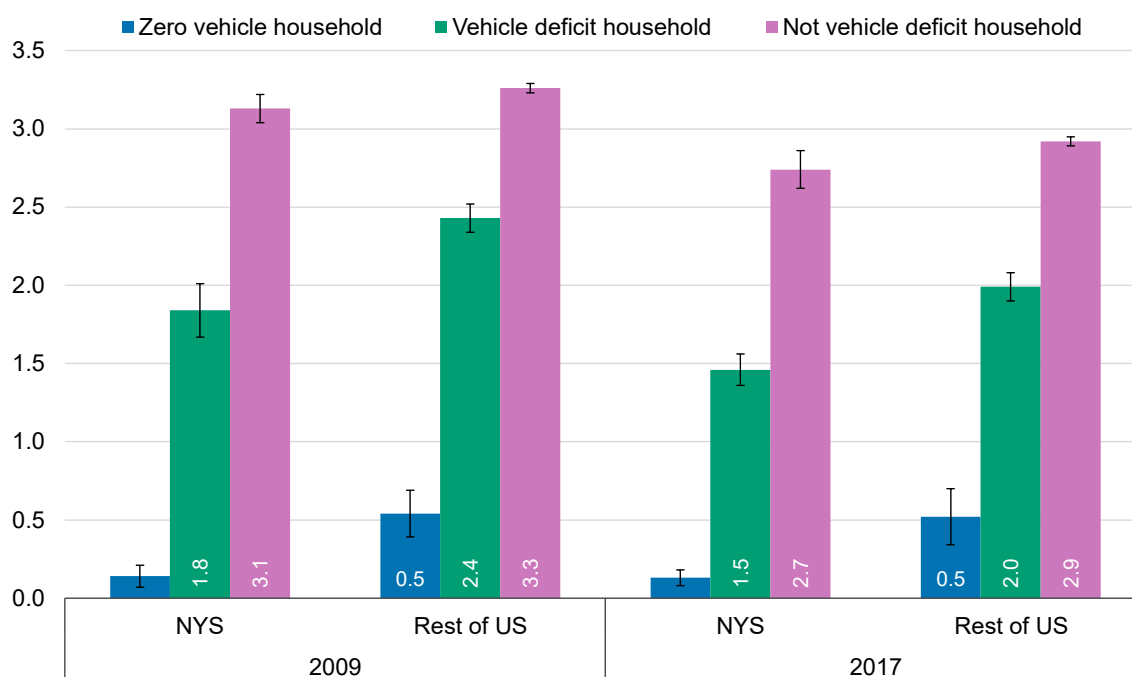


Figure 3-13. Average daily vehicle trips per driver by vehicle ownership type.

Figure 3-14 illustrates the continuation of the discussion on vehicle trips per driver for other regions in NYS. Looking at Manhattan first, zero vehicle households showed minimal driving activity (about 0.2 daily vehicle trips per driver), whereas both vehicle deficit and not-deficit households hovered between 0.6 and 1.1 trips in 2017. In Nassau–Suffolk, zero vehicle households effectively reported no vehicle trips in 2017, down from 0.5 in 2009. That further illustrates how most zero-car suburban dwellers rely on other modes or occasionally borrow or rent a vehicle for very few trips. Across most other regions, the pattern holds: zero vehicle households typically recorded well under 1 daily vehicle trip per driver, whereas vehicle deficit and not-deficit groups ranged around 2–3 trips in the suburbs (e.g., Putnam–Rockland–Westchester at nearly 2.9–3.3) and roughly 2 trips in nonurban settings.

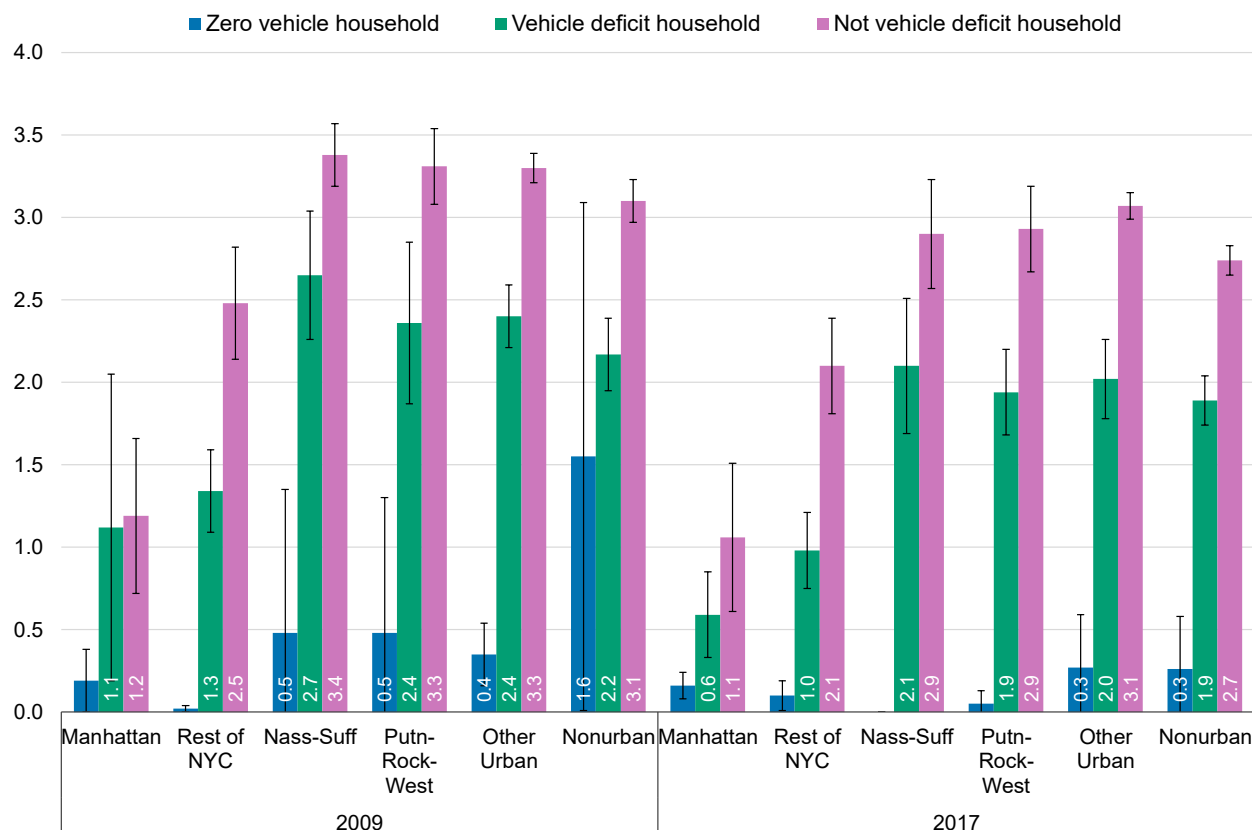


Figure 3-14. Average daily vehicle trips per driver by vehicle ownership type.

In Table 3-5, average daily vehicle trips per driver are further explored exclusively by the vehicle deficit household type in NYS versus the rest of the US over time. In 2009, households with only one vehicle shared among multiple drivers made fewer daily trips per driver than those with more than one vehicle but still insufficient cars overall. For instance, in NYS, one-vehicle-multiple-driver households averaged 1.62 trips versus 2.21 by households with more than one vehicle but still deficit, and the difference is statistically significant. A similar, albeit smaller, gap emerges in the rest of the US (2.30 vs. 2.55 trips). By 2017, both NYS subgroups saw their average vehicle trip rates decline (1.26 and 1.82), but the spread between them stayed statistically significant, indicating households with more than one car still drive more overall. Meanwhile, in the rest of the US, single-vehicle deficit households dropped to 1.98 trips, nearly matching the multivehicle deficit group (1.99), making the difference no longer statistically significant.

Table 3-5. Average daily vehicle trips per driver by vehicle deficit household type.

Year	Region	Estimate		Margin of error	
		One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household	One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household
2009	NYS	1.62	2.21	0.22	0.23
	Rest of US	2.30	2.55	0.13	0.13
2017	NYS	1.26	1.82	0.18	0.20
	Rest of US	1.98	1.99	0.11	0.08

A closer look at vehicle deficit households across the NYS regions reveals some contrasts, as shown in Figure 3-15. In 2009, the most notable contrast appears in Nassau–Suffolk, where one-vehicle deficit households recorded around 3.1 daily car trips per driver, compared with 2.4 for those with multiple vehicles; however, this was not significant. During both years, one vehicle deficit households in the rest of NYC made significantly fewer vehicle trips compared with households in the same category residing in other regions.

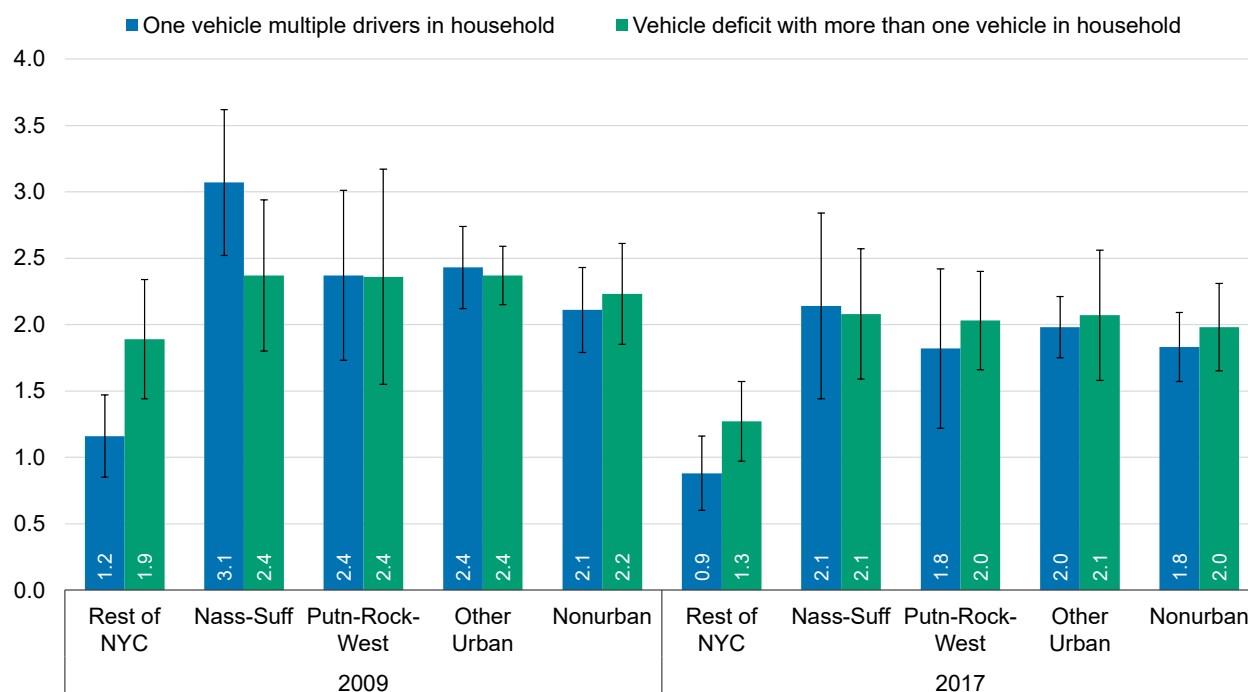


Figure 3-15. Average daily vehicle trips per driver by vehicle deficit household type.

Figure 3-16 illustrates average daily vehicle trips per driver by trip purpose in vehicle deficit households in 2017. Work-related (“earn a living”) trips tended to be modest across most NYS regions—ranging from about 0.1 trips per driver in Manhattan to 0.5 in nonurban areas. The higher figure in certain places (e.g., other urban or nonurban areas) appears statistically significant given the smaller MOE, hinting that suburban or rural drivers in deficit households more often need to commute by car. Family and personal business purpose constituted the highest share of vehicle trips across regions. For instance, households in suburban areas such as Nassau–Suffolk reported around 1.0 trip per driver, again significantly above Manhattan’s 0.3 trips. Meanwhile, social and recreational driving also peaked in suburban regions (around 0.7 in Nassau–Suffolk) compared with about 0.2 in Manhattan. The overall pattern shows that suburban and rural households with fewer cars than drivers still rely on more frequent vehicle trips for everyday errands and social outings, whereas Manhattan and parts of NYC make far fewer vehicle trips, presumably substituting transit, walking, or other modes.

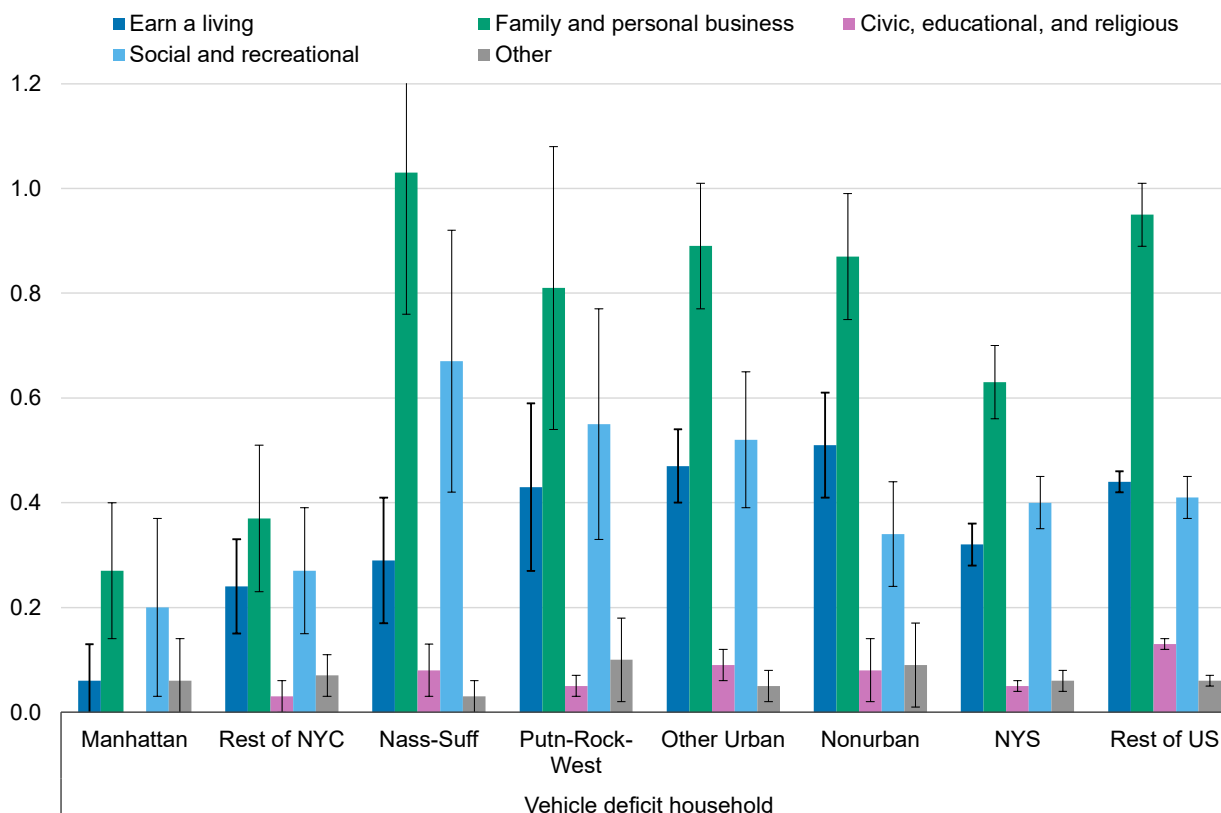


Figure 3-16. Average daily vehicle trips per driver by trip purpose in vehicle deficit households (2017 NHTS data).

3.2.2 Vehicle Miles of Travel

Figure 3-17 shows average daily vehicle miles of travel (VMT) per driver by vehicle ownership type across regions using 2017 NHTS data. Across most regions, drivers in not-vehicle deficit households reported higher daily VMT than that of drivers in vehicle deficit households. In the rest of NYC, for instance, vehicle deficit households averaged about 9 miles, and not-deficit households topped 16 miles—a significant difference. Putnam–Rockland–Westchester also exhibited a clear gap, with deficit households averaging around 18 miles versus nearly 28 miles for not-deficit households, again statistically significant. Nonurban areas feature one of the widest separations: about 19 miles for deficit compared with more than 33 miles for not-deficit, a difference firmly outside the MOE. At the statewide level, New York’s vehicle deficit households (17 miles) trail not-deficit households (26 miles), indicating a clear distinction. A similar pattern exists in the rest of the US (18 vs. 31 miles), also significant.

The numbers for deficit and not-deficit households lie within overlapping intervals for some regions. For example, in Manhattan (about 17 vs. 12 miles) and Nassau–Suffolk (25 vs. 26 miles), a significant difference cannot be determined. Although the exact magnitude varies, not-deficit households in most places tended to average substantially more driving miles compared with deficit households with fewer cars than drivers, except in a few densely populated or car-saturated areas where the MOEs do not confirm a gap.

Among vehicle deficit households, having just one car (i.e., one vehicle multiple drivers) versus multiple cars (but still not enough for every driver) often yields no clear difference in daily VMT. In Nassau–Suffolk, for instance, the two estimates (24.9 vs. 25.0) are identical. Similarly, Putnam–Rockland–

Westchester at about 16 vs. 19 miles and nonurban areas at roughly 18 vs. 20 miles do not confirm a statistically significant difference. A few regions stand out, however. In the rest of the US, single-vehicle deficit households average around 16 miles, and multivehicle deficit households are at about 21 miles, a significant difference. The other urban region shows a particularly high number (31) for one-car households, but an extremely large MOE (about 32), indicating the difference is not significant. Overall, multivehicle deficit households sometimes travel more, but the data in many regions reveal substantial overlaps, meaning the difference is not always statistically reliable.

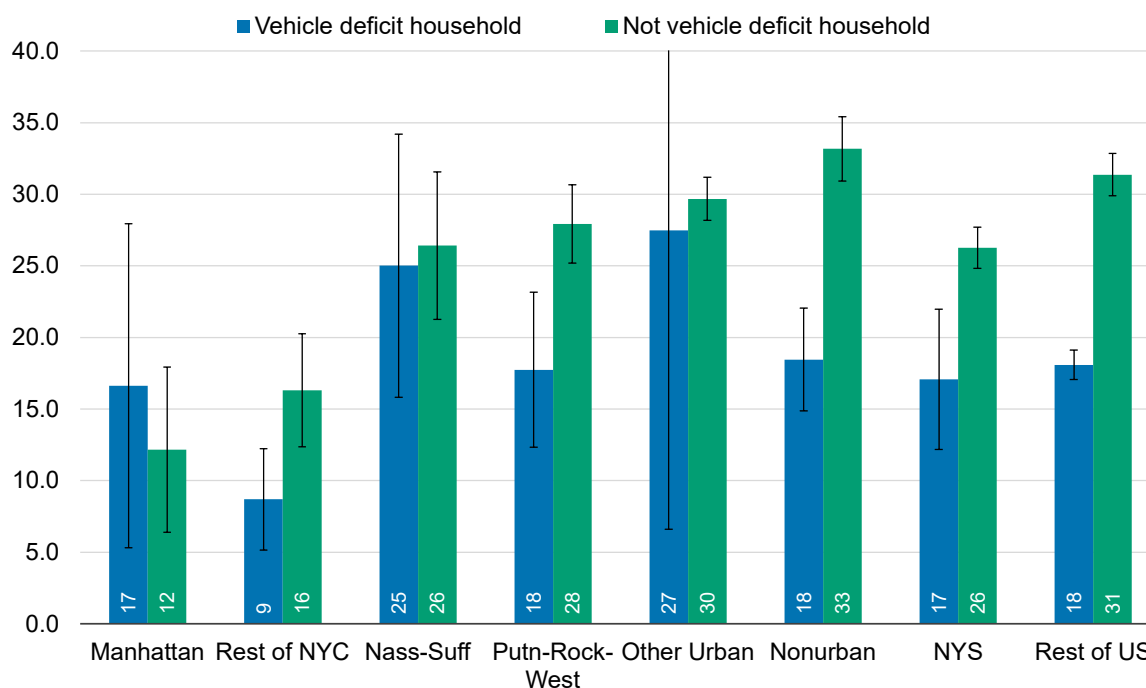


Figure 3-17. Average daily vehicle miles of travel per driver by vehicle ownership type (2017 NHTS data).

Table 3-6. Average daily VMT per driver by vehicle deficit household type (2017 NHTS data).

Region	Estimates		Margin of error	
	One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household	One vehicle multiple drivers in household	Vehicle deficit with more than one vehicle in household
Rest of NYC	8.0	10.8	4.2	5.8
Nass-Suff	24.9	25.0	12.8	15.4
Putn-Rock-West	15.9	19.0	6.0	6.2
Other Urban	31.1	22.8	32.0	8.2
Nonurban	17.7	19.6	5.4	4.1
NYS	15.7	19.4	7.2	3.5
Rest of US	16.1	20.8	1.6	1.9

Note: Manhattan is excluded because of low sample size.

4. USE OF TRANSPORTATION-RELATED TECHNOLOGIES

The 2017 NHTS dataset sheds light on the adoption of various transportation-related technologies that can reduce dependence on conventional car travel. Examples include online shopping; telecommuting; and ride-, car-, and bike-sharing services. This section explores how often these technologies are used, placing special emphasis on zero vehicle households and comparing their activity with other vehicle-ownership categories across the regions. The analysis draws on survey questions asking respondents how many times they engaged with these technologies during the previous 30 days.

4.1 ONLINE SHOPPING

The distribution of online purchases by household vehicle ownership type is shown in Figure 4-1. Zero vehicle households show distinct online shopping patterns compared with households with at least one car. In NYC, for instance, 45% of zero-car households made no online purchases in the past 30 days, which is higher than the 31%–38% “no purchase” share among vehicle-owning households. Zero vehicle households in NYC still shopped online more frequently than their counterparts in other urban (71% with none) or nonurban (75% with none) regions of NYS. This suggests that although zero vehicle households in major cities may rely somewhat more on online deliveries (possibly to offset limited car access), many still do little or no online shopping.

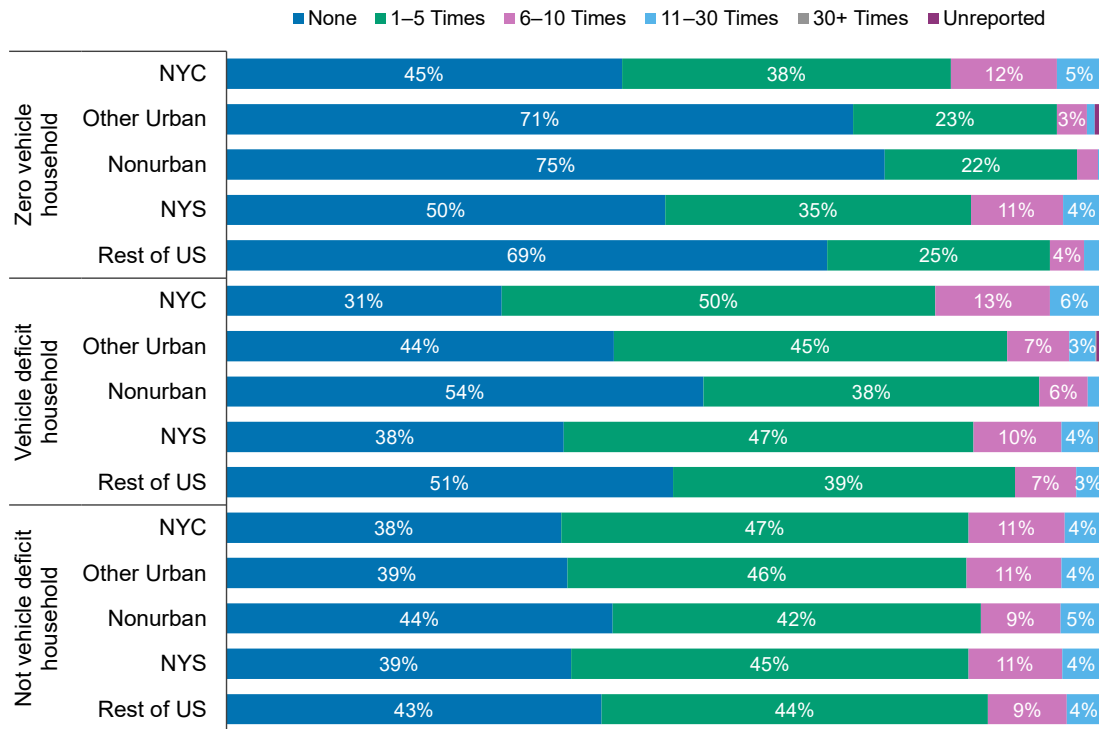


Figure 4-1. Distribution of online purchases by household vehicle ownership type (2017 NHTS data).

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Unlabeled segments represent values lower than 3%.

Outside NYC, zero vehicle households in both other urban and nonurban areas appear the least likely to engage in online purchases, with around three-quarters reporting no online shopping at all. By contrast,

vehicle deficit and not-deficit households in those same areas consistently reported fewer nonshoppers and more frequent purchases (e.g., 6–10 or 11–30 times). Taken together, these data indicate that zero vehicle status does not necessarily lead to high reliance on e-commerce, especially in less dense or less transit-friendly regions. In more urban settings like NYC, however, car-free living does correlate with moderately higher rates of online shopping, yet those rates are still below vehicle-owning households' rates.

When vehicle deficit households are broken down by whether they have one vehicle or more than one, some consistent patterns emerge across regions (Table 4-1). In NYC, both groups show roughly the same share of nonshoppers (31%), but those with multiple vehicles made online purchases a bit more frequently overall. For instance, 15% made purchases online 6–10 times a month versus 13% among single-vehicle deficit households. Outside NYC, the multivehicle deficit group also tended to report lower “no purchase” rates than those of single-vehicle deficit households in the same regions. For example, in other urban areas of NYS, only 40% of multivehicle deficit households did no online shopping, compared with 48% of those sharing a single car. Meanwhile, nonurban families in both deficit categories saw elevated shares of nonshoppers (50%–58%), pointing to potentially limited delivery services or less comfort with online transactions.

Table 4-1. Distribution of online purchases by vehicle deficit household type.

Vehicle deficit household type	Region	None	1–5 times	6–10 times	11–30 times	30+ times	Unreported
One vehicle multiple drivers in household	NYC	31%	49%	13%	6%	1%	0%
	Other Urban	48%	41%	6%	3%	0%	1%
	Nonurban	58%	36%	5%	1%	0%	0%
	NYS	38%	46%	10%	5%	0%	1%
	Rest of US	50%	39%	7%	3%	0%	0%
Vehicle deficit with more than one vehicle in household	NYC	31%	50%	15%	4%	0%	0%
	Other Urban	40%	49%	8%	3%	0%	0%
	Nonurban	50%	42%	6%	2%	0%	0%
	NYS	39%	48%	10%	3%	0%	0%
	Rest of US	52%	39%	6%	2%	0%	0%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

4.2 WORKING FROM HOME

4.2.1 Usually Work from Home

Figure 4-2 shows the distribution of reported households with members who usually work from home by vehicle ownership type using 2017 NHTS data. In NYC, zero vehicle households reported working from home at a rate of 11%, which is slightly higher than that of vehicle deficit households (10%) and not-deficit households (8%). In nonurban areas, however, only about 5% of zero-car households usually worked from home, which is lower than the 8%–9% rate seen among households that own at least one vehicle. That pattern repeats in other urban regions of NYS, where 6% of zero vehicle households worked from home, which is below the roughly 9% share among vehicle-owning households. Interestingly, in the rest of the US, zero vehicle households (13%) slightly surpassed not-deficit households (11%) working

from home. Overall, the rest of the US had a higher share of people working from home than that of NYS in all vehicle ownership categories.

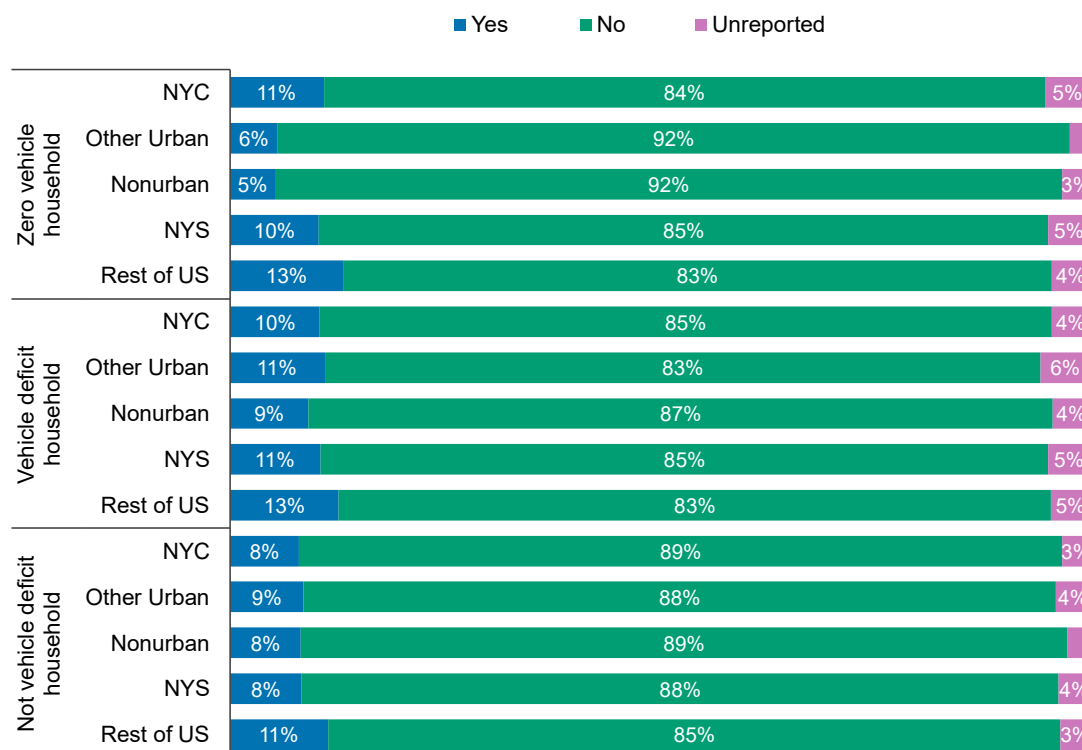


Figure 4-2. Distribution of work from home by household vehicle ownership type (2017 NHTS data). Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values that are 2% or lower.

Looking exclusively at vehicle deficit households, roughly 13%–14% of one-vehicle-multiple-driver households reported typically working from home across NYS, which is on par with or slightly above the rest of the country (Figure 4-3). By contrast, multivehicle deficit households in NYS hovered at just 7% working from home overall, and only 2% reported working from home in NYC specifically. That NYC figure stands in sharp contrast to the 13% working from home among single-vehicle deficit households. This may suggest that households having just one car may have more individuals in telework-friendly jobs or may prefer remote arrangements to avoid sharing limited vehicles.

4.2.2 Option to Work from Home

Figure 4-4 illustrates whether workers have the option to work remotely, regardless of whether they currently do so. Among zero vehicle households, the share of people reporting this option varies noticeably by region. In NYC, nearly a quarter (24%) of zero vehicle respondents said they could work from home if they chose to, well above the shares in nonurban areas (1%) or other urban parts of the state (11%). Interestingly, that 24% figure in NYC is also higher than the 16% seen among not-deficit households in the same city. Overall, this pattern could suggest that households without cars in dense urban centers like NYC often have job types amenable to remote work. Those in suburban or rural settings, however, generally do not have jobs suitable for remote work, regardless of how many vehicles they own. In the rest of the US, zero vehicle households are only modestly above the not-deficit group (20% vs. 16% nationally).

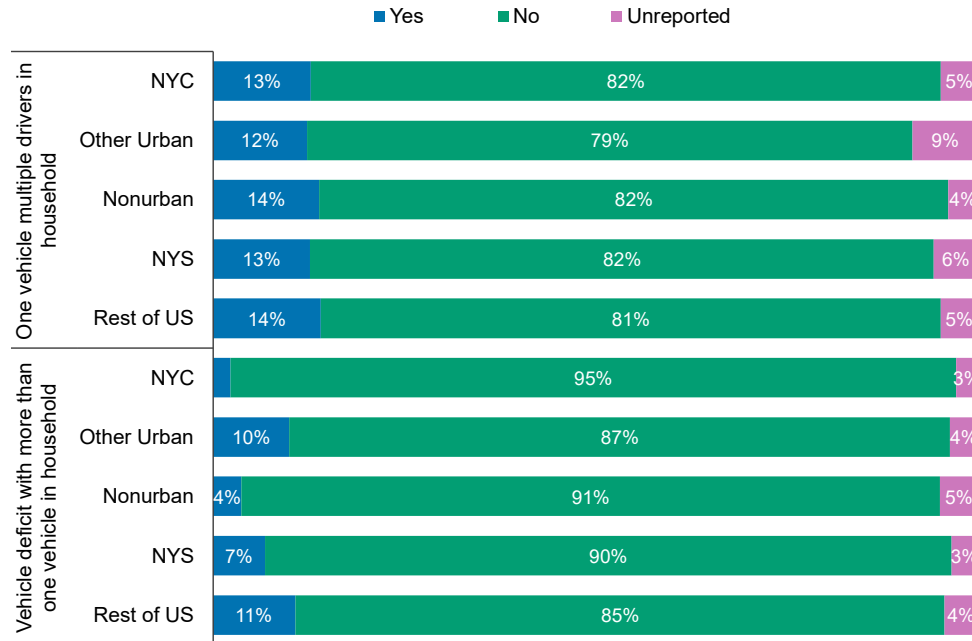


Figure 4-3. Distribution of work from home by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

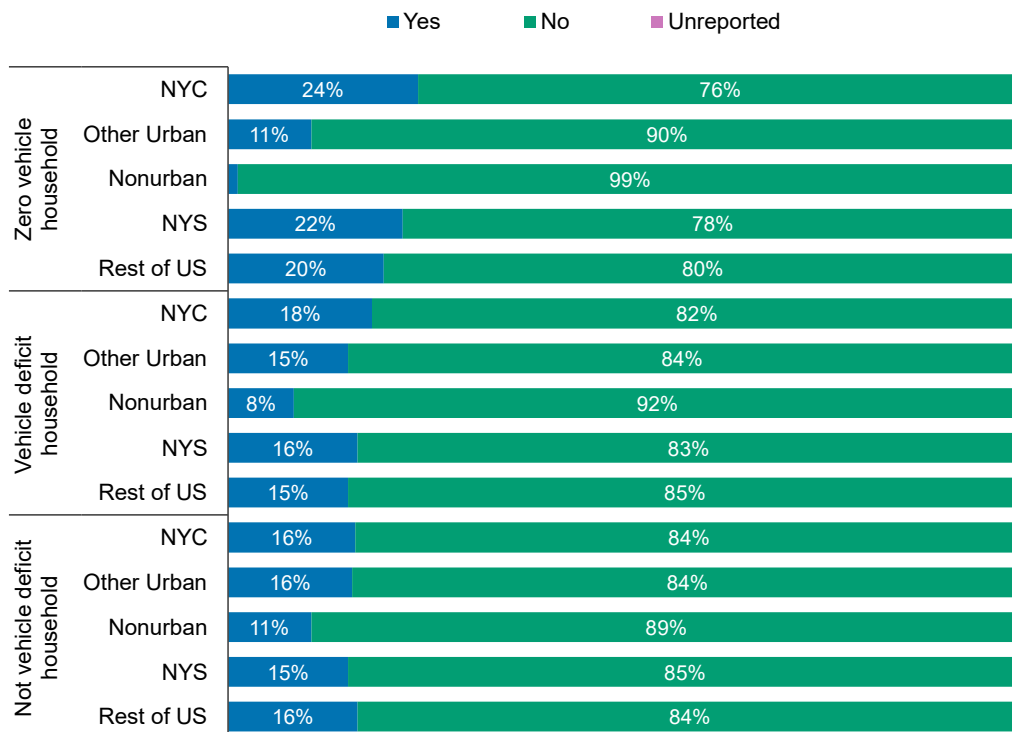


Figure 4-4. Distribution of option to work from home by household vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 2%.

4.2.3 Frequency of Work from Home

Table 4-2 shows the distribution of how many workdays per month participants actually teleworked in cases where teleworking was an option. For zero vehicle households in NYC, about a quarter share (22%) reported no telework days despite having the option, whereas most (61%) teleworked 1–5 days monthly, and around 8% logged 10 or more remote workdays. This pattern suggests that even without a car, many NYC residents work jobs that allow some degree of telecommuting. Outside the city, however, the trend shifts. In NYS other urban areas, 38% of zero vehicle households reported that they never worked from home, and that figure jumps to 87% in nonurban parts of the state. The near absence of teleworking among rural zero vehicle households could stem from more limited job opportunities, technological constraints (e.g., limited broadband access), or industries less amenable to remote work. At a statewide level (NYS), zero vehicle households still posted a higher share of workers (about three-quarters) than those in many nonurban areas who do at least some telework. Zero vehicle households in the rest of the US typically reported 32% with no telework, which is higher than the NYS statewide share.

Table 4-2. Distribution of count of days worked from home per month by household vehicle ownership type.

Vehicle ownership type	Region	None	1–5 times	6–10 times	10+ times
Zero vehicle household	NYC	22%	61%	8%	8%
	Other Urban	38%	31%	19%	13%
	Nonurban	87%	13%	0%	0%
	NYS	23%	60%	9%	9%
	Rest of US	32%	55%	5%	8%
Vehicle deficit household	NYC	29%	46%	23%	3%
	Other Urban	18%	62%	12%	8%
	Nonurban	25%	26%	21%	29%
	NYS	25%	51%	19%	5%
	Rest of US	24%	51%	13%	13%
Not vehicle deficit household	NYC	14%	67%	10%	9%
	Other Urban	26%	50%	16%	8%
	Nonurban	28%	37%	21%	14%
	NYS	23%	53%	15%	9%
	Rest of US	23%	51%	15%	10%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Regarding vehicle deficit household categories (Figure 4-5), nearly one-third (32%) of one-vehicle-multiple-driver households did not work from home in NYC, whereas the no telework share for vehicle deficit households that own multiple vehicles was 25%. In NYS nonurban settings, only 6% reported “none,” but a substantial 30% teleworked 6–10 times per month, and 34% teleworked 10 or more times for the one-vehicle-multiple-driver households. In contrast, days per month that other deficit households spent teleworking were almost evenly split: never (44%), infrequently (22% at 1–5 times), or regularly (24% at 10+ times).

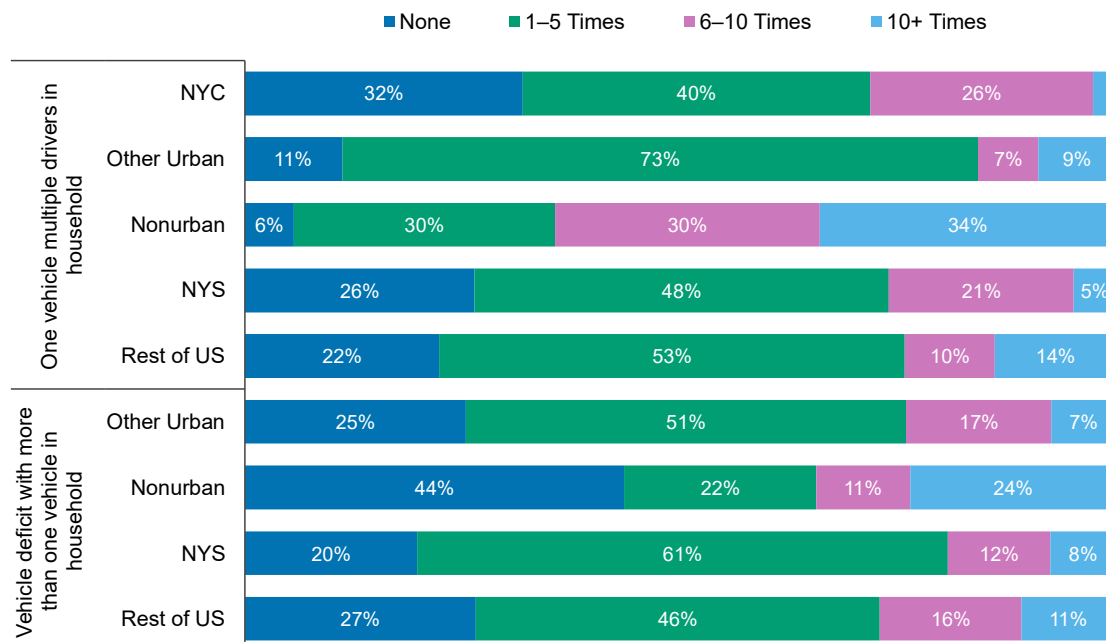


Figure 4-5. Distribution of count of days worked from home per month by vehicle deficit household type.
 Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.
 Unlabeled segments represent values lower than 5%.

4.3 RIDE-SHARING

As shown in Figure 4-6, among zero vehicle households, NYC stands out with about 69% who never used rideshare apps and around 21% reporting 1–5 rides monthly—double or more the proportion seen in other parts of NYS. In nonurban and other urban areas of NYS, nearly all zero vehicle households (97%–99%) refrained from ride-sharing. This contrast likely reflects the abundant availability of services like Uber or Lyft in NYC, compared with limited coverage or higher costs in smaller cities and rural regions. Even nationally, only 16% of zero vehicle respondents reported they used rideshare at least once a month. This could suggest that outside major urban centers, having no vehicle does not necessarily translate to frequent rideshare usage.

Relative to vehicle deficit or not-deficit households, zero vehicle families are more likely to adopt ride-sharing in dense, transit-rich environments like NYC. That said, the pattern reverses outside Manhattan; zero-car households appear less engaged in ride-sharing than those who own cars, possibly because they could not readily afford on-demand services or because service is less accessible. By contrast, vehicle deficit and not-deficit households in NYC still have relatively high “no rideshare” shares (74%–84%), but many did report at least occasional use. In suburban or rural parts of the state, nearly all households, regardless of car ownership, fall into the “no rideshare” category.

4.4 CAR-SHARING

Car-sharing emerges as a viable option even in urban centers, though zero vehicle households are slightly more likely than households with cars to use it, as presented in Table 4-3. In NYC, for instance, 4% of zero vehicle households used carshare once in the past month, and 3% used it two or more times. These shares, while small overall, surpassed those for vehicle-owning groups, of whom 1% or fewer reported multiple uses. This pattern likely reflects the fact that carshare serves as an occasional stand-in for private car access in dense environments with limited parking and robust public transit. Outside NYC, usage rates

dropped even lower. In other urban areas, about 3%–4% of zero vehicle households reported using carshare at least once a month, but nearly all vehicle deficit and not-deficit households reported no usage at all. In nonurban areas, car-sharing was almost nonexistent across every ownership type, likely due to insufficient provider coverage, low population density, and the continued need for personal vehicles in spread-out communities. Even at the national level, 96% of zero vehicle households never used car-sharing.

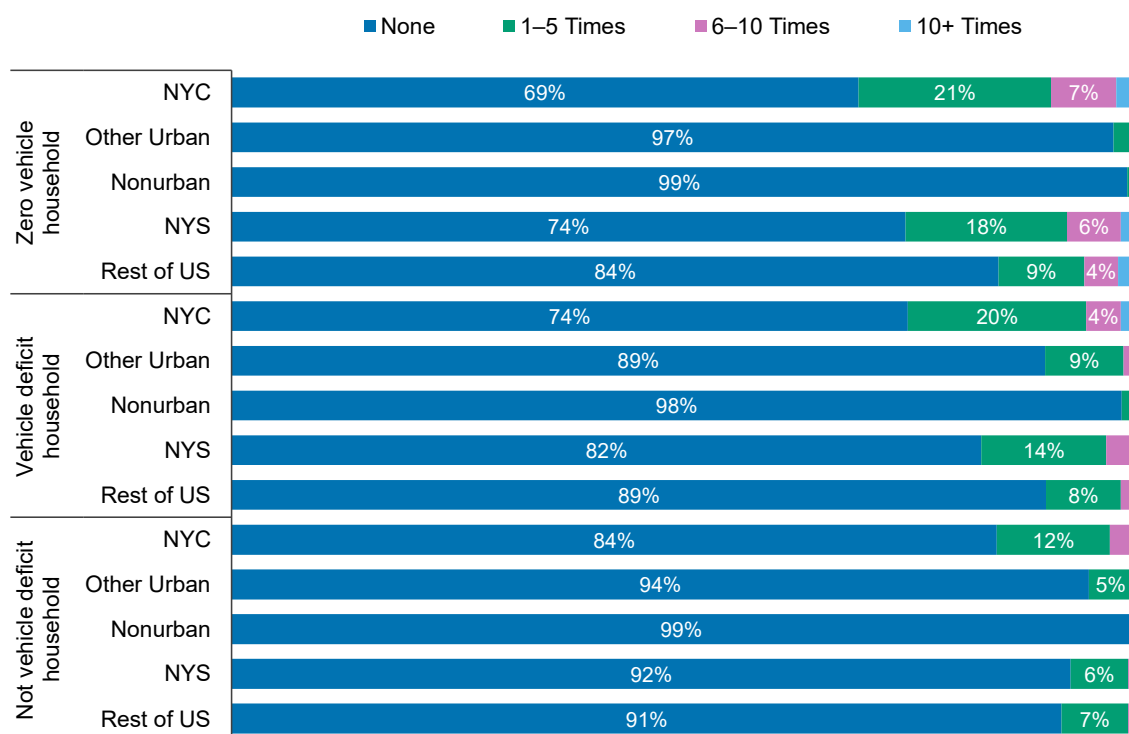


Figure 4-6. Distribution of rideshare app usage by household vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values that are 3% or lower.

4.5 BIKE-SHARING

Shifting from car-sharing to another emerging mobility option, bikeshare usage patterns are examined by household vehicle ownership categories across different regions (Figure 4-7). Among zero vehicle households, NYC again stands out with only 67% saying they never used bikeshare, leaving nearly a third (29%) who used it two or more times in a given period. This is far above the 13% figure for not-deficit households in NYC. The high usage highlights the role bikeshare can play in a transit-rich urban environment where many residents do not own cars. Outside of NYC, however, bikeshare adoption dropped significantly, even for zero vehicle households: in other urban parts of the state, 88% never used bikeshare, and in nonurban areas, that number climbed to 93%.

Comparatively, vehicle deficit and not-deficit households exhibit even higher rates of “none” across most regions, indicating a lower reliance on bikeshare when cars are available. For instance, 84% of not-deficit households in NYC reported never using bikeshare versus 67% for their zero-car counterparts. In nonurban areas, bikeshare usage was nearly nonexistent across all ownership types (close to 100% “none”). These findings could underscore that bikeshare is primarily a big-city phenomenon, especially

appealing to households without a vehicle. Suburban and rural regions, with sparser infrastructure and longer travel distances, saw minimal adoption regardless of vehicle ownership levels.

Table 4-3. Distribution of carshare usage per month by household vehicle ownership type.

Vehicle ownership type	Region	None	1 time	2+ times
Zero vehicle household	NYC	93%	4%	3%
	Other Urban	96%	3%	1%
	Nonurban	99%	1%	0%
	NYS	94%	4%	3%
	Rest of US	96%	1%	2%
Vehicle deficit household	NYC	98%	1%	1%
	Other Urban	99%	0%	1%
	Nonurban	100%	0%	0%
	NYS	99%	1%	1%
	Rest of US	99%	0%	0%
Not vehicle deficit household	NYC	98%	1%	1%
	Other Urban	100%	0%	0%
	Nonurban	100%	0%	0%
	NYS	99%	1%	0%
	Rest of US	99%	0%	0%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

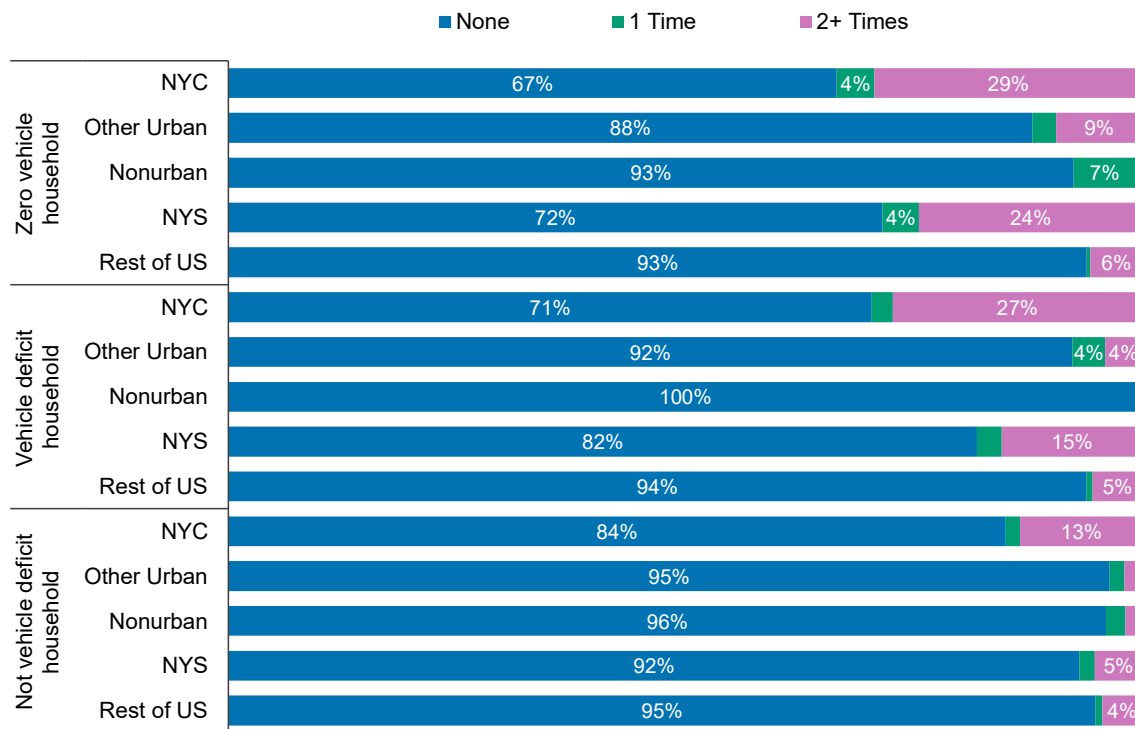


Figure 4-7. Distribution of bikeshare usage per month by household vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values that are 3% or lower.

5. VIEWS OF TRANSPORTATION SERVICES

In the 2017 NHTS recruitment survey, one adult per household was asked how travel costs influence travel behavior. Specifically, respondents were presented with the following question: “How much do you agree or disagree with each of the following statements?” The statements included the following:

- The price of gas affects the number of places I go.
- Getting from place to place costs too much.
- I walk to places to save money.
- I bike to places to save money.
- I use public transportation to save money.

For each statement, participants selected a response from five options: strongly agree, agree, neutral, disagree, or strongly disagree.

The analysis focuses on responses from different geographic areas, including NYC, the NYMTC region, other urban areas in NYS (excluding NYMTC), NYS overall, and the rest of the US.

5.1 PRICE OF GAS AFFECTS AMOUNT OF TRAVEL

Figure 5-1 shows the share of ratings on the statement, “The price of gas affects the number of places I go.” One key regional difference is that in NYC, a sizable portion of zero vehicle households (38%) neither agreed nor disagreed with the statement, a rate much higher than those in other regions. This suggests that gas prices may not be a primary factor influencing their travel decisions, possibly owing to a strong reliance on public transit. In contrast, a larger share of zero vehicle households in nonurban areas either agreed or strongly agreed (53% in total), which implies that even those without a vehicle are indirectly affected, perhaps owing to their dependence on ride-sharing, taxis, or carpooling. For vehicle deficit households, NYC and the NYMTC region show a high percentage of disagreement (29%), whereas in other regions, especially in nonurban areas, agreement is more common (60% combined). This highlights that in rural areas, where driving is necessary and alternatives are limited, gas prices have a stronger effect on travel choices. In contrast, in NYC, even among vehicle deficit households, sufficient nondriving options may be available, reducing sensitivity to gas price fluctuations. In the rest of the US, vehicle deficit household respondents were more likely to strongly agree or agree (55%) than their counterparts in NYC (42%). This aligns with expectations that outside of dense urban environments, fuel costs have a more direct effect on daily mobility. Additionally, among nonurban respondents in vehicle-sufficient households, the share of those who disagreed is notably lower, suggesting that car dependency reduces flexibility in travel behavior when fuel prices fluctuate. Overall, the data suggest that in regions with stronger public transit infrastructure, like NYC, the effect of gas prices on mobility is more muted. However, in areas where driving is a necessity, such as nonurban regions, both zero vehicle households and vehicle deficit households perceive a stronger influence of fuel costs on their ability to travel.

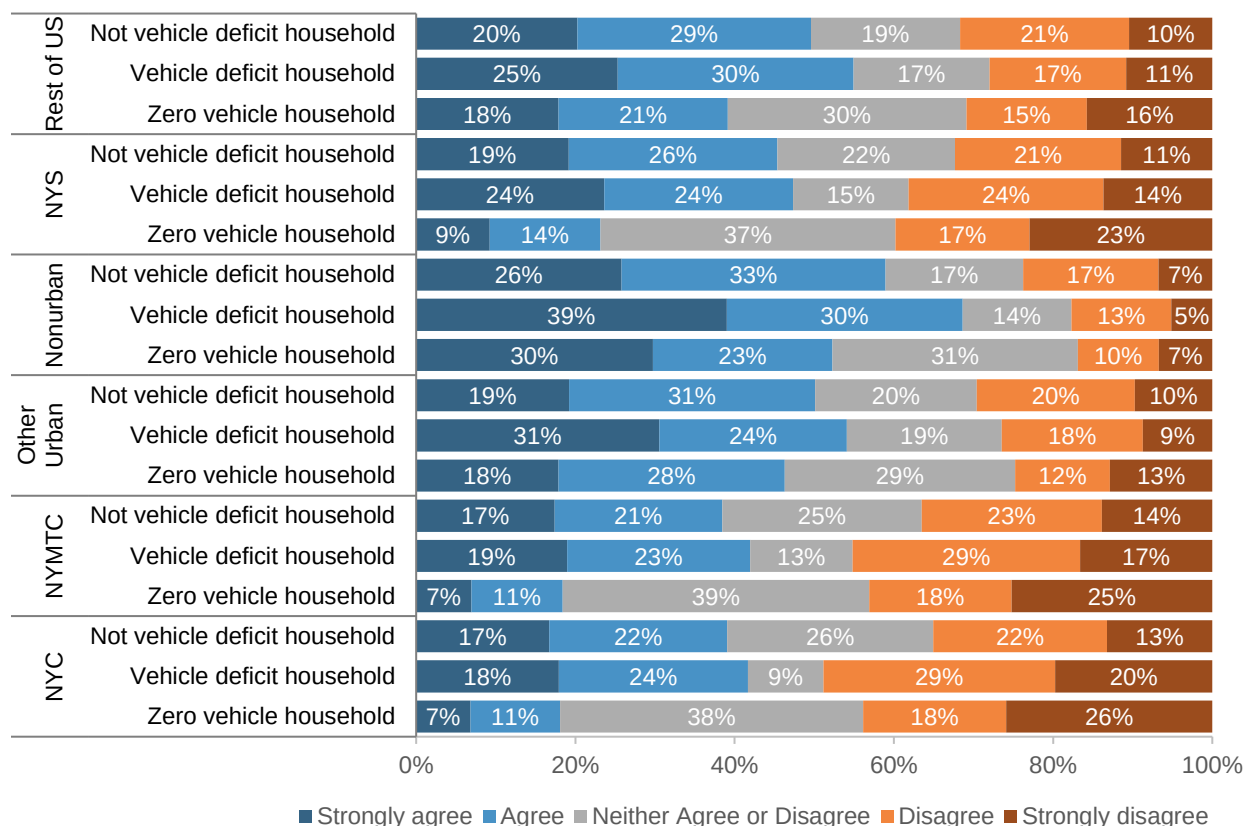


Figure 5-1. Share of ratings on “The price of gas affects the number of places I go” by household vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Figure 5-2 focuses on several types of vehicle deficit households and households with one vehicle and multiple drivers. One major regional difference is that in nonurban areas, respondents in both categories—households with one vehicle and multiple drivers and vehicle deficit households with more than one vehicle—tended to agree more strongly that gas prices affect their mobility. For instance, 40% of respondents in nonurban households with one vehicle and multiple drivers strongly agreed, and 37% in vehicle deficit households with more than one vehicle also strongly agreed. This suggests that in rural areas where alternative transportation options are limited, gas prices play a significant role in restricting travel. In contrast, a higher share of respondents in NYC tended to disagree or strongly disagree with the statement. Among households with one vehicle and multiple drivers, 31% disagreed, whereas among vehicle deficit households with more than one vehicle, the disagreement rate is 19%. Additionally, 18% of the former and 30% of the latter strongly disagreed. This pattern aligns with the expectation that in NYC, where extensive public transit options exist, individuals are less directly affected by fuel costs when making travel decisions.

In the rest of the US, vehicle deficit households with more than one vehicle show higher agreement (24% strongly agree, 32% agree) than their counterparts in urban areas. This suggests that outside dense metropolitan regions, vehicle deficit households—even with multiple vehicles—may struggle to maintain flexibility in travel behavior when gas prices increase. Another trend is that across nearly all regions, households with one vehicle and multiple drivers are slightly more sensitive to gas prices compared with vehicle deficit households with more than one vehicle. This may indicate that in multidriver households with only one vehicle, the need for shared access leads to greater constraints when fuel costs rise, whereas

vehicle deficit households with more than one vehicle may still have some flexibility in adjusting travel behaviors within the household. In summary, gas prices have a stronger influence on travel behavior in nonurban and less transit-served areas, particularly among households with only one vehicle shared among multiple drivers. In dense urban regions like NYC, where alternative mobility options exist, gas price sensitivity is lower, and more people disagreed that fuel costs affected their travel.

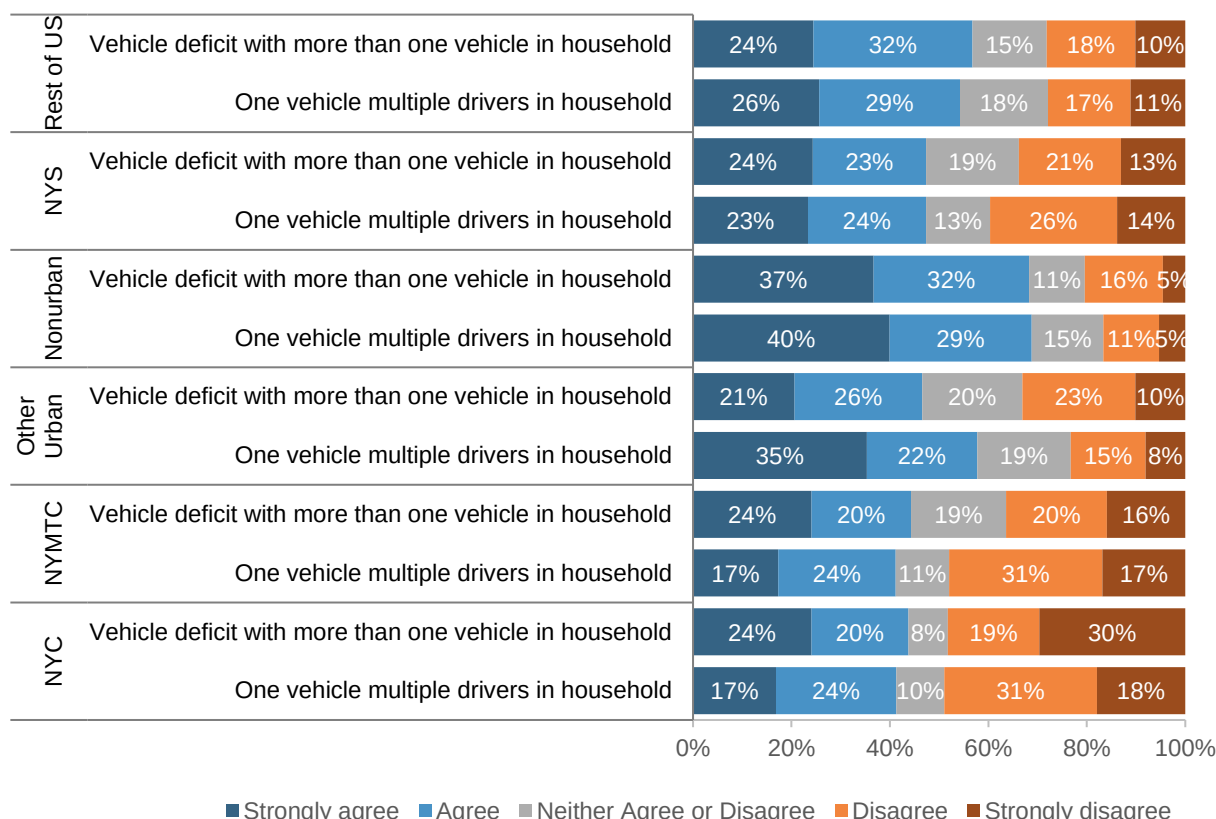


Figure 5-2. Share of ratings on “The price of gas affects the number of places I go” by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

5.2 TRAVEL COSTS TOO MUCH

Regarding the statement, “Getting from place to place costs too much,” zero vehicle households across all regions generally reported higher agreement (strongly agree and agree) with the statement compared with that of vehicle deficit households, as shown in Figure 5-3. This suggests that transportation costs may be particularly burdensome for households without vehicles, likely because of reliance on public transit, taxis, or ride-hailing services, which can be expensive over time. For example, in NYC zero vehicle households, 24% strongly agreed and 35% agreed, meaning nearly 60% of this group perceives transportation costs as high. However, among vehicle deficit households in NYC, only 15% strongly agreed, indicating a slightly lower concern about transportation affordability. This could be because vehicle deficit households may still have occasional access to a private car, mitigating their reliance on costly transportation alternatives. A clear regional divide exists when comparing urban and nonurban areas. In nonurban regions, vehicle deficit households showed particularly high agreement with the statement. For example, in nonurban areas, 42% of vehicle deficit households agreed that transportation costs are too high, compared with 35% of not vehicle deficit households. This suggests that in less

densely populated areas, the cost of owning and maintaining a vehicle can be a significant financial burden because public transportation is often limited or nonexistent, making driving a necessity rather than a choice.

Another observation is that disagreement with the statement tends to be higher among not vehicle deficit households, particularly in nonurban and suburban areas. For example, in nonurban areas, 33% of not vehicle deficit households disagreed with the statement, compared with 28% of vehicle deficit households. This implies that households with sufficient vehicle access may not feel the same cost pressures as those who struggle with transportation access. In the rest of the US, zero vehicle households expressed a stronger perception of high transportation costs, with 33% agreeing and 20% strongly agreeing. However, a difference emerged between vehicle deficit and not vehicle deficit households, with the former group being more likely to agree that transportation is costly.

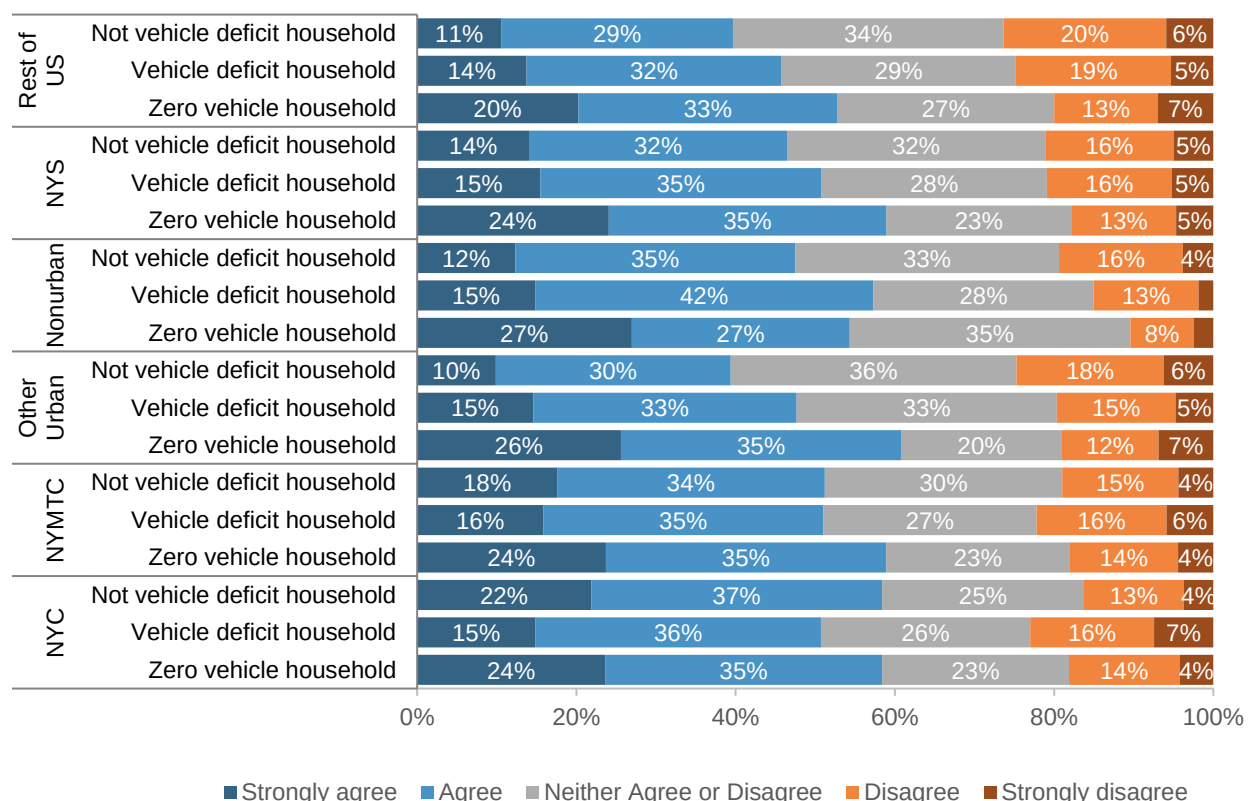


Figure 5-3. Share of ratings on “Getting from place to place costs too much” by household vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 4%.

Among households with one vehicle and multiple drivers and vehicle deficit households with more than one vehicle, nonurban areas exhibited the highest concern over transportation costs, particularly among vehicle deficit households with more than one vehicle (Figure 5-4). In nonurban areas, 22% of respondents in vehicle deficit households with more than one vehicle strongly agreed that transportation costs are excessive, and an additional 40% agreed. In NYC, however, vehicle deficit households with more than one vehicle showed lower concern, with only 9% strongly agreeing that transportation costs too much. This is the lowest percentage in the table, which can likely be attributed to the availability of extensive public transportation options that provide alternatives to car ownership. Households with one

vehicle and multiple drivers in all regions tended to agree at similar rates that transportation costs are high, with agreement percentages ranging from 31% to 44%. However, nonurban areas again stand out—44% of one vehicle multiple driver households agreed with the statement, indicating that car-sharing among multiple drivers can create economic strain, especially in regions with limited transit options.

Another trend is seen in disagreement rates. In nonurban areas, vehicle deficit households with more than one vehicle show the lowest disagreement rates (9% disagree and 2% strongly disagree), reinforcing the idea that transportation affordability is a widespread concern in these areas. On the other hand, in NYC, vehicle deficit households with more than one vehicle have a relatively higher share of disagreement (11% disagree and 13% strongly disagree), indicating that although some residents feel the burden of transportation costs, others—likely benefiting from subsidized or efficient transit—do not perceive it as a major issue. In the rest of the US, vehicle deficit households with more than one vehicle have agreement rates (32% agree and 14% strongly agree) that are comparable to those in urban areas, suggesting that cost concerns for transportation are widespread but particularly heightened in nonurban areas.

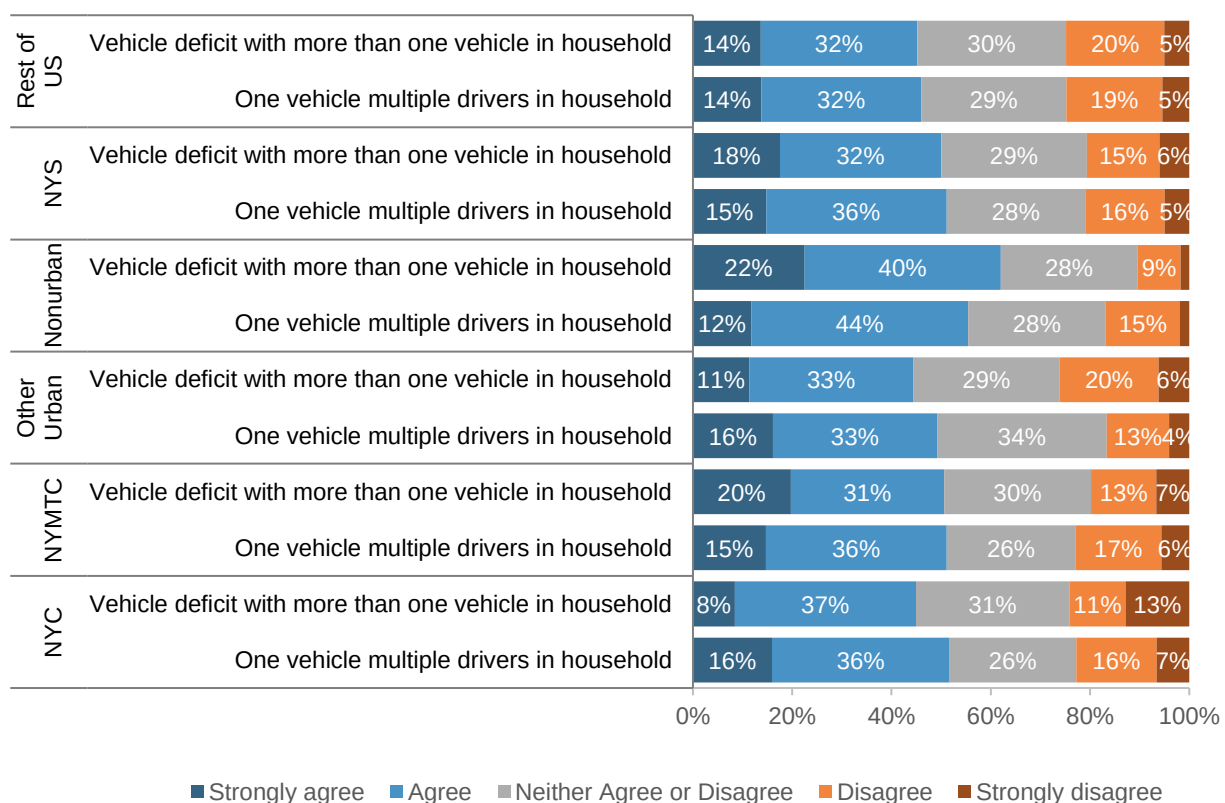


Figure 5-4. Share of ratings on “Getting from place to place costs too much” by vehicle deficit household type.
 Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.
 Unlabeled segments represent values lower than 4%.

5.3 WALK TO PLACES TO SAVE MONEY

Figure 5-5 shows the share of ratings on the statement, “I walk to places to save money.” Zero vehicle households consistently have the highest agreement (strongly agree and agree) with the statement across all regions. For example, in NYC zero vehicle households, 27% strongly agreed and 38% agreed, whereas vehicle deficit and not vehicle deficit households exhibited lower agreement levels. This trend is even

more pronounced in other urban and nonurban areas, where 43% and 34%, respectively, strongly agreed with the statement. This suggests that reliance on walking for financial reasons is significantly higher among those without vehicle access, particularly in urban regions. In contrast, not vehicle deficit households, which presumably have sufficient vehicle access, reported much lower agreement levels. For instance, in the rest of the US, only 4% of not vehicle deficit households strongly agreed with the statement, compared with 26% of zero vehicle households. This pattern highlights the reduced dependency on walking for financial savings among households with sufficient vehicles.

An interesting comparison can be seen between NYC and the rest of the US. Although NYC's zero vehicle households show 27% strongly agreeing and 38% agreeing, the corresponding figures in the rest of the US are lower (26% and 29%, respectively). This suggests that even in NYC, where walking is more feasible owing to urban density and transit accessibility, financial motivations still play a strong role in walking behavior. However, in nonurban areas, a very low percentage of respondents in vehicle deficit and not vehicle deficit households strongly agreed, often below 10%. Disagreement with the statement also varies notably by region and vehicle access. Not vehicle deficit households tended to disagree or strongly disagree at higher rates than those of zero vehicle households, with 30% of not vehicle deficit households in the rest of the US strongly disagreeing compared with only 13% of zero vehicle households. This suggests that walking as a cost-saving measure is much less relevant to those with sufficient vehicle resources, especially in suburban and rural settings.

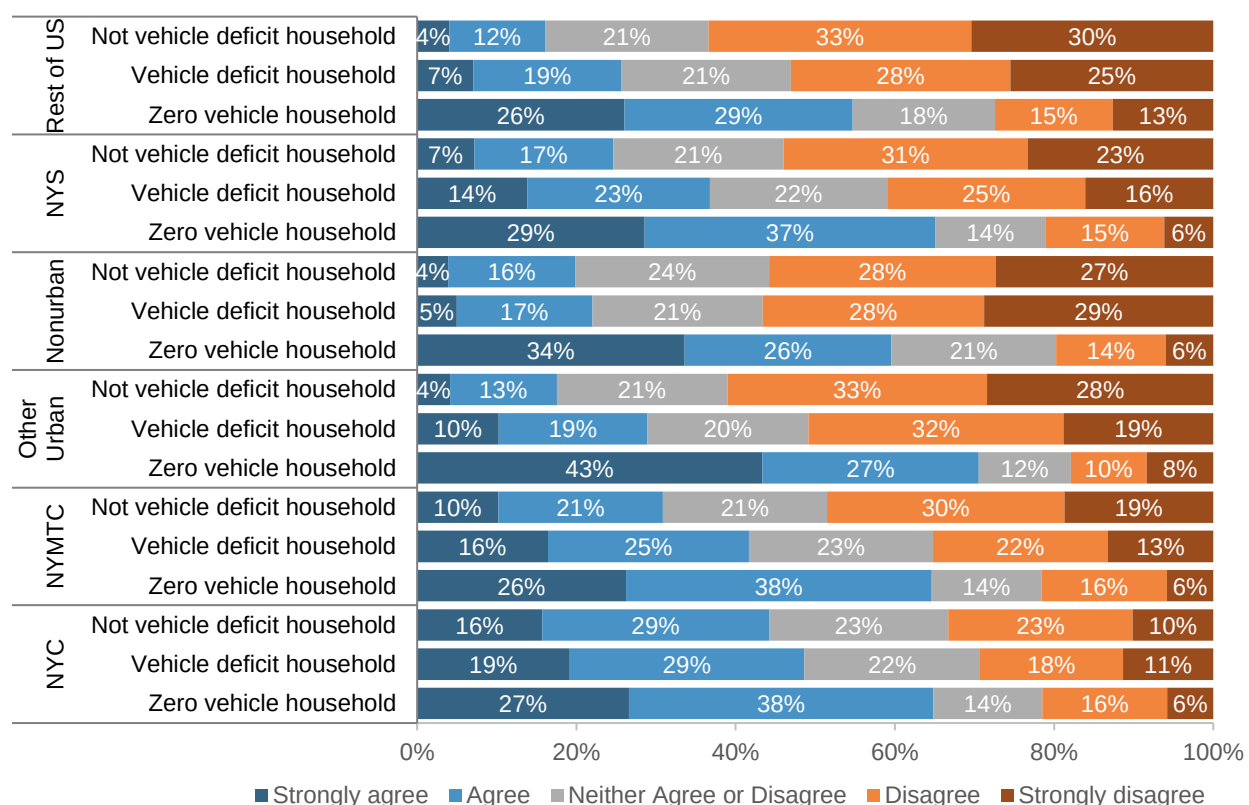


Figure 5-5. Share of ratings on “I walk to places to save money” by household vehicle ownership type.
 Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Figure 5-6 shows the share of ratings on the statement, “I walk to places to save money,” across different household vehicle situations. Individuals in households with one vehicle and multiple drivers are

generally more likely to agree or strongly agree with the statement compared with those experiencing a vehicle deficit (i.e., more drivers than vehicles in the household). For instance, in NYC, 52% of individuals in one vehicle multiple driver households either agreed or strongly agreed, whereas only 28% of those in vehicle deficit households expressed the same sentiment. This pattern is consistent across most regions, highlighting that people in households with some level of vehicle access but constrained sharing arrangements tend to rely on walking for cost-saving more than those facing a more severe vehicle deficit. Another contrast appears in the disagreement levels. In nonurban areas, individuals in vehicle deficit households expressed a high level of disagreement, with 62% either disagreeing or strongly disagreeing, compared with only 48% in urban areas outside NYC. This suggests that in rural settings—where walking is often less feasible because of infrastructure and land-use patterns—individuals are less likely to consider walking as a cost-saving measure, even when they face vehicle constraints. Looking at the NYMTC and the rest of the US, a similar pattern emerges, but with slightly lower agreement levels overall. In NYC and NYMTC, where public transit and walkability are higher, agreement rates are stronger, whereas in nonurban areas, individuals are more resistant to the idea of walking as a financial strategy. The difference is seen in vehicle deficit households, for which disagreement is highest in nonurban settings.

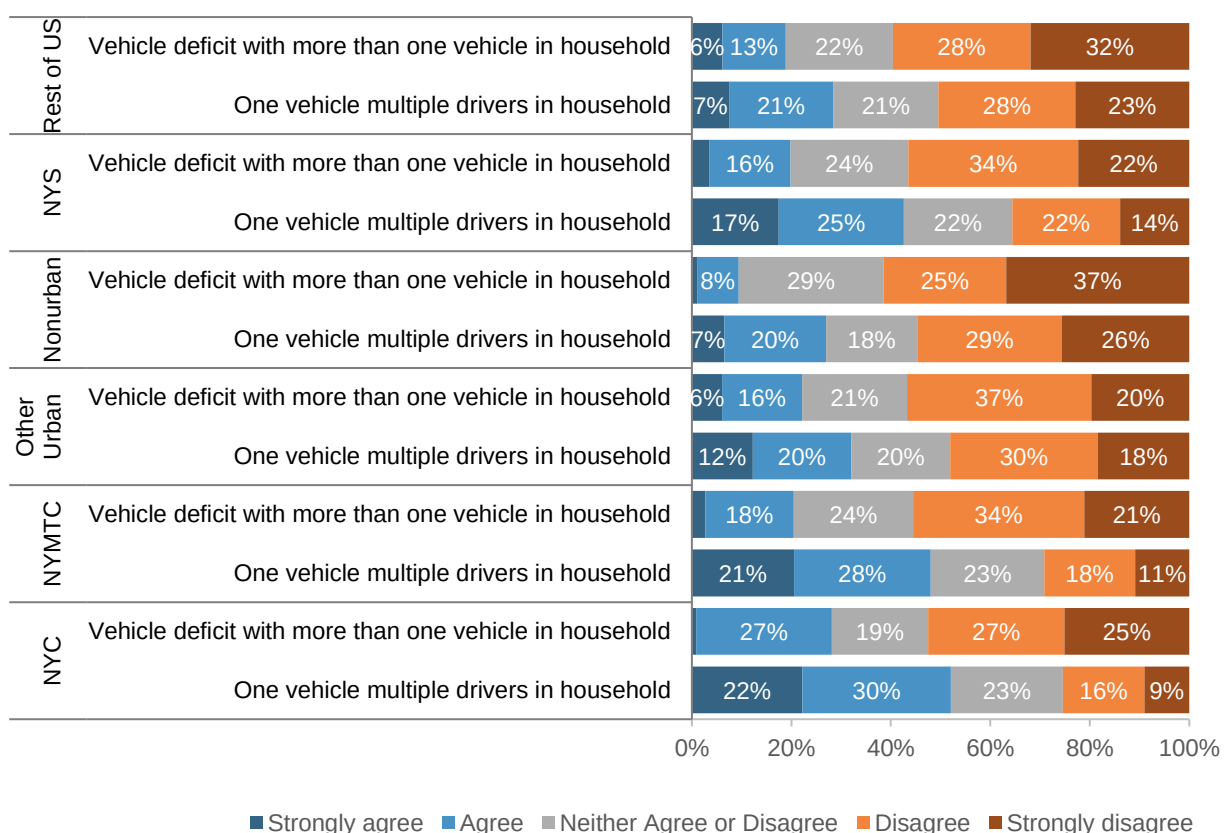


Figure 5-6. Share of ratings on “I walk to places to save money” by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 6%.

5.4 BIKE TO PLACES TO SAVE MONEY

Figure 5-7 shows the share of ratings on the statement, “I bike to places to save money,” across different household vehicle availability categories. A significant portion of respondents across all regions

expressed disagreement with the statement, with a particularly high share of strong disagreement. In most cases, over 30% of individuals in all household vehicle situations strongly disagreed, with nonurban and vehicle-sufficient households showing the highest resistance. For example, in nonurban households without a vehicle deficit, 48% strongly disagreed—the highest percentage across the table—suggesting that biking is not seen as a feasible or attractive money-saving option in these areas. Zero vehicle households, particularly in NYC and urban areas, exhibited relatively higher agreement rates compared with those with vehicles. In NYC, 20% of respondents in zero vehicle households either agreed or strongly agreed compared with just 11% in nondeficit households, indicating that those without cars in dense urban settings are more likely to rely on biking as a financial strategy. Even in these cases, disagreement remains dominant, which may reflect factors such as infrastructure limitations, safety concerns, or personal preferences. Across all regions, urban and NYC residents showed slightly more openness to biking for cost savings than nonurban respondents. However, even in urban areas, the overall low level of agreement suggests that biking for financial reasons is not a widely embraced behavior, possibly because of alternative transportation options such as transit, weather constraints, or a cultural preference for other modes.

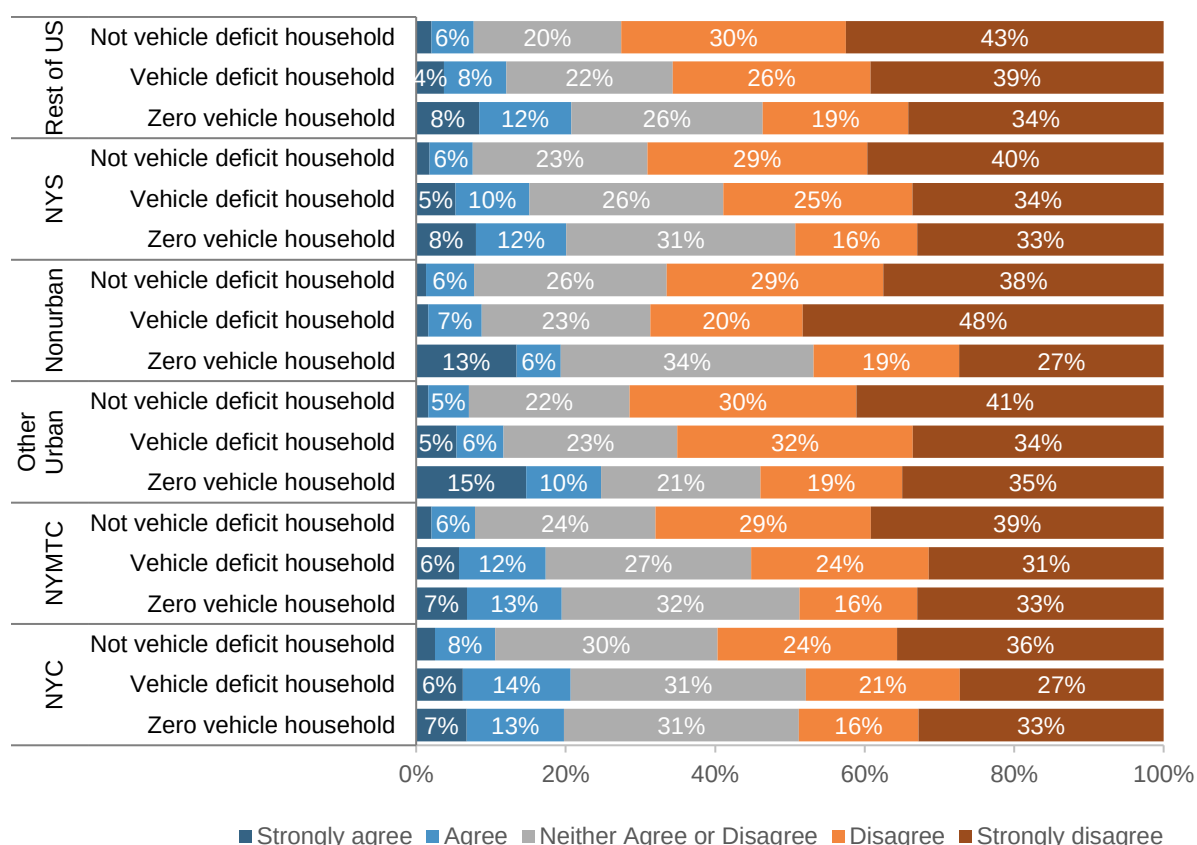


Figure 5-7. Share of ratings on “I bike to places to save money” by vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

In nonurban areas, individuals in vehicle deficit households exhibited some of the highest disagreement levels with this statement, with 45%–49% strongly disagreeing, reflecting the limited feasibility of biking in such regions owing to infrastructure constraints and longer travel distances (Figure 5-8). In urban areas, the trend is slightly different but follows a similar pattern. In NYC, for instance, disagreement is somewhat lower in one vehicle multiple driver households, with 19% disagreeing and 23% strongly

disagreeing, compared with 52% strong disagreement in vehicle deficit households. This suggests that although biking may be a more viable option in dense urban environments, those in vehicle deficit households are still unlikely to use it as a cost-saving strategy. Across all regions, agreement with the statement remained quite low, with agree and strongly agree responses rarely exceeding 15%. The highest agreement is seen in NYC and NYMTC one vehicle multiple driver households, about 23% of which expressed some level of agreement.

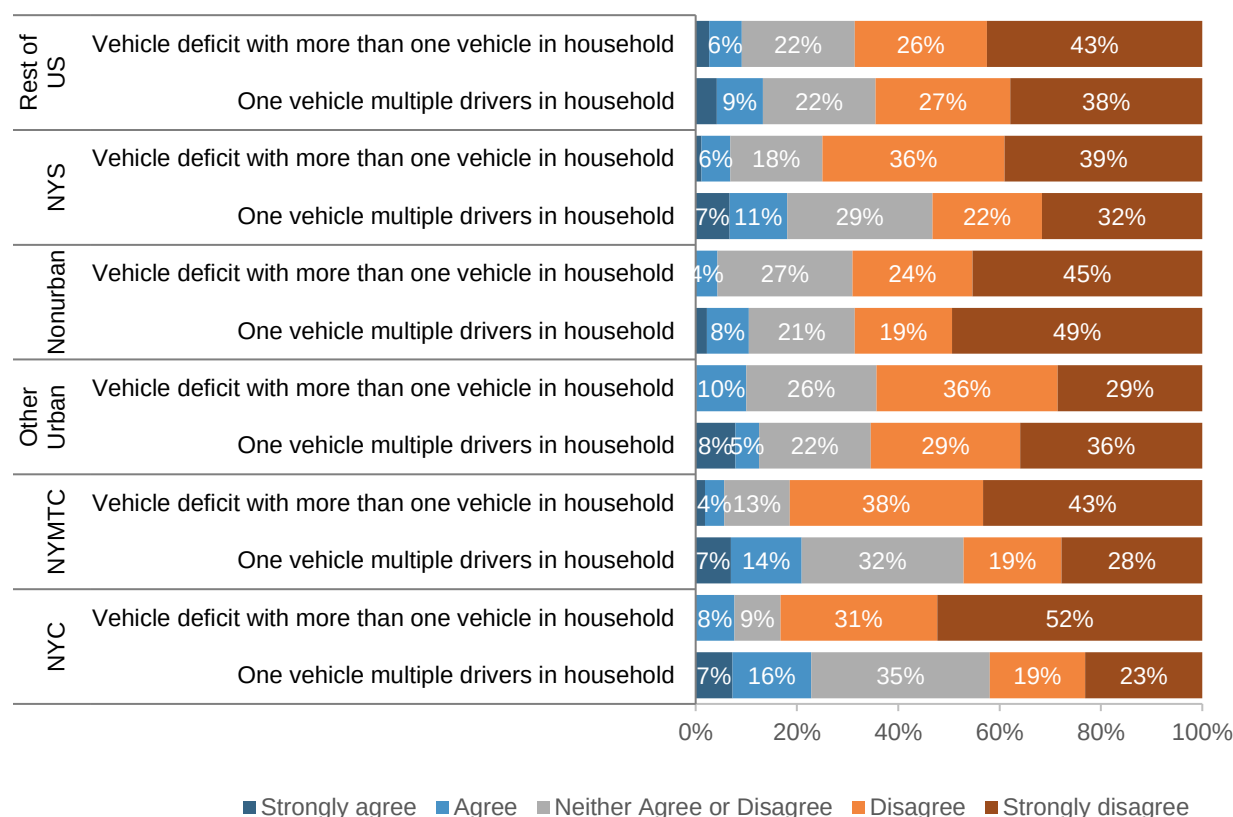


Figure 5-8. Share of ratings on “I bike to places to save money” by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 4%.

5.5 USE PUBLIC TRANSPORTATION TO SAVE MONEY

Figure 5-9 presents the share of ratings on the statement, “I use public transportation to save money,” by household vehicle availability. A strong regional divide is evident, with public transportation more commonly used as a cost-saving strategy in urban areas, particularly in NYC, NYMTC, and other urban settings. In NYC, for example, 42% of individuals in zero vehicle households strongly agreed with the statement, and an additional 33% agreed, indicating widespread reliance on transit for financial savings. A similar pattern is observed in NYMTC and other urban areas, where agreement is significantly higher among those without a vehicle. In contrast, disagreement is substantially higher in nonurban areas and the rest of the US, especially in vehicle deficit and nondeficit households. In nonurban vehicle deficit households, 54% strongly disagreed with the statement, highlighting the limited availability and practicality of public transit in rural areas. Even among zero vehicle households in nonurban settings, disagreement remains high, with 41% either disagreeing or strongly disagreeing. Households with a vehicle deficit, particularly outside urban areas, show lower agreement levels compared with those of zero

vehicle households, likely reflecting their greater dependency on personal vehicles. In NYC and other urban regions, however, even those with a vehicle deficit exhibited moderate agreement, indicating that transit remains a somewhat viable alternative despite household vehicle constraints. In short, the data show a strong reliance on public transportation in transit-rich urban environments and the significant barriers to its use in less urbanized areas, where disagreement with the cost-saving benefits of transit is most pronounced.

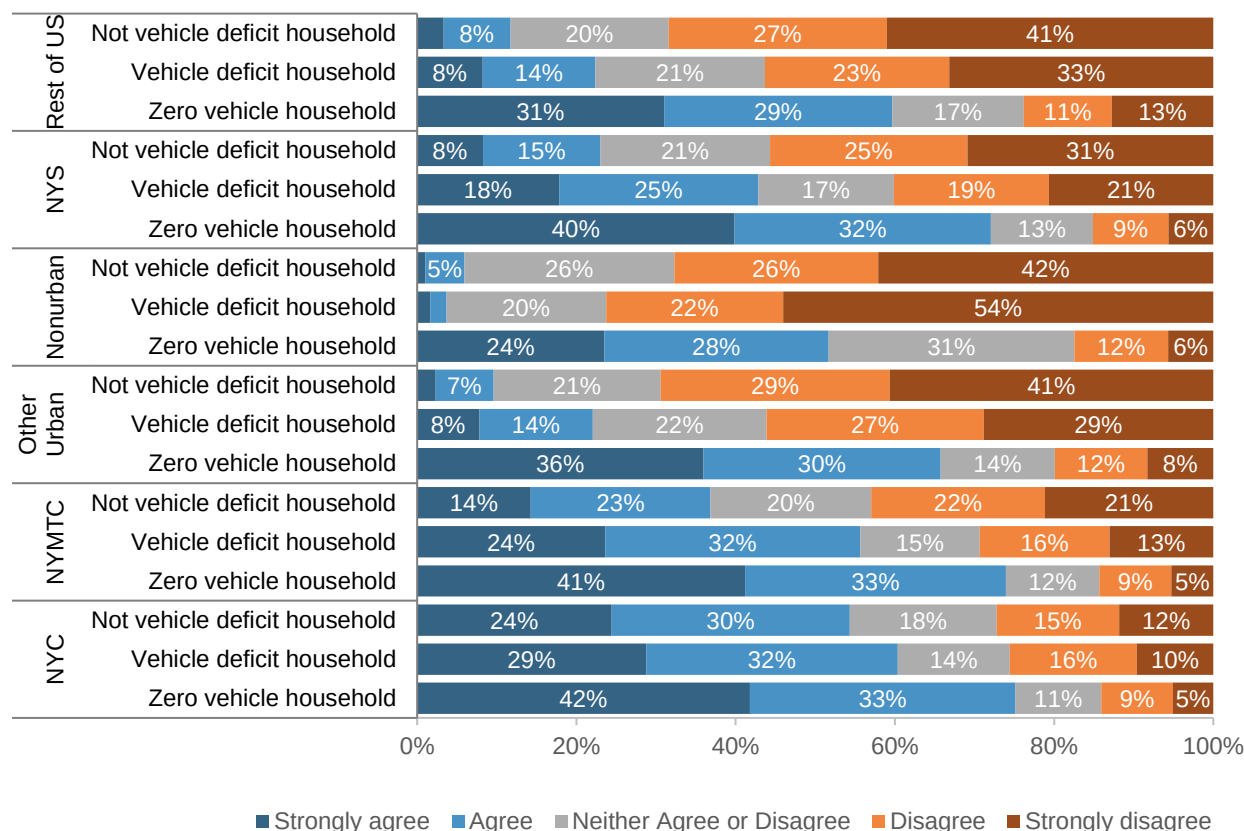


Figure 5-9. Share of ratings on “I use public transportation to save money” by household vehicle ownership type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 5%.

For the two vehicle deficit types (Figure 5-10), a clear pattern emerges to show public transit used more widely as a cost-saving measure in urban areas, especially in NYC and NYMTC. In one vehicle multiple driver households, 64% of respondents either strongly agreed or agreed, and 58% of similar households expressed agreement in NYMTC. However, agreement drops significantly among those in vehicle deficit households, especially in nonurban area and the rest of the US, where access to reliable public transit is likely more limited. A higher disagreement with the statement can be observed in nonurban areas, particularly among vehicle deficit households. In nonurban vehicle deficit households, 58% strongly disagreed and another 21% disagreed, indicating that public transportation is not considered a viable cost-saving alternative in these settings. Similarly, disagreement levels remained high in the rest of the US, with 40% strongly disagreeing in vehicle deficit households. This suggests that the presence of multiple drivers with insufficient vehicle access does not necessarily lead to increased public transit use.

Urban respondents show a more balanced distribution, with significant portions selecting neutral responses or expressing agreement, particularly in one vehicle multiple driver households. However, even

in urban areas, vehicle deficit households show lower agreement rates. This could indicate dissatisfaction with transit service quality or the inconvenience of relying on public transportation when household vehicle constraints exist.

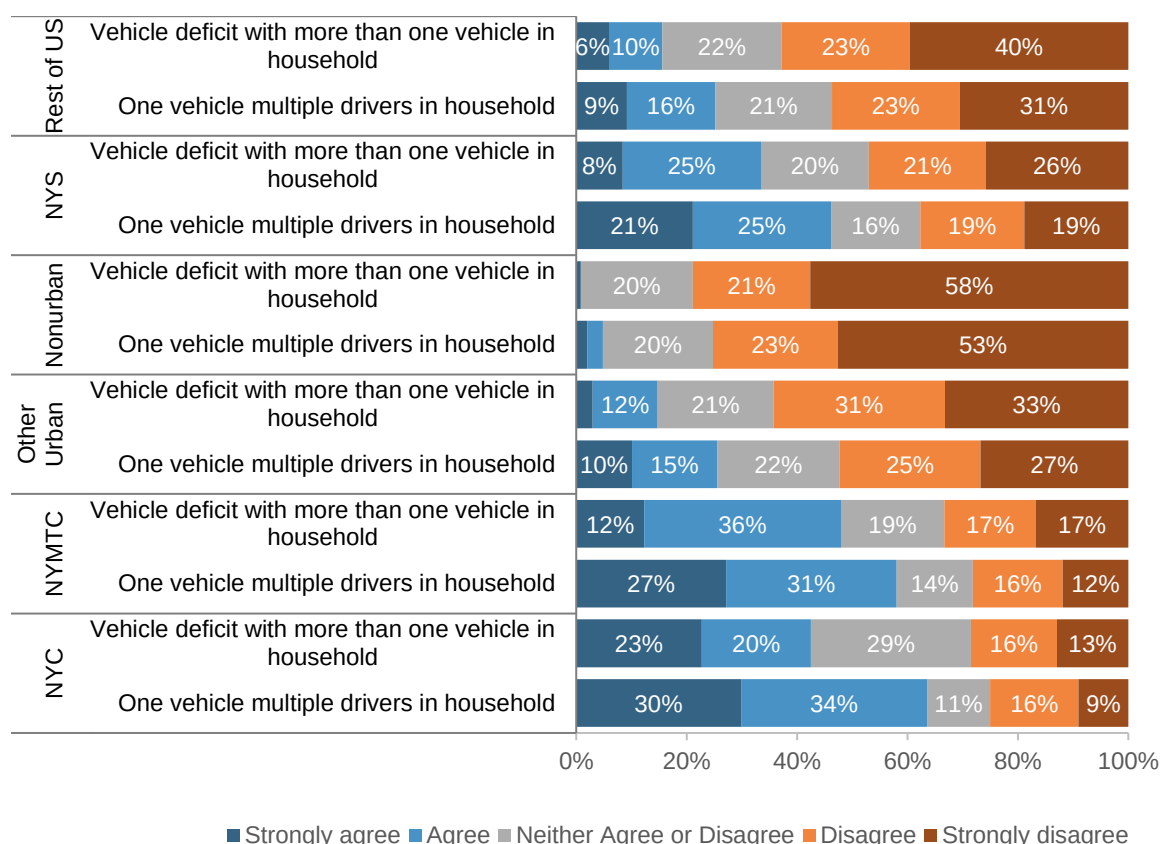


Figure 5-10. Share of ratings on “I use public transportation to save money” by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 6%.

5.6 OVERALL COST CONCERNS

A key finding is that concerns about the cost of travel across different household vehicle ownership categories are more prevalent in zero vehicle households across all regions (Figure 5-11). In NYC, 58% of zero vehicle households reported that getting from place to place costs too much, and similar percentages are seen in NYMTC (59%) and other urban areas (61%). Notably, this concern is slightly lower (54%) in nonurban areas. Among vehicle deficit households, this concern is still substantial but somewhat lower than it is in zero vehicle households. Public transportation as a cost-saving measure is particularly relevant in urban areas, with the highest percentages in NYC (75%) and NYMTC (74%) among zero vehicle households. In contrast, public transit use for saving money is much lower in nonurban vehicle deficit (4%) and not vehicle deficit households (6%), highlighting the lack of transit accessibility in rural areas.

Walking and biking to save money vary by location and household type. In NYC and NYMTC, about 65% of zero vehicle households walk to save money compared with 71% in other urban areas and 60% in nonurban areas. The reliance on biking for financial reasons is relatively low across all regions but is highest in other urban zero vehicle households (25%), suggesting that biking might be a more accessible

or necessary alternative in smaller urban areas. Gas price sensitivity shows a stark urban–rural divide. In NYC and NYMTC, only 18% of zero vehicle households show gas prices affecting the number of places they go because they are less dependent on cars. However, 69% of nonurban vehicle deficit and 59% of not vehicle deficit households reported that gas prices influence their travel, demonstrating a greater reliance on private vehicles in rural areas.

The comparison between NYS and the rest of the US highlights that zero vehicle households in NYS are more likely to use public transportation to save money (72%) compared with 60% in the rest of the US, reflecting the greater availability of transit options in NYS, particularly in urban areas (Figure 5-12). Similarly, walking to save money is more common in NYS (65%) than in the rest of the nation (55%), indicating that walking is a more viable alternative to driving in NYS. For vehicle deficit households, a significant difference is seen in public transportation use, with 43% of NYS respondents relying on it to save money, compared with only 22% in the rest of the US. This suggests that in NYS, even those with limited vehicle access have stronger transit alternatives, whereas in the rest of the country, such households may struggle with fewer options. The price of gas affecting travel decisions is a much greater concern for vehicle deficit households in the rest of the US (55%) than for those in NYS (47%), which could indicate that households outside NYS are more car-dependent, making them more vulnerable to fuel price fluctuations. The percentage of not vehicle deficit households reporting that travel costs too much is slightly lower in the rest of the US (40%) compared with the percentage in NYS (46%), which may be due to higher transportation costs in NYS, especially in urban areas. Additionally, public transportation use for cost savings in this group is much lower nationwide (12%) than it is in NYS (23%), further supporting the idea that public transit is a more practical and utilized option in NYS, even for those who own enough vehicles.

Within each region, some differences arise between households with one vehicle and multiple drivers versus those experiencing a vehicle deficit with more drivers than available vehicles (Figure 5-13). In NYC, individuals in one vehicle multiple driver households expressed greater financial concern regarding transportation costs, with 52% indicating they walk to save money and 64% relying on public transit, compared with only 28% and 43%, respectively, in vehicle deficit households. This suggests that even with constrained vehicle access, those with at least one shared vehicle may have more flexibility in their transportation choices. However, sensitivity to gas prices is similar between both groups, at 41% and 44%, indicating that fuel costs remain a concern across vehicle-limited households. A similar pattern is observed in NYMTC, where those in one vehicle multiple driver households are more likely than those in vehicle deficit households to use transit (58% vs. 48%) and walk (48% vs. 20%) to save money. Notably, concern about transportation costs remains the same at 51% for both groups, yet biking to save money is significantly lower among vehicle deficit households (6% vs. 21%), suggesting that cycling is not a widely used alternative when vehicle access is severely constrained. In other urban areas, the differences between the two household types persist, though they are less pronounced. Individuals in one vehicle multiple driver households also reported higher reliance on transit (26% vs. 15%) and walking (32% vs. 22%) for cost savings.

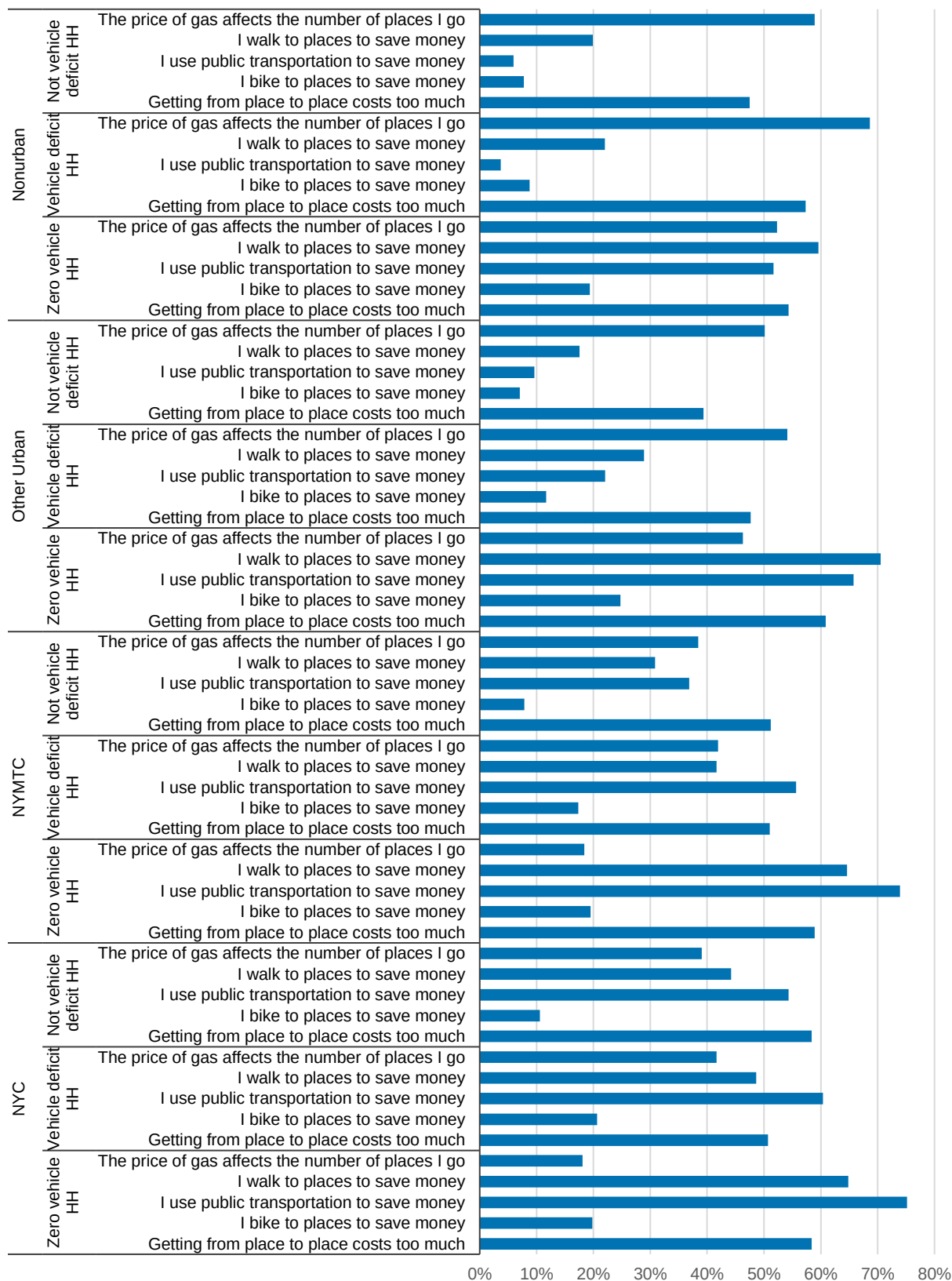


Figure 5-11. Comparison of travel concerns within each region in NYS by vehicle ownership type.

Note: HH = household.

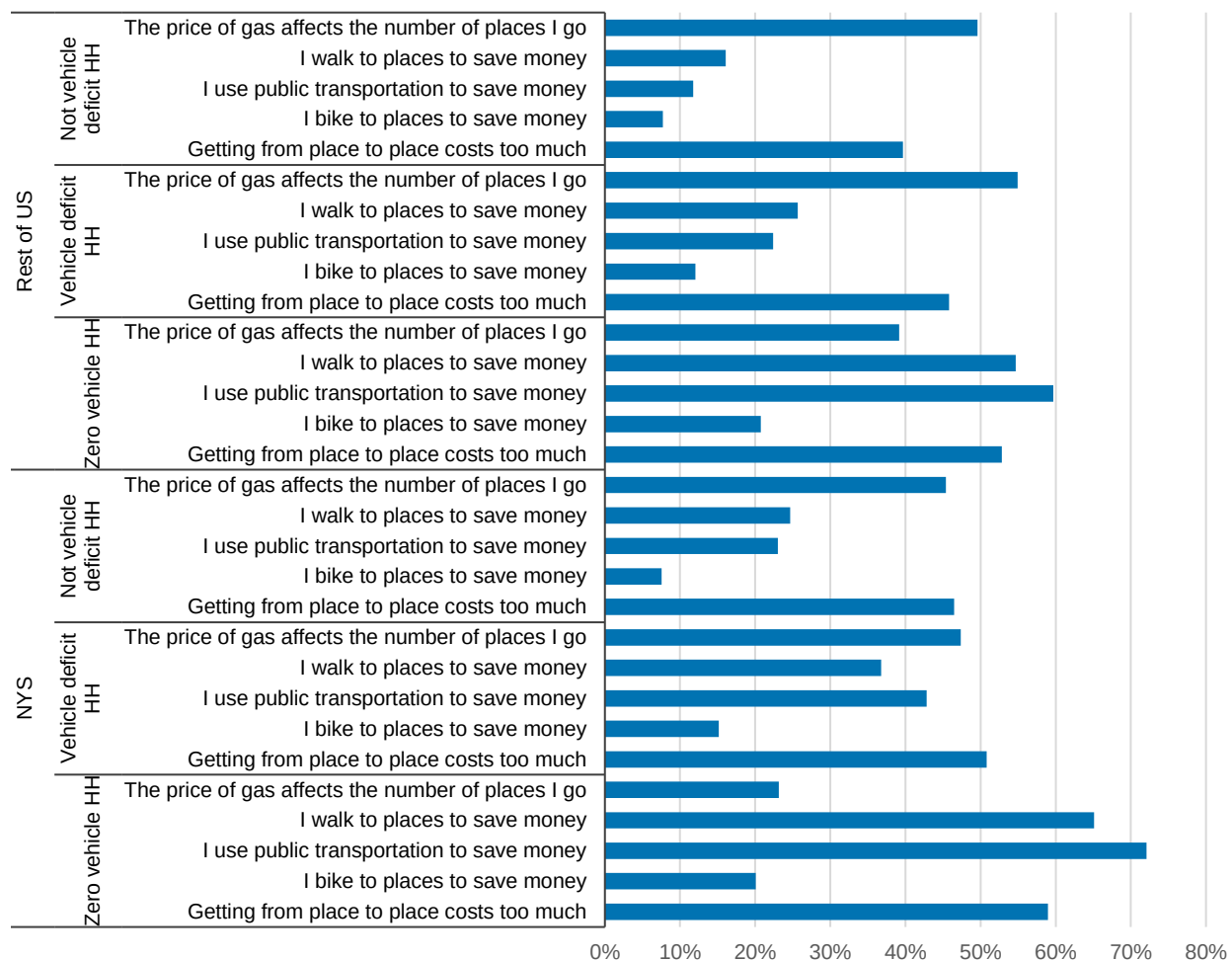


Figure 5-12. Comparison of travel concerns in NYS vs. the rest of the US by vehicle ownership type.
 Note: HH = household.

Gas prices appear to have a greater impact on travel behavior in other urban region compared with more transit-rich areas, with 58% of one vehicle multiple driver households and 47% of vehicle deficit households indicating that fuel costs influence their travel frequency. In nonurban areas—where vehicle deficit households face the most severe transportation constraints—public transit use as a cost-saving measure is nearly nonexistent, with only 1% of vehicle deficit households and 5% of one vehicle multiple driver households using it to reduce expenses. Walking to save money is also significantly lower for vehicle deficit households than for one-vehicle-multiple-driver households (9% vs. 27%), highlighting the challenges of relying on alternative modes in rural areas. Gas prices affect travel decisions for nearly 7 in 10 respondents in both groups, at 69% and 68%, indicating that fuel cost is a dominant financial concern regardless of household vehicle access.

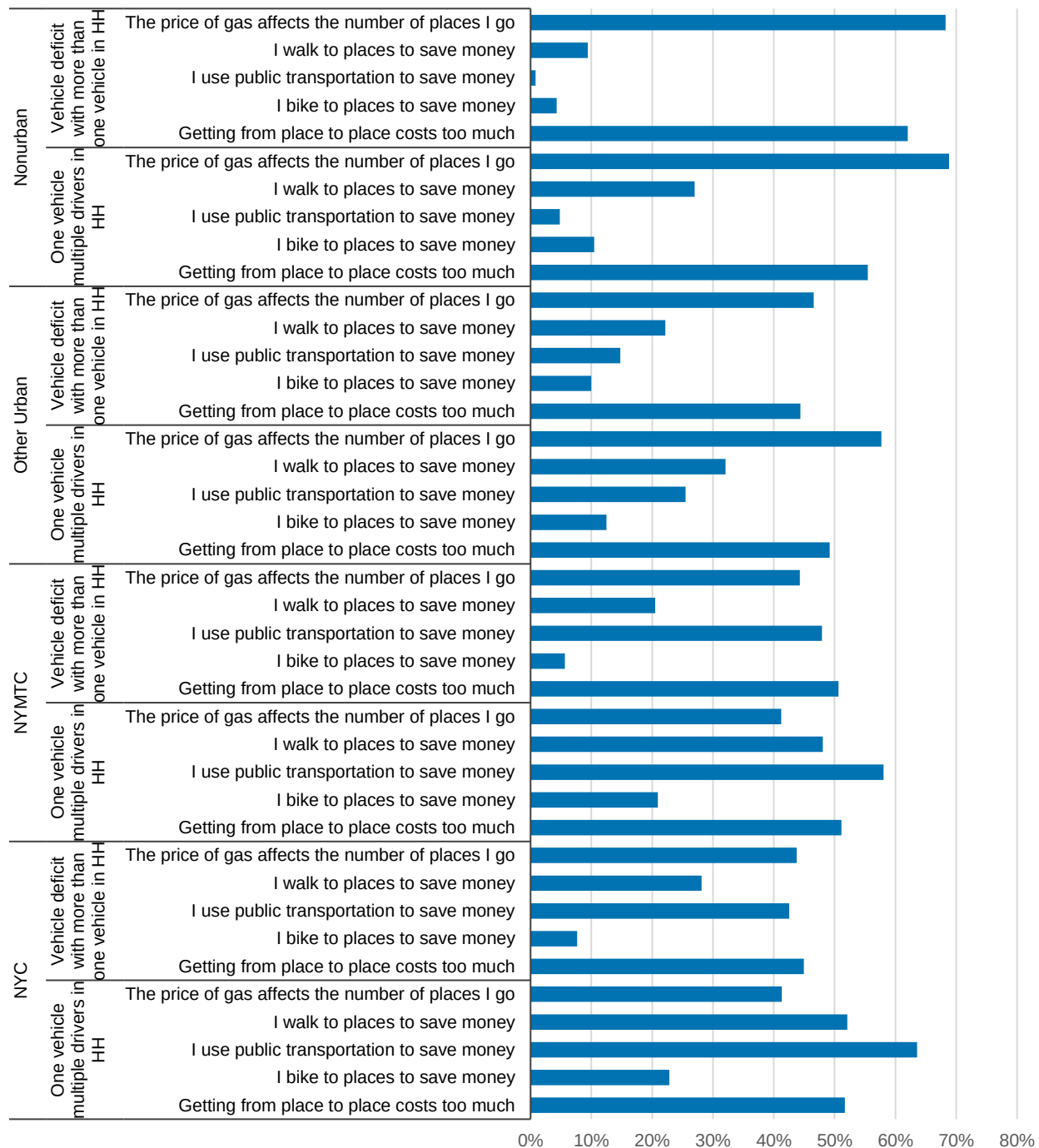


Figure 5-13. Comparison of travel concerns within each region by vehicle deficit household type.

Note: HH = household.

The comparison between NYS and the rest of the US for households with one vehicle and multiple drivers or a vehicle deficit with more than one vehicle highlights key differences in travel cost concerns and cost-saving behaviors (Figure 5-14). For households with one vehicle and multiple drivers, NYS residents expressed greater reliance on public transportation (46%) and walking (43%) to save money, compared with 25% and 28%, respectively, in the rest of the US. This suggests that NYS offers better public transit accessibility, making it a viable cost-saving option even for households with limited vehicle

access. Meanwhile, gas price sensitivity is slightly higher in the rest of the US (54%) than in NYS (47%), indicating that households outside NYS are more affected by fuel costs, likely because of greater car dependency and fewer alternative transportation options. For households with a vehicle deficit and more than one vehicle, reliance on public transit and walking to save money is again higher in NYS (33% and 20%, respectively) than in the rest of the US (16% and 19%, respectively). However, gas price sensitivity is significantly greater in the rest of the US (57%) compared with NYS (47%), reinforcing the idea that fuel costs have a bigger impact on travel decisions outside NYS owing to fewer noncar transportation choices.

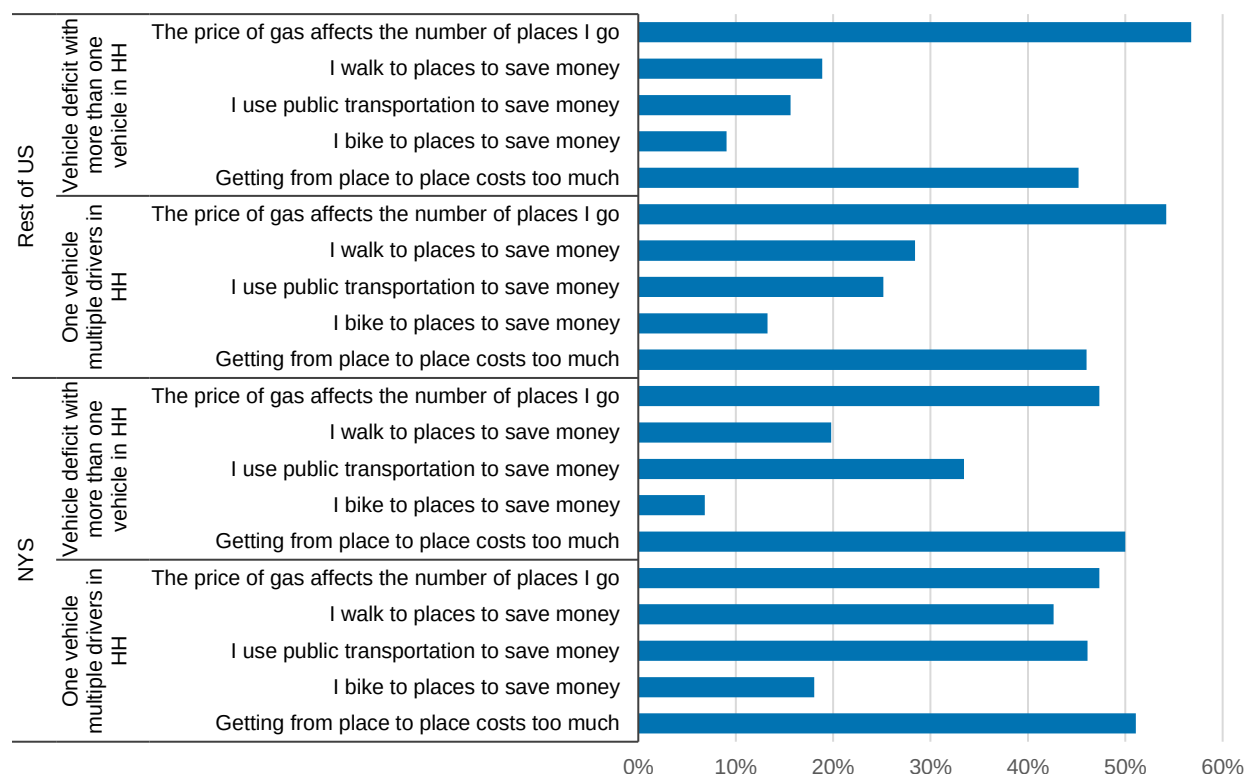


Figure 5-14. Comparison of travel concerns in NYS vs. the rest of the US by vehicle deficit household type.
Note: HH = household.

6. BEHAVIOR DURING COVID-19

The COVID-19 pandemic triggered unprecedented changes in daily life, dramatically altering the way people work, shop, and move around. Understanding how households with no personal vehicles adapted during this period is particularly vital. Using 2022 NHTS data, this section investigates the effect of COVID-19 on travel behaviors across different vehicle-ownership groups, with a focus on online purchasing trends, public transit reliance, and shifts in commuting patterns to in-person workplaces. By analyzing these behaviors, researchers can gauge the degree to which the pandemic reshaped zero vehicle households' mobility choices and whether these changes were temporary adjustments or indicators of more permanent transformations.

6.1 ONLINE SHOPPING

Starting with the question of online shopping behavior during the pandemic, Figure 6-1 illustrates that zero vehicle households in the Mid-Atlantic division reported a relatively high degree of increased online purchasing compared with their pre-COVID-19 habits: 34% indicated they shop online more often now. This figure is less than the online purchasing figures for vehicle deficit households (47%) and not-deficit households (39%). Notably, 19% of zero vehicle households are in the “unknown” category, which may indicate a degree of uncertainty in the distribution. In the rest of the US, zero vehicle households lagged behind vehicle deficit households in the share reporting increased online shopping (31% vs. 40%), though they remained comparable to not-deficit households (39%). Interestingly, 15% of zero vehicle households reported they shop online less often than before, a rate that’s significantly higher than the 4% reported among car-owning groups, and another 16% fell under “unknown.” These numbers hint that many zero-car households have embraced e-commerce more since COVID-19.

Closer examination of vehicle deficit households reveals an interesting distinction between those with just one vehicle versus those that own more than one vehicle but not enough to match every driver’s needs (Table 6-1). In the Mid-Atlantic, multivehicle deficit households reported the highest rate of increased online ordering (47%), surpassing single-vehicle deficit households (34%). The multivehicle group also has a very low “less often” share (2%), indicating that once these households embraced home delivery, they tended to maintain or increase usage. Meanwhile, 19% of single-vehicle deficit households did not specify how their habits changed. In the rest of the US, multivehicle deficit households again lead with 40% reporting they now order online more often, compared with 31% among single-vehicle deficit households. The shares in both deficit categories are lower than those in the Mid-Atlantic division. Overall, these findings underscore that having multiple cars—even if not enough for every driver—often coincides with more online purchases for home delivery, possibly reflecting increased discretionary spending.

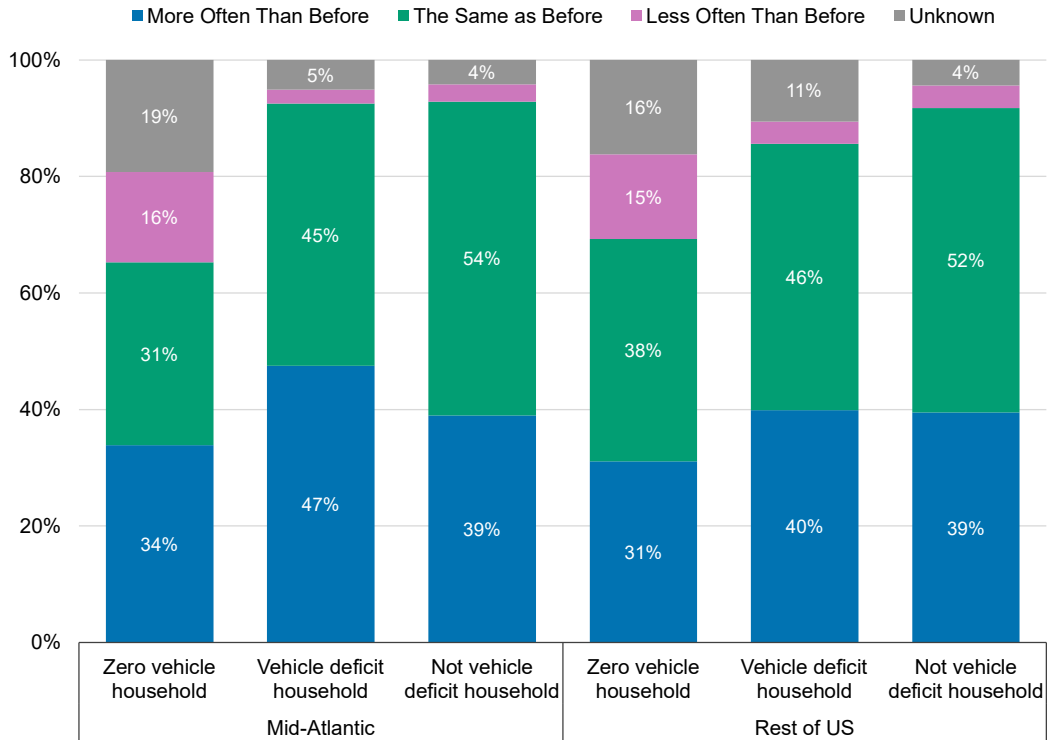


Figure 6-1. COVID-19 effects on online purchases for home delivery by household vehicle ownership type.
 Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.
 Unlabeled segments represent values that are 3% or less.

Table 6-1. COVID-19 effects on online purchases for home delivery by vehicle deficit household type.

Region	Vehicle deficit household type	More often than before	The same as before	Less often than before	Unknown
Mid-Atlantic	One vehicle multiple drivers in household	34%	31%	16%	19%
	Vehicle deficit with more than one vehicle in household	47%	45%	2%	5%
Rest of US	One vehicle multiple drivers in household	31%	38%	15%	16%
	Vehicle deficit with more than one vehicle in household	40%	46%	4%	11%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Turning to whether these pandemic-era online shopping habits are considered permanent or temporary, a clear divergence emerges between zero vehicle households and their car-owning counterparts, as shown in Figure 6-2. In the Mid-Atlantic, zero vehicle households split evenly between those who viewed their extra online buying as temporary (31%) and permanent (31%), but a large segment (38%) was unsure. By contrast, nearly half of vehicle deficit households (49%) believed their changes are permanent, and a substantial 67% of not-deficit households said the same. Meanwhile, in the rest of the US, 45% of zero vehicle households considered their increased online shopping permanent, which was more than those in the Mid-Atlantic division.

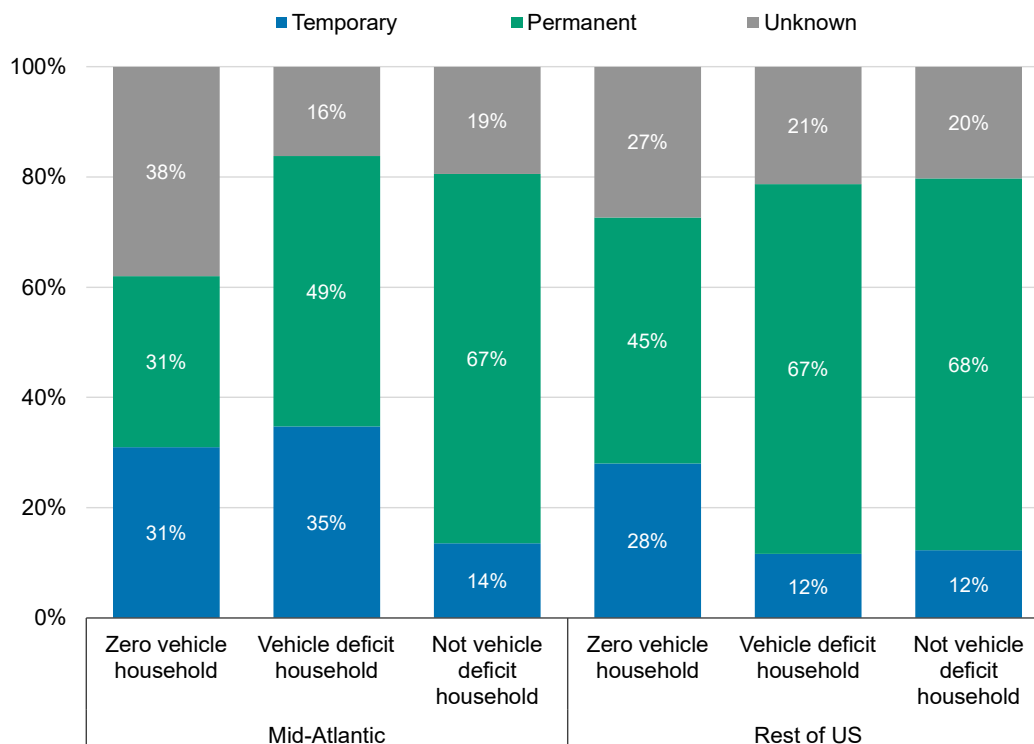


Figure 6-2. Permanent or temporary COVID-19 effects on online purchases for home delivery by household vehicle ownership type.

6.2 PUBLIC TRANSIT USE

Shifting from online shopping to another critical component of mobility, Figure 6-3 shows that many travelers scaled back their public transit use during COVID-19, but not to the same extent across all household types. In the Mid-Atlantic, zero vehicle households remained more reliant on transit even in a pandemic scenario, yet a full 50% said they use transit less often now compared with before. That is notably higher than the 41%–44% among vehicle-owning groups. At the same time, 12% of zero-car households ride transit more frequently. Outside the Mid-Atlantic, the pattern is similar. Among zero vehicle households in the rest of the US, 43% reported using public transit less often, and 5% ride it more. By contrast, not-deficit households had the highest share (56%) who reported their transit usage remained the same, and only about a third (32%) have cut back. Overall, zero-car households appeared more sensitive to pandemic disruptions in their transit habits, whether that meant cutting back significantly or, in some cases, riding more because they had few alternatives.

Looking at transit use among vehicle deficit households, 9% of single-vehicle deficit households in the Mid-Atlantic reported using transit more than before COVID-19 and 46% reported using it less, as illustrated in Figure 6-4. Multivehicle deficit households have a similar share, with 8% using it more often but a somewhat smaller segment reducing their usage (37%).

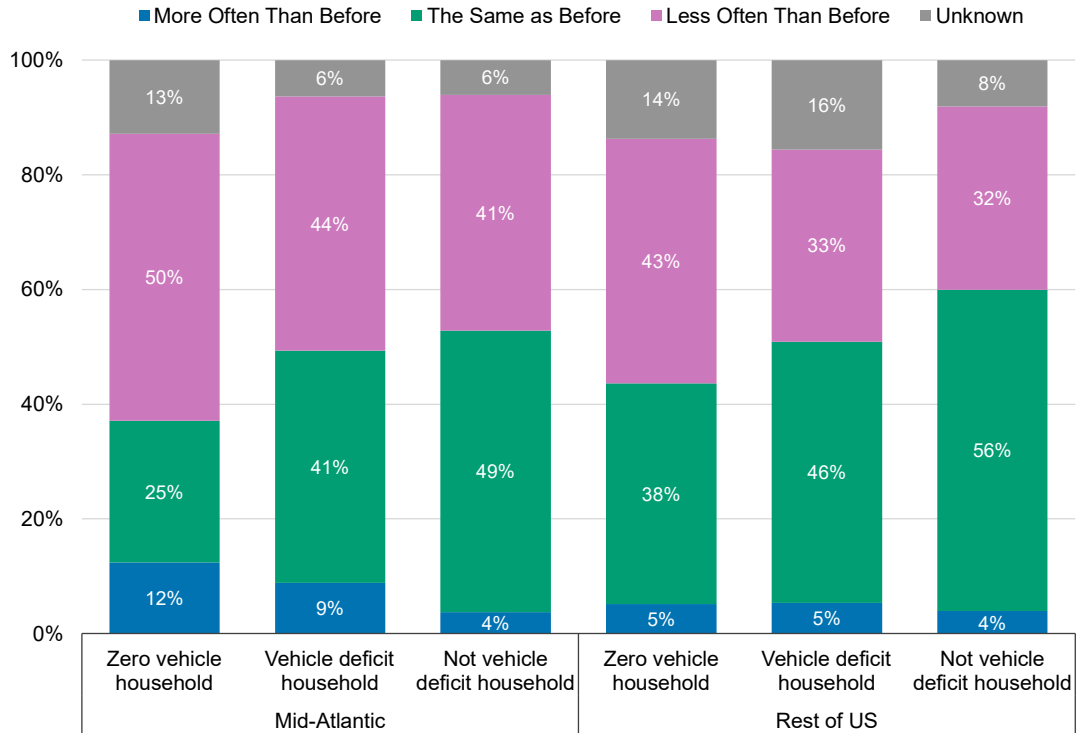


Figure 6-3. Public transit use during COVID-19 by household vehicle ownership type.

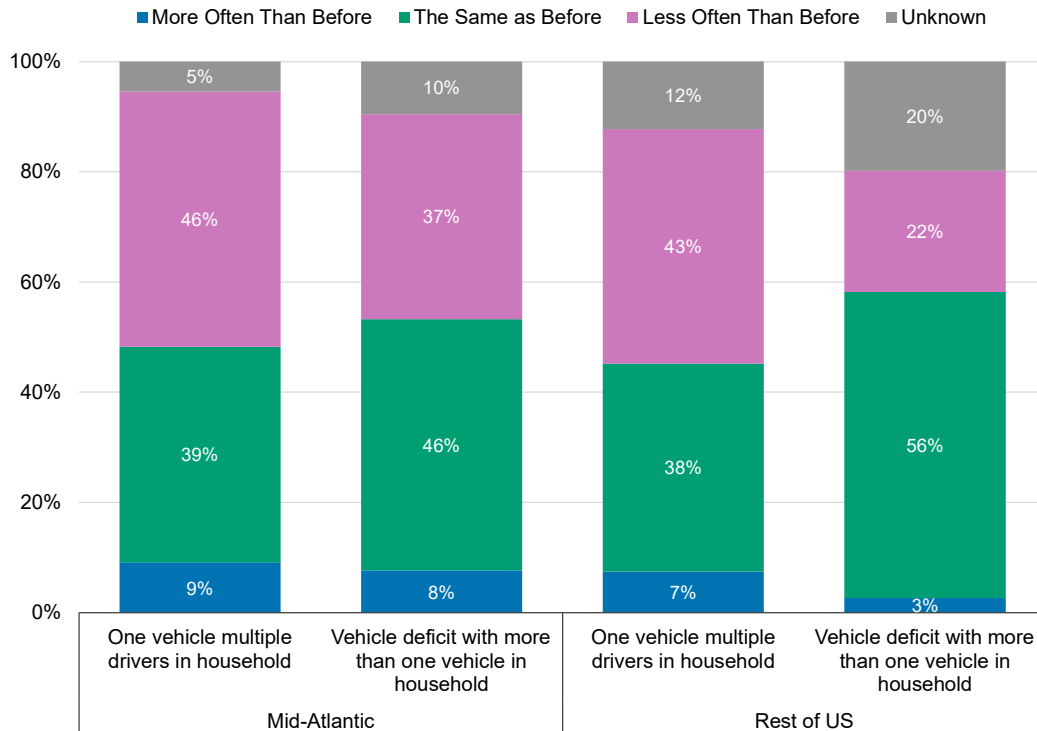


Figure 6-4. Public transit use during COVID-19 by vehicle deficit household type. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Turning to how enduring these pandemic-related transit changes might be, a notably high share of zero vehicle households view their shift as temporary—47% in the Mid-Atlantic and 51% in the rest of the US (Table 6-2). This is higher than any other group, indicating that many car-free riders expected to revert to old transit patterns once concerns about COVID-19 diminished. Still, about a quarter of zero vehicle travelers considered their new behavior permanent, suggesting some level of sustained change even within this transit-dependent group. Meanwhile, vehicle deficit and not-deficit households are more mixed in their perceptions. In the Mid-Atlantic, for instance, 41% of deficit households called their changes temporary, but 31% saw them as permanent, which slightly exceeds the 26% among zero vehicle riders who expected the effects to be permanent.

Table 6-2. Permanent or temporary COVID-19 effects on the use of public transit by household vehicle ownership type.

Region	Vehicle ownership type	Temporary	Permanent	Unknown
Mid-Atlantic	Zero vehicle household	47%	26%	26%
	Vehicle deficit household	41%	31%	28%
	Not vehicle deficit household	34%	30%	36%
Rest of US	Zero vehicle household	51%	23%	26%
	Vehicle deficit household	34%	39%	26%
	Not vehicle deficit household	34%	34%	32%

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

6.3 COMMUTE BEHAVIOR

Figure 6-5 displays changes in how often people commuted to a physical workplace during COVID-19. Zero vehicle households show more volatility than those with sufficient car access, especially in the Mid-Atlantic region, where 13% of zero-car respondents said they go to the workplace more often now than before the pandemic, and 35% reported commuting less frequently. That spread is notably wider than that among not-deficit households, in which only 3% commute more often and 31% less often, but a solid majority (64%) have kept the same schedule as before. In the rest of the US, zero vehicle households still stand out with 7% increasing their physical commute frequency and 27% decreasing it, which is similar for not-deficit households. Meanwhile, vehicle deficit groups in both regions settle in between these two extremes: 15% in the Mid-Atlantic and 10% in the rest of the country said they commute more often, whereas roughly a third reported commuting less.

Examining vehicle deficit households exclusively reveals differing perspectives on whether changes to in-person work are temporary or permanent (Table 6-3). In the Mid-Atlantic, one fifth of single-vehicle deficit households said the shift is temporary, 48% deemed it as permanent, and nearly a third were unsure. Multivehicle deficit households showed a lower temporary rate (4%) but a very high “unknown” share (50%), suggesting that many remained uncertain about how their return-to-work patterns would evolve.

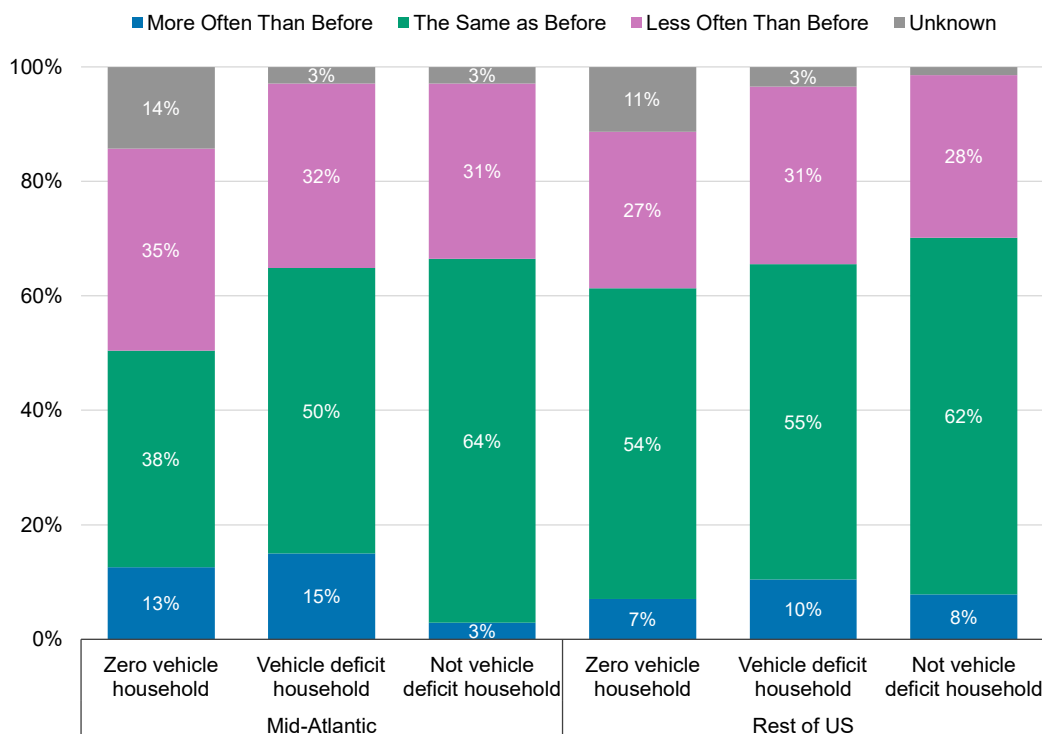


Figure 6-5. COVID-19 effects on travel to a physical work location by household vehicle ownership type.

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

Moving to the rest of the US, one vehicle deficit households appear more confident that the change is here to stay, with over half (54%) labeling their new schedule permanent. For multivehicle deficit households, that share rises to 57%—the highest of any group—and only 17% remain undecided. Overall, households having more than one car but still in deficit seemed more inclined to embrace their altered commute routines as “the new normal,” especially outside the Mid-Atlantic, whereas single vehicle deficit households in the region were somewhat more split but still leaned strongly toward a lasting shift.

Table 6-3. Permanent or temporary COVID-19 effects on travel to a physical work location by vehicle deficit household type.

Region	Vehicle deficit household type	Temporary	Permanent	Unknown
Mid-Atlantic	One vehicle multiple drivers in household	20%	48%	32%
	Vehicle deficit with more than one vehicle in household	4%	46%	50%
Rest of US	One vehicle multiple drivers in household	16%	54%	30%
	Vehicle deficit with more than one vehicle in household	26%	57%	17%

7. CONCLUSIONS

The motivation for this study stemmed from the need to understand the travel behavior and mobility constraints of zero vehicle and vehicle deficit households—groups that face unique challenges in transportation access. With an increasing share of households falling into these categories across urban and suburban areas, their experiences provide valuable insights into mobility, transportation infrastructure, and policy needs. Using data from 2009, 2017, and 2022 NHTS, this study examined key aspects of travel behavior, including trip rates, distances, mode choices, and trip purposes, as well as the role of emerging transportation technologies and the effects of the COVID-19 pandemic. The analysis primarily focused on NYS, a region with a diverse mix of urban, suburban, and rural environments, enabling a nuanced understanding of how limited vehicle access influences daily travel decisions across different geographic contexts.

The findings reveal that the distribution of zero vehicle and vehicle deficit households varies significantly across NYS. Nearly two-thirds of Manhattan’s population (64%–67%) ages five and older belonged to zero vehicle households over the years; this is the largest share of any region. The share of vehicle deficit households declined in other parts of NYC in 2017 compared with 2009. Suburban regions like Nassau, Suffolk, Putnam, Rockland, and Westchester counties tended to have a lower presence of vehicle deficit households. In Manhattan, nearly 100% of vehicle deficit households typically shared just one vehicle among multiple drivers. The socioeconomic profiles of these households also varied widely; urban zero vehicle households exhibited a diverse range of incomes, whereas those in suburban and rural areas were predominantly lower income. This suggests that although urban households may choose to forgo vehicle ownership owing to transit accessibility, those in nonurban areas often face vehicle constraints due to financial limitations.

As expected, zero vehicle households make fewer daily trips than vehicle-owning households make. Their trip-making patterns have remained relatively stable over time, whereas vehicle deficit and not-deficit households have shown declining trip rates since 2009, yet they have significantly higher trip rates than those of zero vehicle households. In 2017, zero vehicle households in Manhattan made significantly more daily trips (3.7) than their counterparts in other NYS regions made, reflecting the accessibility of transit and walkable infrastructure. Across NYS, trip distances for zero vehicle households were significantly shorter than those of vehicle-owning households, with public transit and walking serving as their dominant travel modes—especially in NYC. In suburban and rural areas, vehicle deficit households remained heavily reliant on private vehicles. Notably, zero vehicle households in dense urban environments allocated a higher share of their trips to work and social or recreational activities. In 2017, vehicle deficit households in NYS traveled significantly fewer miles (17 VMT) compared with not-deficit households (26 VMT), underscoring the mobility constraints associated with limited vehicle access.

An exploration of transportation-related technologies revealed that zero vehicle households, particularly in NYC, engage in online shopping at higher rates than those in nonurban regions. However, many zero vehicle households still do not shop online, suggesting that e-commerce is not a universal substitute for in-person travel. Adoption of ride-, car-, and bike-sharing remains limited among zero vehicle households outside major cities, with NYC standing out as the primary location where these services provide meaningful mobility alternatives. Public transit remains a crucial mode of travel for zero vehicle households, especially for cost-saving reasons, with 75% in NYC using it to reduce transportation expenses. However, in rural areas, the lack of public transit options limits their role in addressing affordability concerns.

The COVID-19 pandemic significantly influenced the travel behavior of zero vehicle and vehicle deficit households, particularly in terms of transit use and work commutes. Although online shopping increased for all household types, vehicle deficit households reported the highest rates of increased e-commerce

reliance. Public transit use declined across all groups, with half of zero vehicle households in the Mid-Atlantic division reducing their transit trips, though they remained more reliant on transit than vehicle-owning households. Additionally, zero vehicle households showed more volatility in their commute patterns, with some increasing their in-person work trips and others reducing them. This suggests that car-free households experienced greater disruptions in their work-related travel routines compared with those who had stable vehicle access.

Future research should further investigate the long-term effects of changing mobility patterns among zero vehicle and vehicle deficit households. Specifically, analyzing how transit service adjustments, remote work policies, and e-commerce trends continue to shape travel demand for these groups could provide valuable insights for planners and policymakers. Additionally, more granular studies at the neighborhood level could help identify localized transportation gaps and opportunities for targeted investments in public transit, active transportation infrastructure, and shared mobility services. Finally, exploring the intersection of vehicle ownership constraints with broader issues such as housing affordability, land use patterns, and access to essential services could provide a more holistic understanding of mobility and inform policies that support a more resilient transportation system.

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APPENDIX A. GLOSSARY OF NHTS TERMS

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This glossary provides the most commonly used terms in the National Household Travel Survey (NHTS) and the terms used in this report, along with definitions. These definitions are provided to assist the user in interpreting the NHTS data.

Adult	For NHTS, this is defined as a person 18 years or older.
Block Group	A subdivision of a census tract that contains 600–3,000 people. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Census Tract	A small subdivision of a county, generally having a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people. The geographic size of the tract may vary considerably, depending on population density. Tracts were designed to be homogeneous regarding to population characteristics, economic status, and living conditions when they were first delineated. Because the first tracts were delineated for the 1890 Census, current tracts may be far from homogeneous. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Destination	For travel day trips, the destination is the point at which there is a break in travel, except if the break is only to change vehicles or means of transport.
Driver	A driver is a person who operates a motorized vehicle. If more than one person drives the vehicle on a single trip, the person who drives the most miles is classified as the principal driver.
Elderly	A person age 65 years or older.
Elderly Household	Households with one or more persons age 65 years or older.
Employed	A person is considered employed if that person worked for pay, either full time or part time, during the week before the interview. This includes persons who work at home and/or have more than one job.
Household	A group of persons whose usual place of residence is a specific housing unit; these persons may or may not be related to each other. The total of all US households represents the total civilian noninstitutionalized population. A household does not include group quarters (i.e., 10 or more persons living together, none of whom are related).
Household Income	Household income is the money earned by all family members in a household, including those temporarily absent. Annual income consisted of the income earned 12 months preceding the interview. Household income includes monies from all sources, such as wages and salary, commissions, tips, cash bonuses, income from a business or farm, pensions, dividends, interest, unemployment or workers' compensation, social security, veterans' payments, rent received from owned property (minus the operating costs), public assistance payments, regular gifts of money from friends or relatives not living in the household, alimony, child support, and other kinds of periodic money income other than earnings. Household income

	excludes in-kind income such as room and board, insurance payments, lump-sum inheritances, occasional gifts of money from persons not living in the same household, withdrawal of savings from banks, tax refunds, and the proceeds of the sale of one's house, car, or other personal property.
Household Members	Household members include all people, whether present or temporarily absent, whose usual place of residence is in the sample unit. Household members also include people staying in the sample unit who have no other usual place of residence elsewhere.
Household Vehicle	A household vehicle is a motorized vehicle that is owned, leased, rented, or company owned and available to be used regularly by household members during the 2 week travel period. Household vehicles include vehicles used solely for business purposes or business-owned vehicles if they are driven home and can be used for the home-to-work trip (e.g., taxicabs and police cars). Household vehicles include all vehicles that were owned or available for use by members of the household during the travel period, even though a vehicle may have been sold before the interview. Vehicles excluded from household vehicles are (1) those that were not working and were not expected to be working within 60 days and (2) those that were purchased or received after the designated travel day.
Journey-to-Work Trips (Commute Trips)	Includes travel to and from a place where one reports for work. Does not include any other work-related travel. Does not include trips for persons who work at home.
Means of Transportation	A mode of travel used for going from one place (origin) to another (destination). A means of transportation includes private and public transit modes, as well as walking. The following transportation modes, grouped by major mode, are included in the NHTS data. Active modes—include modes by which a person must actively move from one place to the next. • Walk: This category includes walking and jogging. • Bicycle: This category includes bicycles of all speeds and sizes that do not have a motor. Private vehicle—a stipulation for being a private vehicle is that the vehicle is privately operated, including rental cars. • Car: Includes cars and station wagons. Leased and rented cars are included if privately operated and not used for picking up passengers in return for fare. • Sport utility vehicle: Includes vehicles that are a hybrid of design elements from a van, pickup truck, and station wagon. Examples include a Ford Explorer, Jeep Cherokee, or Nissan Pathfinder. • Van: Includes vans or minivans designed to carry 5–13 passengers or to haul cargo. • Pickup truck: Includes vehicles with an enclosed cab that usually accommodates two to three passengers and has an open cargo area in the rear. Late-model pickups often have a back seat that allows for total seating of four to six passengers. Pickup trucks usually have a wheelbase the same size as that of a full-size station wagon. This category also includes pickups with campers.

- Motorcycle/moped: This category includes large, medium, and small motorcycles and mopeds.
- RV (motor home, ATV, snowmobile): An RV or motor home includes a self-powered recreational vehicle that is operated as a unit without being towed by another vehicle (e.g., a Winnebago motor home). This category also includes ATVs and snowmobiles.

Rental car—Includes Zipcar and Car2Go, in addition to commercially rented cars for private use.

- Golf cart/Segway: This includes all electric or gas-operated vehicles designed for use on a golf course, but whose use has recently extended to within smaller, often gated, communities. NOTE: Travel taken via golf cart is NOT included in vehicle trips and vehicle miles of travel.

Public transportation—as used in Federal Highway Administration publications and analysis of NHTS data, typically includes the following (indicated in *italics* in the following text): public or commuter bus, commuter rail, and subway/elevated rail/light rail/streetcar.

Bus—This category includes the following:

- *Public or commuter buses*: local public transit buses available to the public and buses used for short-distance public transport purposes (e.g., city bus or public bus), school buses
- Private/charter/tour/shuttle buses: private buses operating on a fixed schedule between population centers and buses that shuttle passengers from one fixed place to another (e.g., airport shuttles)
- City-to-city buses: buses that run from one urban center to another (e.g., Greyhound)

Train—This category includes the following:

- *Amtrak/commuter rail*: trains that run from one urban center to another
- *Subway/elevated rail/light rail/streetcar* (also known as rail rapid transit): a high-capacity system operated on a fixed-rail or guideway system on a private right of way, and vehicles that run on a fixed-rail system powered by electricity obtained from an overhead power distribution system

Other modes—This category includes the following:

- School buses
- Paratransit/dial-a-ride
- Taxi/limo: use of a taxicab by a passenger for fare, including limousines. In 2017, this category also includes ride-sharing such as Uber and Lyft.
- Airplane: includes commercial airplanes and smaller planes available for use by the public in exchange for a fare. Private and corporate planes and helicopters are also included.
- Boat/ferry/water taxi: includes travel by passenger line ferries
- Something else: includes any type of transportation not previously listed (e.g., skateboards, roller blades, sailboats, cruise ships)

**Metropolitan
Statistical Area
(MSA)**

Except in the New England states, a metropolitan statistical area is a county or group of contiguous counties that contains at least one city of 50,000 inhabitants or more, or “twin cities” with a combined population of at least 50,000. In addition, contiguous counties are included in an MSA if, according to certain criteria, they are socially and economically integrated with the central city. In the New England states, MSAs consist of towns and cities instead of counties. The source used for the 2017

	NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from the Census 2010 definition).
Motorized Vehicle	Motorized vehicles include all vehicles licensed for highway driving. Snowmobiles and minibikes are specifically excluded.
New York City (NYC)	New York City is defined in this report as the five-county area: Bronx, Kings (Brooklyn), Queens, New York (Manhattan), and Richmond (Staten Island).
Occupancy	Occupancy is the number of persons, including driver and passenger(s), in a vehicle.
Occupancy Rate	NHTS occupancy rates are generally defined as the mileage-weighted averages of the number of persons on a vehicle trip.
Origin	Origin is the starting point of a trip.
Passenger	For a specific trip, a passenger is any occupant of a motorized vehicle, other than the driver.
Person Miles of Travel (PMT)	PMT is a primary measure of person travel. When one person travels 1 mile, 1 person mile of travel results. When two or more persons travel together in the same vehicle, each person makes the same number of person miles as the vehicle miles. Therefore, four persons traveling 5 miles in the same vehicle results in 20 person miles ($4 \times 5 = 20$).
Person Trip	A person trip is a trip by one or more persons in any mode of transportation. Each person is considered as making one person trip. For example, four persons traveling together in one auto are counted as four person trips.
Population Density	Category of population density (persons per square mile) in the Census Block Group of the household's home location: 0–99, 100–499, 500–999, 1,000–1,999, 2,000–3,999, 4,000–9,999, 10,000–24,999, and 25,000–999,999.
Privately Owned Vehicle (POV)	A privately owned or privately operated vehicle. In either case, the vehicle is not available to the public for a fee (i.e., it is not a bus, subway, taxi, or other vehicle for which a fee is charged for use).
Travel Day	A travel day is a 24 h period from 4:00 a.m. to 3:59 a.m. designated as the reference period for studying trips and travel by members of a sampled household.
Travel Day Trip	A travel day trip is defined as any time the respondent went from one location to another by private motor vehicle, public transportation, bicycle, walking, or other means during the NHTS-assigned reporting travel day. However, a separate trip is not counted in the following two instances: 1. When the sole purpose for the trip is to get to another vehicle or mode of transportation to continue to the destination. 2. Travel within a shopping center, mall, or shopping areas of four to five blocks is to be considered as travel to one destination.
Vehicle	In the 2017 NHTS, the term vehicle includes autos, passenger vans, sport utility vehicles, pickups and other light trucks, RVs, motorcycles, and mopeds owned or available to the household.

Vehicle Miles of Travel (VMT)	VMT is a unit to measure vehicle travel made by a private vehicle, such as an automobile, van, pickup truck, or motorcycle. Each mile traveled is counted as one vehicle mile regardless of the number of persons in the vehicle.
Vehicle Occupancy	Vehicle occupancy is the number of persons, including driver and passenger(s) in a vehicle; it also includes persons who did not complete a whole trip.
Vehicle Occupancy Rate	NHTS vehicle occupancy rates are generally calculated as person miles divided by vehicle miles.
Vehicle Trip	A trip by a single POV, regardless of the number of persons in the vehicle.
Work-Related Travel	Trips related to business activities except travel to the place of work; for example, a plumber drives to a wholesale dealer to purchase supplies for his business, or a company executive travels from his office to another building to attend a business meeting. Business, out-of-town trips, and professional conventions are also included.
Worker	See “Employed.”

